The Causal Effect of WIC on Breastfeeding: Partial Identification under Misclassification of Treatment Assignment [†‡]

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Abstract

Breastfeeding promotion and support is a core service of the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), which serves approximately half of all U.S. infants; however, program benefits include vouchers for infant formula, encouraging substitution away from breastfeeding among participating mothers. To estimate WIC's effect on breastfeeding, I formalize a partial identification approach to address misclassification of treatment assignment due to measurement error in the running variable, a common challenge in regression discontinuity designs that invalidates standard assumptions. Leveraging the program's income eligibility cutoff and data from the Pregnancy Risk Assessment Monitoring System (PRAMS), I bound the local average treatment effect (LATE) on breastfeeding initiation and duration. I find that WIC participation significantly reduces breastfeeding duration by 4 to 12 weeks, representing 30% to 90% of the mean among non-participants. In contrast, the effect on initiation is consistently null, suggesting that price-subsidized formula acts as a potent substitute for continued breastfeeding rather than a deterrent to the initial attempt. My findings are robust across various bandwidths and smoothness calibrations, providing informative bounds in a setting where conventional point identification fails due to coarse measurement of the running variable.

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1 Introduction

The Special Supplemental Nutrition Program for Women, Infants, and Children (WIC)—the third largest federally funded nutrition program—serves approximately half of all infants in the United States and 6.3 million participants monthly. While the program improves birth outcomes and maternal health during pregnancy (Carlson and Neuberger, 2017; Bitler and Currie, 2005; Kreider et al., 2016; Figlio et al., 2009; Hoynes et al., 2011), its effects on infant feeding practices remain unclear. Mandated to promote breastfeeding and support childhood nutrition, WIC provides both breastfeeding support services and quantity vouchers for infant formula to participating mothers. The existing academic literature provides no consensus on the program's net effect on breastfeeding: studies have found no effect (Jiang et al., 2010), substantial decreases (Ryan and Zhou, 2006; Bullinger and Gurley-Calvez, 2016; Topolyan and Xu, 2017), and even increases in breastfeeding (Joyce et al., 2008; Chatterji and Brooks-Gunn, 2004). Identifying WIC's causal effect poses a methodological challenge due to the non-random nature of program participation and limitations in the available survey data.

I examine how WIC participation affects breastfeeding initiation and duration among new mothers. Understanding these effects is critical due to the well-documented health benefits associated with breastfeeding for both infants and mothers, including reduced risks of childhood obesity, Type II diabetes, and certain cancers (Dieterich et al., 2013; Bartick and Reinhold, 2010).[1] Among income-eligible mothers, WIC participants are less likely to initiate breastfeeding (87% vs. 91%) and breastfeed for shorter durations (8.7 vs. 10.8 weeks) than non-participants. With WIC's broad reach, even modest changes in breastfeeding behavior may have substantial public health implications. In 2018, subsidies for infant formula accounted for 49.1% of the program's $4.5 billion in food spending before manufacturer rebates (18.2% of $2.8 billion after rebates) (Kline et al., 2020). The potential for WIC's formula subsidies to discourage breastfeeding—through reducing the relative price of formula—may undermine the program's nutritional objectives (Rose et al., 2006; Joyce et al., 2008).

Two problems have confounded previous attempts to identify WIC's causal effects on breastfeeding. First, participation in WIC is endogenous: unobserved infant feeding preferences affect a mother's decision to enroll in WIC. Mothers with stronger preferences for formula feeding may be more likely to enroll in the program precisely because it provides formula subsidies, and as a result, simple comparisons of observed differences between participants and non-participants offer a biased measure of the program's effect. Second, few datasets contain detailed measures of infant feeding outcomes, and the existing datasets are either small, retrospective, or suffer from

[1] While numerous observational studies document positive associations between breastfeeding and later health outcomes, the causal interpretation of these relationships is debated. See Oster (2013) for a popular discussion.

measurement error in key variables. My primary data source, the Pregnancy Risk Assessment and Monitoring System (PRAMS), is large and collected contemporaneously with infant feeding decisions but suffers from one important limitation common to survey data: household income is reported in rounded brackets, creating uncertainty regarding eligibility near WIC's income cutoff, set nationally at 185% of the federal poverty level (FPL).[2] This bracketed reporting complicates empirical designs, such as regression discontinuity designs (RDD), that rely on precise measurement of income and treatment assignment to leverage policy discontinuities as a source of local exogeneity, motivating the need for an alternative identification strategy.[3]

I extend the regression discontinuity framework by developing a novel partial-identification approach that accommodates mismeasurement in the running variable and misclassification in treatment assignment, enabling estimation of the local average treatment effect (LATE) of WIC participation on breastfeeding outcomes. My approach mirrors traditional RDD, using WIC's income eligibility threshold as a discontinuity in program participation. In contrast to conventional designs, my method requires neither precise knowledge of the assignment variable, continuity of potential outcomes at the threshold, nor locally exogenous treatment assignment. Instead, I impose minimal and empirically supported shape restrictions of two flexible forms: (1) sign restrictions and (2) magnitude restrictions on the slope of the conditional expectation function. My insight is that observations far from the threshold—where treatment status is known with certainty despite bracketed income—can be used to bound treatment effects near the threshold, where classification is uncertain. By imposing restrictions on the shape of conditional expectation functions across income, I extrapolate from regions with certain assignment to the local neighborhood around the cutoff. The resulting estimates are valid even under non-classical measurement error. This framework is well suited to applications where eligibility is measured with error or only partially observed, making it broadly applicable to the evaluation of means-tested programs using survey data.

To contextualize my findings, I implement three conventional benchmarks. First, I estimate a traditional fuzzy RDD using bracket midpoints as a proxy for true income, with the 185% FPL cutoff as an instrument for WIC participation. Second, I estimate a donut RDD, which excludes observations near the threshold to mitigate misreporting. Third, I implement a simulation-based RDD that explicitly models within-bracket income uncertainty by drawing uniformly within reported brackets. All three approaches rely on strong assumptions: the first two assume continuity of counterfactual outcomes and locally exogenous treatment assignment using mismeasured in-

[2]Two commonly used alternatives—the National Health and Nutrition Examination Survey (NHANES) and the WIC Infant and Toddler Feeding Practices Study (ITFPS)—are similarly limited. NHANES is small and retrospective, sampling only 5,000 individuals nationwide, and collects infant feeding data only for children under age five. ITFPS, while more detailed, samples only infant WIC participants, precluding comparison across program participation.

[3]Even when income is reported as a continuous value, misreporting or recall error can lead to misclassification of treatment assignment near program eligibility thresholds.

come, and the simulation imposes untestable distributional assumptions on within-bracket income.

My shape-restricted partial-identification estimates provide evidence that WIC participation significantly reduces duration of breastfeeding but does not affect the initiation of breastfeeding. Under weak and empirically grounded assumptions, I find that WIC participation reduces breastfeeding duration by between 4.5 and 12.3 weeks. The bounds for initiation range from -4.8 to +8.3 percentage points, failing to identify any non-null effect on initiation. The duration bounds suggest WIC's effect on compliers substantially exceeds the average difference observed between eligible and ineligible mothers. These bounds are robust to a range of slope restrictions and remain negative even under conservative assumptions about the rate of change in outcomes across income levels.

In contrast, all three conventional approaches exhibit considerable sensitivity to modeling choices and specification decisions. The simulation-based estimation, although conceptually appealing, produces extreme imprecision with first-stage F-statistics below standard thresholds. The traditional RDD fails to consistently identify the sign of the treatment effect at narrower bandwidths. The donut design yields statistically significant estimates but requires excluding all observations within approximately 52 percentage points of the threshold, imposing the strong assumption that conditional expectations observed outside this exclusion zone extend unchanged to the cutoff. My partial identification approach addresses these challenges by providing transparent bounds under minimal, empirically supported assumptions rather than imposing untestable restrictions.

My partial-identification approach contributes to the literature on inference under imperfect data, particularly regression discontinuity designs (RDD) affected by misclassified treatment assignment and measurement error in the running variable. While recent advances address various measurement issues in regression discontinuity designs—including heaping bias, discrete running variables, and rounding error—the specific problem of misclassified treatment assignment remains distinct and underexplored (Dong, 2015; Barreca et al., 2016; Kolesár and Rothe, 2018). A key limitation of existing methods is their reliance on correctly observed treatment status, which does not hold in many applied contexts with survey data. Standard polynomial selection methods are ill-suited for settings with error-prone running variables (Gelman and Imbens, 2019; Imbens and Wager, 2019; Pei et al., 2022), and existing solutions rely upon dropping misclassified observations, modeling manipulation patterns, or leveraging auxiliary datasets (Dong, 2015; Dong and Kolesár, 2023; Gerard et al., 2020; Davezies and Le Barbanchon, 2017). In my setting, none of these solutions are viable: income is reported in broad intervals; mismeasurement is bidirectional; and there are no comparable auxiliary data with which to validate assignment and treatment.

Building on the framework of Manski and Pepper (2018), who introduced bounded-variation assumptions for difference-in-differences designs, I develop a shape-restricted partial-identification

method tailored to RDDs with interval-reported running variables and misclassified treatment assignment. This approach offers a middle ground between the strong continuity and local randomization assumptions of traditional RDD and the conservatism of worst-case nonparametric bounds. My method is specifically designed to address the challenges of fuzzy designs with mismeasurement of treatment assignment, offering applied researchers a transparent and tractable framework for bounding treatment effects when point identification is not credible.

This paper is organized as follows. Section 2 describes the institutional details of WIC eligibility and benefit structure, and introduces the PRAMS data. Section 3 formalizes the identification challenges arising from interval-reported income and presents my shape-restricted partial-identification approach, including the theoretical assumptions and supporting evidence. Section 4 details the implementation of each empirical strategy, the partial-identification method and conventional benchmarks. Section 5 presents my main findings, with a particular focus on my partial-identification bounds under varying slope restrictions in comparison with the conventional benchmarks. Section 6 concludes the paper.

2 Institutional Setting and Data

2.1 WIC Program Overview

The Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) is a federal assistance program that aims to improve the health and nutrition of low-income pregnant and postpartum women, infants, and children under age five. In 2021, WIC served an average of 6.3 million participants per month, including approximately half of all infants born in the United States.[4] Eligibility requires: (1) household income below 185% of the Federal Poverty Level (FPL), (2) categorical eligibility as a pregnant or postpartum woman, infant, or child under five, (3) documented nutritional risk, and (4) state residency.[5] Participants in SNAP, Medicaid, or TANF are considered adjunctively income-eligible and are exempted from WIC's standard income requirement, enabling participation at higher incomes. Although cross-program participation is substantial—in 2020, 74% of WIC participants were enrolled in Medicaid and 28% in SNAP—only 3.5% reported household incomes above the program's standard income eligibility threshold of 185% of the Federal Poverty Level (USDA Food and Nutrition Service, 2022).

WIC provides a combination of nutrition education, referrals to public assistance programs and healthcare providers, and supplemental food packages. Food benefits include quantity vouchers—restricted to specific brands, package sizes, and other characteristics that vary by state—for des-

[4] WIC Program Overview (https://www.ers.usda.gov/topics/food-nutrition-assistance/wic-program/)
[5] WIC Eligibility (https://www.fns.usda.gov/wic/about-wic)

ignated food items, as well as cash-value vouchers for fresh produce. Throughout the first year of life, the quantity of infant formula provided is determined by the mother's self-reported feeding practice: *fully formula feeding*, *partially breastfeeding*, or *fully breastfeeding*. As shown in Panel A of Table 1, fully formula-feeding mothers receive up to 806 fluid ounces of infant formula monthly during the first three months, while fully breastfeeding mothers receive no formula. Partially breastfeeding mothers receive an intermediate allocation of up to 364 fluid ounces. These differences in formula provision create substantial pecuniary incentives: a mother who reports fully formula feeding receives formula worth approximately $160–$185 per month at retail prices, while a mother who reports fully breastfeeding receives none.[6] Although food packages may be adjusted if feeding practices change, such transitions are biologically constrained, and the cessation of breastfeeding is often irreversible due to declining milk production.[7]

WIC is federally mandated to promote and support breastfeeding as the preferred method of infant feeding. In line with this mandate, WIC provides educational programs and offers "anticipatory guidance, counseling, and breastfeeding educational materials" to participating pregnant mothers.[8] Postpartum, additional support includes: (1) education and counseling through group classes or individual sessions; (2) breastfeeding supplies, such as breast pumps and storage containers; and (3) peer counseling from experienced mothers (National WIC Association, 2019). The 2009 Food Package Revision—occurring before the period of my analysis—was designed, in part, to strengthen pecuniary incentives for breastfeeding. Following the reform, fully breastfeeding mothers became eligible for more comprehensive food benefits and extended postpartum participation—up to one year, compared to six months for mothers who do not breastfeed.

Despite these breastfeeding support services and policy reforms, the program's provision of infant formula creates countervailing incentives. The high cost of formula feeding—estimated at $1,600 to $2,100 during the first year—makes infant formula the most financially valuable component of the WIC food packages (Centers for Disease Control and Prevention, 2023; USDA Food and Nutrition Service, 2017).[9] By providing infant formula in-kind, the program reduces its relative price, thereby creating price incentives that may discourage breastfeeding initiation and encourage earlier cessation. Although the 2009 reforms increased both the initiation and exclusive duration of breastfeeding among WIC participants (Whaley et al., 2012), overall breastfeeding rates remain low: only 22% of WIC infants are still breastfeeding at 6 months, well below the CDC's 12-month recommendation (USDA Food and Nutrition Service, 2022). Across all levels of household in-

[6]Based on retail prices of $0.20–$0.23 per prepared fluid ounce in 2022 dollars. Prices vary by brand, formula type, and region.

[7]Regular milk expression is required to maintain supply and extended interruptions may result in permanent supply reduction.

[8]Breastfeeding Priority Programs (https://www.fns.usda.gov/wic/breastfeeding-priority-program)

[9]U.S. Surgeon General's U.S. 2011 Call to Action to Support Breastfeeding, adjusted to 2022 dollars (https://www.hhs.gov/surgeongeneral/reports-and-publications/breastfeeding/factsheet/index.html)

come, WIC participants are less likely to initiate breastfeeding and breastfeed for shorter durations than income-eligible non-participants (Figure 1).

The documented association between early introduction of infant formula and adverse health outcomes raises concerns about the health benefits and public value of WIC's in-kind provision of infant formula. Reduced breastfeeding duration has been linked to increased rates of childhood obesity, maternal obesity, breast cancer, and Type II diabetes (Dieterich et al., 2013; Bartick and Reinhold, 2010). These associations are economically meaningful: childhood obesity alone imposes substantial healthcare costs, and maternal health conditions affect labor force participation and productivity. Although WIC participation improves birth outcomes—including reduced rates of low birth weight, preterm birth, and infant mortality—and enhances maternal nutrition during pregnancy (Carlson and Neuberger, 2017; Bitler and Currie, 2005; Kreider et al., 2016; Figlio et al., 2009; Hoynes et al., 2011), the program's provision of infant formula may undermine these gains by reducing breastfeeding, which confers its own protective health benefits. Given the potential negative effects of reduced breastfeeding, the net public value of WIC's infant food packages is ambiguous.

2.2 PRAMS Data

This study uses data from the Pregnancy Risk Assessment and Monitoring System (PRAMS), a surveillance project conducted jointly by the Centers for Disease Control and Prevention (CDC) and participating state health departments. PRAMS collects state-specific, population-based data on maternal attitudes, behaviors, and experiences before, during, and shortly after pregnancy.[10] Mothers are surveyed 2–6 months postpartum, with annual state-level sample sizes ranging from approximately 1,000 to 3,000 respondents. The dataset links birth certificate records with survey responses collected through personalized mailing packets, use of incentives and rewards, and telephone follow-up for mail non-respondents (Shulman et al., 2018). To ensure adequate representation of key subpopulations, participating states employ stratified sampling that oversamples "mothers of infants with low-birthweight, those living in high-risk geographic areas, and racial/ethnic minority groups" (Shulman et al., 2018).

My analysis focuses on PRAMS Phases 6–8 (2009–2022), a period selected for two key reasons: the consistent structure of the WIC infant food packages following its October 2009 revision and the systematic collection of WIC participation data. The PRAMS questionnaire includes core questions asked across all participating states (55–60% of content), standardized CDC-developed questions, and state-specific additions (Shulman et al., 2018). From an initial sample of 508,387 mothers with live births, I construct the analytical sample by excluding observations with missing

[10] PRAMS Overview (https://www.cdc.gov/prams/index.htm). State participation in PRAMS is voluntary, and as of 2023, California, Idaho, North Carolina, and Ohio are the only states not participating.

values for any of four key variables: breastfeeding initiation, breastfeeding duration, WIC participation during pregnancy, and household income relative to the Federal Poverty Level (FPL). This yields a final sample of 424,796 observations. Additional details on sample construction, including variable-specific sample sizes, and geographic coverage appear in Appendix E.

I examine two primary breastfeeding outcomes: initiation (a binary indicator) and duration (weeks of breastfeeding). Treatment is defined as WIC participation during pregnancy, recorded on birth certificates. PRAMS provides survey weights designed to produce estimates representative of each state's population of women with live births. I present results both with and without survey weights to assess sensitivity. Table 2 summarizes key variables by eligibility status, and detailed variable definitions appear in Appendix E.2.

2.3 Income Bracketing and Measurement

PRAMS reports income in rounded brackets that vary in both width and number across survey phases. Each bracket defines upper and lower bounds for household income, with the highest income bracket left open-ended (e.g., income $>$ \$100,000). I combine these brackets with household size to calculate upper (u_i) and lower (ℓ_i) bounds on each respondent's household income as a percentage of the Federal Poverty Level (FPL), adjusted for state- and year-specific thresholds. Additional details on income bracket structure across PRAMS phases appear in Appendix E.3.

The coarseness of bracketed income reporting creates a fundamental challenge: for many observations near the 185% FPL eligibility threshold, the reported income bracket *straddles* the threshold, meaning the lower bound falls below 185% FPL and the upper bound exceeds it. For these "straddling" observations, true eligibility status is unknown, as the respondent's actual income could place them on either side of the cutoff. Formally, observation i straddles the threshold if $\ell_i \leq 185 < u_i$.

The magnitude of straddling is substantial. Figure 2 displays the distribution of reported income brackets near the threshold, showing both total observations and straddlers by income bin, with kernel density estimates on each side of the cutoff. The average bracket width is approximately 52 percentage points of FPL. Among observations with bracket midpoints within 10 percentage points of the eligibility threshold, 97% have bounds that straddle the cutoff, rendering treatment assignment uncertain. Comparing observation counts with and without straddling observations reveals the severity of this measurement problem: the majority of observations near the threshold cannot be definitively classified as eligible or ineligible (Figure 3).

Despite this measurement challenge, the relationship between income and WIC participation exhibits a clear pattern. Household income and WIC participation have a negative relationship: as income increases, average program participation decreases. Figure 4 documents both this re-

lationship and a distinct discontinuity at the 185% FPL eligibility threshold—participation drops sharply as income crosses the cutoff from below. The pattern and threshold discontinuity persist when varying bandwidth and excluding straddling observations. Within eligibility groups, the relationship appears approximately linear, although the slope differs between eligible and ineligible households. This approximately linear negative gradient is consistent with patterns observed in the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC) (Flood et al., 2022).

Breastfeeding outcomes also vary systematically with income. On average, both initiation and duration increase with household income (Figures 5 and 6). The figures reveal positive income gradients for both outcomes that persist when excluding straddling observations. Within eligibility groups, the income-breastfeeding relationship appears approximately linear, although slopes may differ between eligible and ineligible households.

The high prevalence of straddling near the eligibility threshold creates challenges for causal identification using regression discontinuity methods. Given the inability to precisely classify observations as eligible or ineligible, standard RDD approaches that rely on exact treatment assignment may yield biased estimates, an issue discussed in more detail in Section 3.

2.4 Summary Statistics

Table 2 summarizes characteristics of the PRAMS sample by eligibility status. I classify households into three groups based on their FPL bounds: *eligible* households have an upper bound at or below 185% FPL; *ineligible* households have a lower bound above 185% FPL; and *straddlers* have bounds that cross the threshold. This classification reveals sharp differences across the eligibility cutoff that motivate and support the regression discontinuity design.

Eligible and ineligible households differ substantially along socioeconomic dimensions. Eligible households are less likely to have a college-educated mother (8.4% versus 63.5%), less likely to be married (37.4% versus 85.8%), and more likely to have Medicaid coverage (85.9% versus 14.3%). These differences reflect the strong correlation between income and other socioeconomic characteristics. Straddlers exhibit intermediate characteristics: 23.0% have college-educated mothers and 62.0% are married, falling between the eligible and ineligible groups.

Employment and leave patterns also differ markedly across eligibility groups. Although employment rates before pregnancy are comparable (61.3% for eligible versus 69.1% for ineligible), access to paid leave diverges sharply: only 23.3% of eligible mothers report taking paid leave, compared to 61.9% of ineligible mothers. This disparity is consequential for breastfeeding, as mothers without paid leave face greater pressure to return to work quickly, limiting opportunities for sustained breastfeeding. Eligible households also exhibit higher rates of low birth weight

(25.2% versus 20.4%) and preterm delivery (20.4% versus 17.3%), conditions that can complicate breastfeeding initiation and continuation.

WIC participation varies sharply across the eligibility threshold, consistent with the program's income-based targeting. Among eligible households, 74.7% participate in WIC during pregnancy, compared to only 10.6% of ineligible households. Straddlers participate at intermediate rates (50.0%), reflecting their ambiguous eligibility status.

Breastfeeding outcomes also exhibit a clear difference across eligibility groups. Eligible households have lower breastfeeding initiation rates (88.1% versus 96.1%) and shorter durations (9.2 versus 12.7 weeks) than ineligible households. Straddlers again fall between the two groups, with 93.5% initiation and 11.0 weeks average duration. These patterns are also evident in the binned scatter plots (Figures 5 and 6), which show breastfeeding outcomes increasing with income and a visible discontinuities at the eligibility threshold.

The sharp differences across the eligibility threshold—in both WIC participation and breastfeeding outcomes—suggest that the 185% FPL cutoff provides useful identifying variation. However, the straddler population presents a challenge. These 29,763 households—representing 6% of the sample—cannot be definitively classified as eligible or ineligible. Section 3 develops a partial identification framework that addresses this challenge by bounding treatment effects without requiring precise eligibility classification.

3 Identification

This section formalizes my partial-identification framework and outlines the identification challenges posed by interval-reported income. I impose empirically grounded shape restrictions on the conditional expectation functions to bound the limits of mean participation and outcomes at the threshold. Under these assumptions, the local average treatment effect (LATE) is partially identified.

3.1 Notation and the Identification Conundrum

I estimate the causal effect of WIC participation ($T \in \{0, 1\}$) on breastfeeding outcomes (Y). Under the potential outcomes framework, the observed outcome is $Y = Y_0 + T_Z(Y_1 - Y_0)$, where Y_0 and Y_1 represent outcomes under untreated and treated states, and $Z \in \{0, 1\}$ denotes treatment assignment.

WIC eligibility is determined by household income relative to the Federal Poverty Level (FPL). Let $X^\star$ denote household income relative to the 185% FPL threshold, centered so that the cutoff is at zero. True eligibility assignment is $Z = \mathbb{1}[X^\star \leq 0]$. PRAMS reports income in intervals $[\ell_i, u_i]$

such that $X_i^\star \in [\ell_i, u_i]$. Based on these intervals, I partition the population into three mutually exclusive sets:

$$\begin{aligned} \mathcal{X}_E &= \{i : u_i \leq 0\} && \text{(Certain Eligibility)} \\ \mathcal{X}_I &= \{i : \ell_i > 0\} && \text{(Certain Ineligibility)} \\ \mathcal{S} \quad &= \{i : \ell_i \leq 0 < u_i\} && \text{(Ambiguous Status/Straddlers)} \end{aligned}$$

Households in $\mathcal{X}_E$ are certainly eligible ($Z = 1$), households in $\mathcal{X}_I$ are certainly ineligible ($Z = 0$), and for straddlers in $\mathcal{S}$, true eligibility depends on the unobserved location of $X^\star$ within the bracket.

Following Dong and Kolesár (2023), I define the conditional probability of treatment under assignment $z \in \{0, 1\}$ as $p_z^\star(x) := \mathbb{E}[T_z | X^\star = x]$ and the observed treatment probability as

$$p^\star(x) := \mathbb{E}[T|X^\star = x] = \mathbb{1}[x > 0]p_0^\star(x) + \mathbb{1}[x \leq 0]p_1^\star(x)$$

Similarly, let $g_t^\star(x) := \mathbb{E}[Y_t | X^\star = x]$ denote the conditional mean of the potential outcomes, where $t \in \{0, 1\}$ denotes treatment, and define the conditional mean of the observed outcomes as

$$g^\star(x) := \mathbb{E}[Y|X^\star = x] = g_0^\star(x) + p^\star(x)[g_1^\star(x) - g_0^\star(x)]$$

The conventional fuzzy regression discontinuity design relies on the following assumptions:

$$\begin{aligned} &\text{(i) Monotonicity:} \quad \mathbb{P}(T_1 \geq T_0 | X^\star = 0) = 1, \quad \text{with } \mathbb{P}(T_1 > T_0 | X^\star = 0) > 0 \\ &\text{(ii) Continuity:} \quad p_z^\star(x) \text{ and } g_t^\star(x) \text{ are continuous at } x = 0 \text{ for } z, t \in \{0, 1\} \end{aligned}$$

Under these assumptions, the object of interest, the local average treatment effect (LATE) at the threshold, is identified as:

$$\tau = \frac{\lim_{x \uparrow 0} g^\star(x) - \lim_{x \downarrow 0} g^\star(x)}{\lim_{x \uparrow 0} p^\star(x) - \lim_{x \downarrow 0} p^\star(x)}$$

However, interval reporting induces measurement error in the running variable, as true household income is only known to fall within the reported bracket. I observe a noisy proxy $X = X^\star - e$, where e denotes measurement error.[11] Unlike the setting analyzed by Dong and Kolesár (2023), measurement error not only affects the precision with which I observe the running variable but also may misclassify treatment assignment for straddlers. When using the proxy X, I observe assignment $\tilde{Z} = \mathbb{1}[X \leq 0]$ rather than true assignment $Z = \mathbb{1}[X^\star \leq 0]$ for observations where X and $X^\star$ lie on opposite sides of the cutoff.

[11] In my context, I will use the midpoint of each household's reported FPL interval as the proxy for true income

Under the potential outcomes framework, the observed conditional means based on X are given by:

$$\tilde{p}(x) := \mathbb{E}[T \mid X = x] = \int_{-\infty}^{\infty} p^{\star}(x^{\star}) f_{X^{\star}|X}(x^{\star}|x) dx^{\star}$$
$$\tilde{g}(x) := \mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} g^{\star}(x^{\star}) f_{X^{\star}|X}(x^{\star}|x) dx^{\star}$$

The conventional regression discontinuity estimator using the noisy measure is:

$$\tilde{\tau} = \frac{\lim_{x \uparrow 0} \tilde{g}(x) - \lim_{x \downarrow 0} \tilde{g}(x)}{\lim_{x \uparrow 0} \tilde{p}(x) - \lim_{x \downarrow 0} \tilde{p}(x)}$$

This estimator does not generally recover the true LATE. The discontinuities in $\tilde{g}(x)$ and $\tilde{p}(x)$ at $X = 0$ reflect an unobserved, weighted average of income-eligible and ineligible households. The resulting bias depends on misclassification probabilities and differences in potential outcomes at the threshold. Without knowledge of the distribution of measurement error e, neither the direction nor magnitude of bias is identified. Thus, standard RDD estimators do not yield valid causal estimates, motivating my partial-identification approach to establish bounds without requiring assumptions on the error distribution.

3.2 Partial Identification Framework

My strategy focuses on projecting from regions of certain assignment. Within $\mathcal{S}$, eligible and ineligible units are indistinguishably mixed, precluding local comparison. I instead rely on shape restrictions—monotonicity and smoothness—and observations in the certainty regions $\mathcal{X}_E$ and $\mathcal{X}_I$ to define feasible paths for the conditional expectation functions $p^{\star}(x)$ and $g^{\star}(x)$ near the income eligibility threshold.

Figure 9 illustrates this concept. Observations in the certainty regions serve as anchors, and shape restrictions define a feasible cone of potential values for the unobserved limits at the cutoff. This approach exchanges point identification for valid set identification under empirically supported constraints.

I identify LATE using three assumptions:

Assumption 1 (Positive First Stage)**.** Income eligibility increases treatment take-up at the threshold:

$$\tau_T := \lim_{x \uparrow 0} p^{\star}(x) - \lim_{x \downarrow 0} p^{\star}(x) > 0$$

Assumption 2 (Monotonicity in Income)**.** For any $x^-, x^+ \in \mathbb{R}$ with $x^- \leq x^+$ such that $0 \notin [x^-, x^+]$:

$$\text{(i)} \quad p^\star(x^-) \geq p^\star(x^+) \quad \text{(WIC participation is weakly decreasing in income)}$$
$$\text{(ii)} \quad g^\star(x^-) \leq g^\star(x^+) \quad \text{(Breastfeeding is weakly increasing in income)}$$

Assumption 3 (Lipschitz Smoothness)**.** Let $\mathcal{R}_1 = [-\delta_1, 0]$ and $\mathcal{R}_0 = (0, \delta_0]$ denote neighborhoods on the eligible and ineligible sides of the threshold, respectively. For $z \in \{0, 1\}$ and all $x_1, x_2 \in \mathcal{R}_z$:

$$\text{(i)} \quad |p^\star(x_1) - p^\star(x_2)| \leq \lambda_z \cdot |x_1 - x_2|$$
$$\text{(ii)} \quad |g^\star(x_1) - g^\star(x_2)| \leq \kappa_z \cdot |x_1 - x_2|$$

where $\kappa_z, \lambda_z > 0$ are assignment-specific Lipschitz constants.

The Lipschitz constants κ_z and λ_z control the maximum rate of change in outcomes and participation. These constants are calibrated empirically from observed slopes in $\mathcal{X}_E$ and $\mathcal{X}_I$, with details provided in Section 4.

3.3 Empirical Support for Assumptions

These assumptions are grounded in economic theory and supported by empirical regularities in the data. Assumption 1 reflects that removing a constraint on program participation—moving from income ineligibility to eligibility—should weakly increase participation rates. Empirically, Figure 4 shows a discontinuous increase in WIC participation at the eligibility threshold, with consistently higher participation rates observed among income-eligible households. This pattern is consistent with administrative data and prior analyses: WIC Participant Characteristics Reports and survey-based analyses document that program take-up is concentrated below the 185% FPL threshold, with participation rates declining steadily as income rises (USDA Food and Nutrition Service, 2022; Hoynes and Schanzenbach, 2015).

Assumption 2 imposes monotonicity in income for both treatment take-up and outcomes. As income rises, program benefits represent a decreasing share of household resources, decreasing the value of participation. In addition to the discontinuity at the eligibility threshold, Figure 4 shows a decrease in the average rate of participation as income increases. With respect to infant feeding practices, mothers in higher-income households have greater access to paid leave, flexible work arrangements, professional lactation support, and familial resources that facilitate breastfeeding

(Table 2). Existing research documents strong socioeconomic gradients in breastfeeding behaviors, with higher-income and higher-education mothers initiating and sustaining breastfeeding at higher rates (American College of Obstetricians and Gynecologists, 2021; Diaz et al., 2023). Figures 5 and 6 reveal increasing average rates for the initiation and duration of breastfeeding as household income rises, both across the income distribution and within assignment groups.

Assumption 3 places limits on the rate at which these conditional expectations may change with household income. Changes in participation and breastfeeding behaviors are driven by structural and socioeconomic forces—program eligibility, workplace policies, and access to information—that evolve smoothly with income. Determinants such as paid leave, flexible work arrangements, health literacy, and social support are closely tied to income and accumulate gradually across the income distribution (American College of Obstetricians and Gynecologists, 2021; Diaz et al., 2023). Figures 4, 5, and 6 display smooth relationships without abrupt changes or irregularities, and within-partition patterns appear approximately linear. The gradual evolution of these characteristics across income levels provides empirical support for the bounded-slope restriction in local neighborhoods. I assess the sensitivity of my results to these calibrations in Section 5.

3.4 Main Identification Result

Under Assumptions 1–3, I derive bounds on the unobserved limits of the conditional expectation functions at the threshold. Define the left and right limits as:

$$p^\star(0^-) := \lim_{x\uparrow 0} p^\star(x), \quad p^\star(0^+) := \lim_{x\downarrow 0} p^\star(x)$$
$$g^\star(0^-) := \lim_{x\uparrow 0} g^\star(x), \quad g^\star(0^+) := \lim_{x\downarrow 0} g^\star(x)$$

so that the discontinuities in treatment take-up and outcomes are $\tau_T = p^\star(0^-) - p^\star(0^+)$ and $\tau_Y = g^\star(0^-) - g^\star(0^+)$, respectively.

I bound these limits by projecting from observed conditional expectations in the certainty regions. Let x^- denote any point in $\mathcal{X}_E$ and x^+ denote any point in $\mathcal{X}_I$. For certainly eligible households, monotonicity and Lipschitz smoothness imply:

$$p^\star(0^-) \in [p^\star(x^-) - \lambda_1 \cdot |x^-|, \quad p^\star(x^-)]$$
$$g^\star(0^-) \in [g^\star(x^-), \quad g^\star(x^-) + \kappa_1 \cdot |x^-|]$$

For certainly ineligible households:

$$p^\star(0^+) \in [p^\star(x^+), \quad p^\star(x^+) + \lambda_0 \cdot x^+]$$
$$g^\star(0^+) \in [g^\star(x^+) - \kappa_0 \cdot x^+, \quad g^\star(x^+)]$$

Combining these one-sided projections yields bounds on the discontinuities:

$$\tau_T \in \left[p^\star(x^-) - p^\star(x^+) - \lambda_0 \cdot x^+ - \lambda_1 \cdot |x^-|, \quad p^\star(x^-) - p^\star(x^+)\right]$$
$$\tau_Y \in \left[g^\star(x^-) - g^\star(x^+), \quad g^\star(x^-) - g^\star(x^+) + \kappa_1 \cdot |x^-| + \kappa_0 \cdot x^+\right]$$

Denote the resulting intervals as $[\tau_T^{LB}, \tau_T^{UB}]$ and $[\tau_Y^{LB}, \tau_Y^{UB}]$. The ratio of these bounds partially identifies LATE.

Proposition 3.1 (Partial Identification of LATE)**.** *Under Assumptions 1–3, the local average treatment effect $\tau = \tau_Y/\tau_T$ is partially identified within:*

$$\tau \in \begin{cases} \left[\dfrac{\tau_Y^{LB}}{\tau_T^{UB}}, \dfrac{\tau_Y^{UB}}{\tau_T^{LB}}\right] & \text{if } \tau_Y^{LB} \geq 0 \\ \left[\dfrac{\tau_Y^{LB}}{\tau_T^{LB}}, \dfrac{\tau_Y^{UB}}{\tau_T^{UB}}\right] & \text{if } \tau_Y^{UB} < 0 \\ \left[\dfrac{\tau_Y^{LB}}{\tau_T^{LB}}, \dfrac{\tau_Y^{UB}}{\tau_T^{LB}}\right] & \text{if } \tau_Y^{LB} < 0 < \tau_Y^{UB} \end{cases}$$

Since $\tau_T > 0$ by Assumption 1, the sign of the effect is determined by τ_Y. When the upper and lower bounds on the outcome are both positive or both negative, the widest valid interval is obtained by dividing the more extreme numerator by the less extreme denominator. When the outcome bounds straddle zero, I conservatively divide by τ_T^{LB} in both expressions to ensure coverage.

My approach departs from the traditional regression discontinuity design (RDD) assumption of exogeneity, which requires that households cannot precisely manipulate their income to fall just below the eligibility threshold. Instead, I exclude observations near the threshold, where manipulation is most likely, and restrict attention to households with unambiguous treatment assignment. Households capable of sorting to just below the threshold in response to incentives are thus omitted from my identification strategy. By imposing shape restrictions on the conditional expectation functions (CEF), I extrapolate from manipulation free regions to bound treatment effects at the threshold. In so doing, I trade the local randomization assumption of the standard RDD for smoothness and monotonicity restrictions on the CEFs. As long as the first stage is positive, the lo-

cal average treatment effect (LATE) is well-defined, and the resulting bounds remain valid—even in the presence of sorting around the eligibility threshold. Unlike conventional RDD approaches, my method does not require dense support or local randomization near the eligibility threshold. Identification is achieved through shape restrictions rather than localized comparisons, and my bounding procedure avoids reliance on narrow bandwidths and remains valid even with sparsely populated neighborhoods around the cutoff.

4 Empirical Strategy

This section outlines my empirical strategy for estimating the causal effect of WIC participation on breastfeeding outcomes. I begin with my primary partial-identification approach, which bounds the local average treatment effect (LATE) under flexible, empirically supported assumptions. To benchmark these results, I also implement three alternative designs—a traditional fuzzy regression discontinuity design (RDD), a "donut" RDD, and a simulation-based approach—each of which provides point identification under progressively stronger assumptions.

4.1 Partial Identification Implementation

I implement the partial identification strategy using a four-step procedure that estimates the components of LATE bounds defined in Proposition 3.1. This framework uses households with unambiguous treatment assignment—those whose income brackets lie entirely above or below the eligibility threshold–and shape restrictions on the CEF to bound treatment effects at the threshold.

Step 1: Binning and Sample Means

For a given bandwidth h, I partition observations across the sample space into non-overlapping bins of width h, tiling outward from the eligibility threshold over the income proxy X_i. Within each bin, I compute sample means of treatment take-up and the outcome variable. These binned means serve as nonparametric estimates of the conditional expectation functions $p^\star(x)$ and $g^\star(x)$ and form the basis for all subsequent steps.

Step 2: Reference Bin Selection

I identify reference bins—the nearest bins on each side of the threshold for which treatment assignment is unambiguous on average. On the left, I select the first bin b_h^- where the average household is certainly income-eligible ($\frac{1}{n_b}\sum_{i\in b} u_i < 0$). On the right, I select the first bin b_h^+ where the average household is certainly ineligible ($\frac{1}{n_b}\sum_{i\in b} \ell_i > 0$). Each reference bin must contain at least

100 observations. Under Assumption 2, the sample means in these bins provide one-sided monotonicity bounds on the unobserved conditional expectations at the threshold. As WIC participation is weakly decreasing and breastfeeding outcomes are weakly increasing in income, $\hat{p}(b_h^-)$ bounds $p^\star(0^-)$ from above, and $\hat{g}(b_h^-)$ bounds $g^\star(0^-)$ from below. Analogous inequalities hold on the ineligible side.

Step 3: Calibration of Shape Constraints

To determine how far the conditional expectations may deviate between the reference bins and the threshold, I calibrate the Lipschitz constants λ (treatment) and κ (outcomes) from observed patterns in the certainty regions $\mathcal{X}_E$ and $\mathcal{X}_I$. Separately in each region, I estimate local linear fits across the binned means of treatment take-up and outcomes. The fitted slopes capture the rate of change in the conditional expectation functions across the income distribution. I set each Lipschitz constant equal to a specific empirical quantile q of the absolute fitted slopes, reporting results for $q \in \{0.50, 0.75, 0.90\}$. This calibration ensures that the smoothness restrictions imposed near the threshold are consistent with variation observed elsewhere in the income distribution.

Step 4: Projection and Bound Construction

I project from the reference bin means to the eligibility threshold using both shape restrictions. Monotonicity constrains the direction: $p^\star(0^-)$ cannot exceed $\hat{p}(b_h^-)$, and the calibrated Lipschitz constant λ limits how far below it may fall over the distance from b_h^- to the threshold. Together, these constraints define a feasible interval for $p^\star(0^-)$. Analogous intervals are constructed for $p^\star(0^+)$, $g^\star(0^-)$, and $g^\star(0^+)$. Differencing the left and right limit intervals yields bounds $[\hat{\tau}_T^{LB}, \hat{\tau}_T^{UB}]$ and $[\hat{\tau}_Y^{LB}, \hat{\tau}_Y^{UB}]$ on the first-stage and reduced-form discontinuities, which combine to form the identified set for LATE as defined in Proposition 3.1.

Inference and Parameters

To account for sampling uncertainty in this multi-step procedure, I construct confidence intervals using the method of Imbens and Manski (2004), which is valid for set-identified parameters. I perform 1,000 subsampling draws to estimate the sampling distribution of the bounds and construct 95% confidence intervals for the true parameter.

The choice of bandwidth h involves a trade-off between the two shape restrictions. Wider bandwidths yield more observations per reference bin, producing more precisely estimated monotonicity bounds. However, wider bins also mean fewer bins tile the certainty regions, reducing the number of local linear fits available for calibrating the Lipschitz constants. Conversely, narrower

bandwidths provide richer slope information for calibration but noisier reference bin means. Unlike conventional RDD, where MSE-optimal bandwidths assume precise measurement, the presence of interval censoring renders standard selectors theoretically ambiguous. I therefore use a fixed baseline bandwidth of $h = 10$ percentage points of the FPL, which focuses the analysis on the population within approximately ± 10 percentage points of the threshold (175–195% FPL). To assess sensitivity, I report results across a "ladder" of bandwidths ranging from $h = 5$ to $h = 30$.

4.2 Benchmark Designs

I benchmark the partial identification results against three alternative approaches: a standard fuzzy RDD, a donut RDD that excludes observations with ambiguous treatment assignment, and a simulation-based RDD that models income uncertainty via Monte Carlo draws. The first two approaches use the bracket midpoint $X_i = (\ell_i + u_i)/2$ as the running variable. The simulation instead draws point-identified income values from within each bracket. All three exploit the discontinuity in WIC eligibility at the threshold.

Conventional Fuzzy RDD

I estimate a standard fuzzy RDD, instrumenting for WIC participation T_i with income eligibility $Z_i = \mathbb{1}[X_i \leq 0]$:

$$Y_i = \alpha + \tau T_i + f^+(X_i) + Z_i \cdot f^-(X_i) + \varepsilon_i \tag{1}$$

where f^+ and f^- are polynomials of order p allowing the relationship between income and outcomes to vary on each side of the threshold.

Identification requires two standard assumptions:

(i) Monotonicity: $\mathbb{P}(T_1 \geq T_0 \mid X = 0) = 1$ and $\mathbb{P}(T_1 > T_0 \mid X = 0) > 0$

(ii) Continuity: $p_z(x)$ and $g_t(x)$ are continuous at $x = 0$ for $z, t \in \{0, 1\}$

Monotonicity requires that income eligibility weakly increases WIC participation for all households, with a strictly positive effect for some. Continuity requires that both the conditional probability of participation, $p_z(x) = \mathbb{E}[T_z \mid X = x]$, and conditional mean outcomes, $g_t(x) = \mathbb{E}[Y_t \mid X = x]$, are continuous at the threshold. Under these conditions, τ identifies the local average treatment effect among compliers at the eligibility cutoff. As discussed in Section 3, the systematic misclassification of eligibility for households near the eligibility threshold threatens identification.

Donut RDD

To address the ambiguity of treatment assignment near the threshold, I implement a donut RDD that excludes a large subset of observations near the cutoff. Let $\mu = \frac{1}{N}\sum_{i=1}^{N}(u_i - \ell_i)$ denote the average bracket width, approximately 52 percentage points in my sample. I retain only households whose brackets fall entirely below the threshold ($u_i < -\mu$) or entirely above it ($\ell_i > \mu$), and estimate equation (1) on the subsample $|X_i| \in [\mu, \mu + h]$.

This approach eliminates the primary source of misclassification bias but strengthens the identifying assumptions. While the standard fuzzy RDD requires continuity only at the threshold, the donut RDD must extrapolate relationships estimated from observations approximately $\mu \approx 52$ percentage points away. This requires the stronger assumption that the conditional expectations observed outside the exclusion zone extend unchanged through it to the threshold.

Simulation RDD

As a third benchmark, I implement a simulation-based RDD that explicitly models income uncertainty using Monte Carlo methods. For each household with reported income bracket $[\ell_i, u_i]$, I draw simulated income values from a uniform distribution:

$$X_{i,s}^{\star} \sim \text{Uniform}(\ell_i, u_i),$$

where $s \in \{1, \ldots, 1000\}$ indexes the simulation draws. Each draw yields a complete dataset with point-identified income values, from which I assign treatment eligibility $Z_{i,s} = \mathbb{1}[X_{i,s}^{\star} \leq 0]$ and estimate equation (1) as a standard fuzzy RDD. This produces 1,000 treatment effect estimates $\{\hat{\tau}_s\}_{s=1}^{1000}$. I report the mean estimate $\bar{\tau} = \frac{1}{1000}\sum_{s=1}^{1000}\hat{\tau}_s$ as the simulation-based LATE, with 95% confidence intervals defined by the 2.5th and 97.5th percentiles of the empirical distribution.

This approach retains all observations—including those nearest the threshold—and explicitly accounts for uncertainty in treatment assignment. However, it has two important limitations. First, the uniform distributional assumption is untestable: the true within-bracket distribution of income is unknown, and results are conditional on this modeling choice. Second, the width of the income brackets—approximately 52 percentage points on average—means that each household's simulated income can vary enormously across draws, injecting substantial noise into both the running variable and the eligibility instrument. This noise attenuates the first stage and inflates standard errors, particularly at narrower bandwidths where fewer observations anchor the estimates. The simulation RDD is therefore not a robustness check for the partial identification bounds but a separate benchmark that offers point-identified comparisons under strong structural assumptions.

I estimate all three benchmark designs across the same bandwidth ladder as the partial identifi-

cation analysis ($h \in \{5, 10, 15, 20, 25, 30\}$), with the linear specification ($p = 1$) as the primary specification and quadratic ($p = 2$) results reported in the Appendix. The standard RDD restricts the sample to $|X_i| \leq h$, the donut RDD to $|X_i| \in [\mu, \mu + h]$, and the simulation RDD to $|X^\star_{i,s}| \leq h$.

5 Results

I present results in six parts. Section 5.1 reports the main partial identification estimates, which reveal a robust negative effect of WIC on breastfeeding duration and a consistently null effect on initiation. Section 5.2 examines the sensitivity of these findings to the calibration of the Lipschitz smoothness constraints. Section 5.3 compares the partial identification bounds to conventional RDD benchmarks, where weak instruments and specification sensitivity limit the informativeness of point-identified alternatives. Section 5.4 explores heterogeneity across demographic subgroups, Section 5.5 reports robustness checks including covariate balance; pre-pandemic samples; and placebo thresholds, and Section 5.6 discusses the implications of the findings.

5.1 Main Partial Identification Estimates

I find that WIC participation reduces the duration of breastfeeding but has no identifiable effect on breastfeeding initiation. Table 3 presents the main unweighted results across the bandwidth ladder, using the median calibration quantile ($q = 0.50$).

The first-stage bounds confirm a strong discontinuity in program participation at the 185% FPL eligibility threshold. At the baseline bandwidth ($h = 10$), program eligibility increases the probability of WIC participation by 14.0 to 25.1 percentage points. The first-stage bounds remain strictly positive across the entire bandwidth ladder, with the lower bound ranging from 5.3 percentage points at $h = 5$ to 27.3 percentage points at $h = 30$.

The partial identification estimates for breastfeeding duration reveal a large negative treatment effect. At the baseline bandwidth, WIC participation reduces breastfeeding duration by between 4.5 and 12.3 weeks. The identified bounds remain strictly negative at all bandwidths, ranging from $[-12.7, -1.7]$ weeks at $h = 5$ to $[-4.7, -0.7]$ weeks at $h = 30$. For bandwidths between 10 and 25 percentage points, the 95% Imbens-Manski confidence intervals also exclude zero. Figure 10b presents results across a wider range of bandwidths, revealing similar results for both bounds and confidence intervals.

In contrast, the estimated effect of WIC on breastfeeding initiation is consistently null. At the baseline bandwidth, the effect on initiation is bounded between -4.8 and 8.3 percentage points, with a 95% confidence interval of $[-15.3, 21.8]$. The identified bounds encompass zero at every bandwidth. Figure 10a reinforces this pattern, with the identification region straddling zero across

the entire range examined.

The results are qualitatively similar when using survey weights. Weighted estimates, provided in Table 14 of the Appendix, yield a larger effect size for the reduction in duration at the baseline bandwidth (-14.0 to -5.8 weeks) but maintain the same patterns of sign and significance across the range of bandwidths examined. In summary, I find no evidence that WIC affects breastfeeding initiation and abundant evidence that it reduces the duration of breastfeeding.

5.2 Sensitivity to Calibration Quantile

The identified bounds for breastfeeding duration remain strictly negative across all calibration quantiles. Table 3 reports sensitivity to the calibration quantile q, which governs the permitted slopes in the unobserved region near the threshold. Increasing q from the median (0.50) to the 75th and 90th percentiles loosens the Lipschitz constraints and widens the bounds. At the baseline bandwidth ($h = 10$), the bounds shift from $[-12.3, -4.5]$ weeks at $q = 0.50$ to $[-12.8, -4.3]$ at $q = 0.75$ and $[-13.2, -4.2]$ at $q = 0.90$. The identification region remains entirely negative in every case, and the sign of the duration effect is robust to more extreme counterfactual slopes than those generally observed in the certainty regions.

The statistical significance of the duration effect is sensitive to q at the baseline bandwidth. The 95% Imbens-Manski confidence interval excludes zero at $q = 0.50$ but includes zero at $q = 0.75$ and $q = 0.90$ for $h = 10$ (intervals of $[-17.5, 0.9]$ and $[-18.3, 0.4]$, respectively). This sensitivity is confined to the baseline bandwidth, and at $h = 15$ and $h = 25$, the confidence intervals exclude zero even at $q = 0.90$. Figures 16b and 17b in the Appendix confirm that the negative duration effect persists across the bandwidth range under these more flexible shape restrictions.

For breastfeeding initiation, the null finding holds at all calibration quantiles. At $h = 10$, the identified bounds at $q = 0.90$ are $[-5.2, 9.2]$ percentage points, and the bounds and 95% confidence intervals encompass zero at every bandwidth, ruling out that the null result is an artifact of overly restrictive smoothness assumptions.

5.3 Benchmark Comparisons

The conventional benchmark designs highlight both the strengths and limitations of point-identifying methods in this setting. Tables 4 and 5 compare the partial identification bounds to point estimates from the standard fuzzy RDD, donut RDD, and simulation RDD.

For breastfeeding duration, the standard RDD estimates a reduction of 4.1 weeks at the baseline bandwidth ($h = 10$), which falls within the partial identification bounds (Table 5); however, the sign and magnitude of the standard RDD estimates are unstable across the range of bandwidths examined. At $h = 5$, where the first-stage F-statistic drops below 10, the sign reverses to a positive

12.1 weeks. The donut RDD produces strong first-stage F-statistics (276–1,509) across the entire bandwidth ladder. At $h = 10$ ($F = 541$), the donut estimates a reduction of 3.9 weeks, and estimates remain stable from -2.7 weeks at $h = 5$ to -4.8 weeks at $h = 30$. The donut duration estimates fall within or near the partial identification bounds at all bandwidths. The simulation RDD suffers from weak instruments across the entire bandwidth ladder, with F-statistics never exceeding 5.9. The wide range from which FPL values are drawn—approximately 52 percentage points on average—injects substantial noise into the simulated running variable, attenuating the first stage and producing implausible estimates with standard errors orders of magnitude larger than the point estimates.

The initiation benchmarks exhibit similar patterns. The standard RDD estimates a negative effect of 11.4 percentage points at the baseline bandwidth, outside the partial identification bounds of $[-4.8, 8.3]$ (Table 4), and reverses sign at $h = 5$ ($+28.7$ percentage points, $F = 4.1$). The donut RDD estimate at $h = 10$ is -3.1 percentage points ($F = 537$, not statistically significant), falling within the partial identification bounds; at wider bandwidths ($h \geq 15$), the donut estimates become statistically significant at -5.8 to -7.3 percentage points. The simulation RDD produces implausible point estimates driven by F-statistics near 1.

The standard RDD is sensitive to bandwidth selection and fails to identify the sign of the treatment effect at narrow bandwidths. The simulation RDD is uninformative throughout, with F-statistics near 1. The donut design excludes observations near the threshold and strengthens the first stage, producing stable estimates that corroborate the partial identification duration findings; however, the donut increases reliance on observations farther from the cutoff where the identifying assumptions are more tenuous. For narrower estimation windows, the effective bandwidth for the donut RDD is larger than any bandwidth examined with the standard RDD.

5.4 Heterogeneity Analysis

Tables 6 and 7 present the partial identification estimates by parity, education, race/ethnicity, and employment status. The identified bounds for breastfeeding duration are strictly negative for most subgroups at the baseline bandwidth, though the 95% Imbens-Manski confidence intervals exclude zero only for the education subgroups and non-hispanic (NH) Black mothers. No subgroup yields a statistically significant effect on initiation.

By parity, first births show duration bounds of $[-7.4, -5.0]$ weeks and initiation bounds of $[-5.9, -3.5]$ percentage points at the baseline bandwidth. Higher-parity births show wider bounds for both duration ($[-15.4, -1.8]$ weeks) and initiation ($[-49.3, -14.1]$ percentage points). The confidence intervals include zero in all four cases.

By education, both mothers with a high school education or less and those with some college

show strictly negative duration bounds ($[-9.0, -4.2]$ and $[-13.4, -5.6]$ weeks, respectively), with confidence intervals that also exclude zero ($[-11.1, -1.2]$ and $[-20.6, -1.2]$). For initiation, both groups exhibit bounds that encompass zero.

By race and ethnicity, NH Black mothers show the largest duration reduction ($[-13.5, -7.8]$ weeks), with a confidence interval of $[-18.7, -5.3]$ that excludes zero. The initiation bounds for this group are positive ($[+0.7, +20.4]$ percentage points), though the confidence interval includes zero. Hispanic mothers show negative duration bounds ($[-38.6, -8.7]$ weeks), but the confidence interval ($[-40.8, +0.5]$) marginally includes zero. NH White mothers show bounds that include zero for both outcomes.

By employment status, mothers who are not employed show statistically insignificant duration bounds of $[-7.6, -2.9]$ weeks. The results for employed mothers are implausible, with duration bounds of $[-28.2, +30.4]$ weeks and initiation bounds of $[-101.0, -0.9]$ percentage points. More generally, these results reflect data limitations for this subsample, where observations are unevenly distributed across the range of household FPL, limiting the feasibility of the partial identification approach.

5.5 Robustness Checks

5.5.1 Covariate Balance

Table 8 tests for sorting at the 185% FPL threshold by comparing maternal characteristics within a 10 FPL percentage point bandwidth of the cutoff. Most covariates show no statistically significant discontinuity, including maternal age, marital status, and race/ethnicity (all $p > 0.1$). The one exception is a marginal imbalance in the proportion of high school graduates (RD $= 6.56$ percentage points, $p = 0.081$). The partial identification framework does not require perfect local quasi-randomization to recover informative bounds, limiting the concern that this marginal imbalance may introduce.

5.5.2 Pre-Pandemic Sample (2009–2020)

Tables 9 and 10 replicate the analysis using only the pre-pandemic sample (2009–2020), excluding years in which WIC operations and breastfeeding behaviors may have been affected by pandemic-era policy changes. At the baseline bandwidth, the duration bounds are $[-12.7, -5.2]$ weeks, with a 95% confidence interval of $[-16.7, -1.8]$ that excludes zero. For initiation, the pre-pandemic bounds ($[-3.3, 8.8]$ percentage points) continue to encompass zero. Taken together, I find no evidence that my main findings are driven by pandemic-era observations.

5.5.3 Placebo Threshold Tests

Tables 11 and 12 report placebo threshold tests at 135% and 235% FPL. At 135% FPL, the first-stage relationship is weak, yielding wide bounds for both initiation ($[-254.6, 98.0]$ percentage points) and duration ($[-60.5, 72.4]$ weeks) at the baseline bandwidth. At 235% FPL, a positive first-stage relationship is present, likely reflecting adjunctive eligibility through other programs. The LATE bounds for both initiation ($[-1.3, 4.8]$ percentage points) and duration ($[-4.5, 1.0]$ weeks) encompass zero.

The feasible percentage—the share of subsampling draws that yield a positive first stage and thus well-defined LATE bounds—provides additional evidence. While the true 185% threshold maintains near 100% feasibility, the 135% placebo has rates of 56–70%, reflecting the absence of a first-stage discontinuity at this threshold. At 235% FPL, feasibility is higher (65–94%), but the LATE bounds remain centered near zero. These results suggest that the effects on breastfeeding outcomes are specific to the true eligibility threshold and are not driven by the partial identification framework.

5.6 Discussion

The central finding is that WIC participation reduces breastfeeding duration but has no identifiable effect on initiation. The duration bounds are strictly negative across the bandwidth ladder, and the 95% Imbens-Manski confidence interval excludes zero at the baseline bandwidth and across a wide range of bandwidths. The initiation bounds encompass zero at all specifications. Taken together, WIC's provision of free infant formula does not deter mothers from initiating breastfeeding but does reduce its duration.

The magnitude of the duration effect is substantial, 4.5 to 12.3 weeks—among individuals not participating in WIC, mean breastfeeding duration in the sample is approximately 13 weeks. As WIC serves roughly half of all U.S. infants, even modest per-mother effects may aggregate to a large impact on public health. My findings are consistent with the negative WIC-breastfeeding association documented by Ryan and Zhou (2006), Bullinger and Gurley-Calvez (2016), and Topolyan and Xu (2017) but provide causal estimates under weaker identification assumptions than prior work.

The estimated effects apply to compliers at the 185% FPL threshold—mothers whose WIC participation is determined by eligibility status, with the heterogeneity analysis (Section 5.4) providing some characterization. The duration effect is statistically significant for mothers across education levels and for non-Hispanic Black mothers, although the results for employed mothers and non-Hispanic White mothers are less informative. These effects are local to the 185% threshold and cannot be extrapolated to very low-income households or higher-income households.

These findings speak to the broader WIC literature that has documented lower breastfeeding among participants but has faced challenges in isolating causal effects. By accommodating bracket-induced uncertainty directly and focusing on compliers at the eligibility threshold, the partial identification results demonstrate a negative local effect on duration without invoking continuity or local randomization assumptions. The estimated 4.5 to 12.3-week reduction moves many participants further from the American Academy of Pediatrics' recommendation of six months of exclusive breastfeeding, highlighting a tension between WIC's breastfeeding promotion efforts and the pecuniary benefit of the formula subsidy for mothers near the eligibility margin.

Three properties distinguish my approach from conventional methods. First, the framework explicitly addresses misclassification of treatment assignment. When observed assignment $\tilde{Z} = \mathbb{1}[X \leq 0]$ differs from true assignment $Z = \mathbb{1}[X^\star \leq 0]$, the conventional estimator captures an unidentified weighted average of eligible and ineligible households. My bounds rely only on certainly eligible households (where $u < 0$) and certainly ineligible households (where $\ell > 0$), eliminating misclassification bias. Second, I replace the continuity assumption required for point identification with weaker shape restrictions—monotonicity and smoothness—that are theoretically motivated and empirically supported. Third, the bounds avoid reliance on local randomization near the eligibility threshold, where measurement error and potential manipulation pose the greatest threat. The donut RDD corroborates the duration findings but requires extrapolating from observations far from the threshold, while the standard and simulation RDDs suffer from weak instruments or sign instability (Section 5.3). The framework makes the strength of identification transparent: conclusions depend on the plausibility of the imposed shape restrictions, and researchers can assess robustness by varying the calibration quantile q.

Several limitations apply. Partial identification yields bounds, not point estimates, so the data cannot distinguish between a 4.5-week and a 12.3-week reduction. The Lipschitz constants are calibrated from the data, and statistical significance at the baseline bandwidth is sensitive to this calibration (Section 5.2). Although bounds may widen under more conservative assumptions, the framework provides valid inference in the presence of misclassified treatment assignment—a property that conventional RDD cannot guarantee in this setting. Income brackets prevent precise threshold assignment, WIC participation and breastfeeding duration are self-reported, and the duration measure does not distinguish between exclusive and any breastfeeding.

6 Conclusion

I develop a partial identification framework for regression discontinuity designs with a coarsely measured running variable, replacing the continuity and precise measurement assumptions required for point identification with shape restrictions—monotonicity and Lipschitz smoothness.

Applied to the effect of WIC on breastfeeding, the method yields statistically significant bounds that exclude zero for duration (a reduction of 4.5 to 12.3 weeks at the baseline bandwidth) but not for initiation. The duration result is robust across the bandwidth ladder, calibration quantiles, and the pre-pandemic sample.

My framework applies to any RDD setting where the running variable is reported in brackets, rounded, or otherwise measured with error, issues common to survey data used in the evaluation of public programs. The calibration quantile q provides a built-in sensitivity analysis, and the bounds avoid reliance on local randomization near the threshold, where measurement error and potential manipulation are most severe. Researchers can adjust the shape restrictions to match their level of confidence in the underlying assumptions, relaxing them when greater caution is warranted.

These findings suggest that despite WIC's investments in breastfeeding promotion through counseling and peer support programs, the in-kind transfer of infant formula creates stronger countervailing incentives. While my partial identification approach cannot pinpoint precise effect magnitudes, it provides policymakers with credible bounds on the trade-offs inherent in WIC's current benefit structure.

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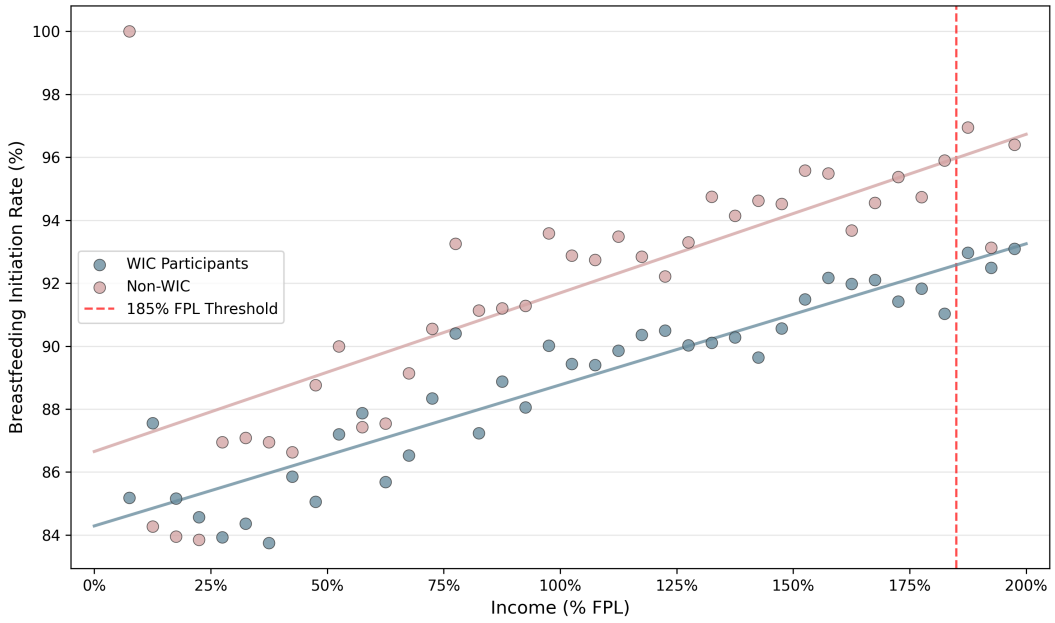
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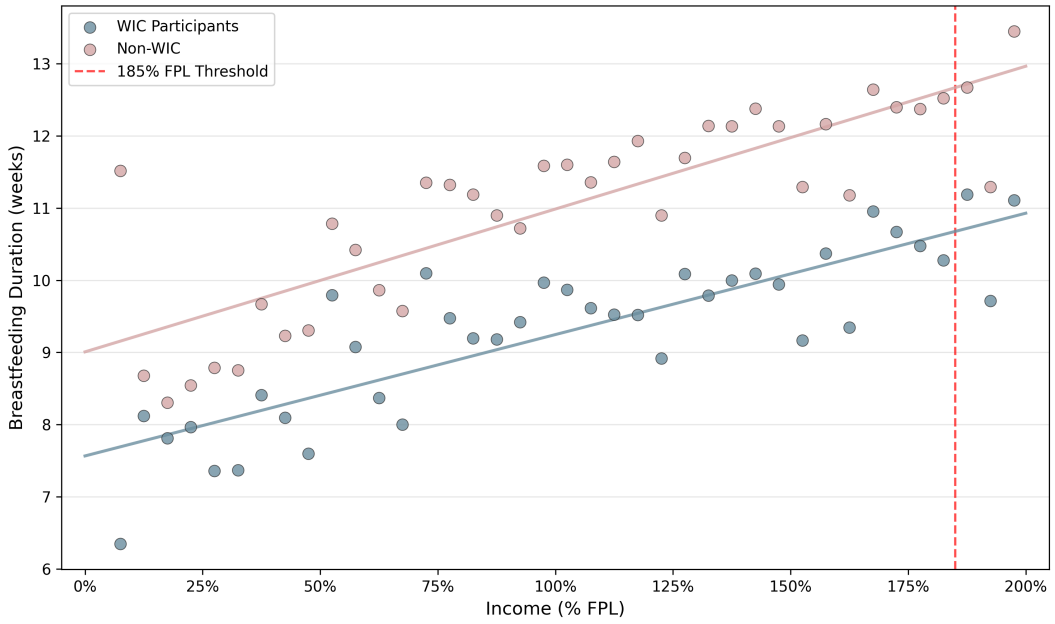
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A Figures



(a) Breastfeeding Initiation



(b) Breastfeeding Duration

Figure 1: Breastfeeding Outcomes by WIC Participation Status

Notes: Panel (a) displays breastfeeding initiation rates and Panel (b) displays breastfeeding duration (weeks) by household income as a percentage of the Federal Poverty Level (FPL). Binned scatter plots with 50 equally-spaced bins show separate series for WIC participants and non-participants. Lines represent linear fits within each group. The vertical dashed line indicates the 185% FPL eligibility threshold. At all income levels, WIC participants exhibit lower initiation rates and shorter breastfeeding durations than non-participants. Sample: PRAMS 2009–2022.

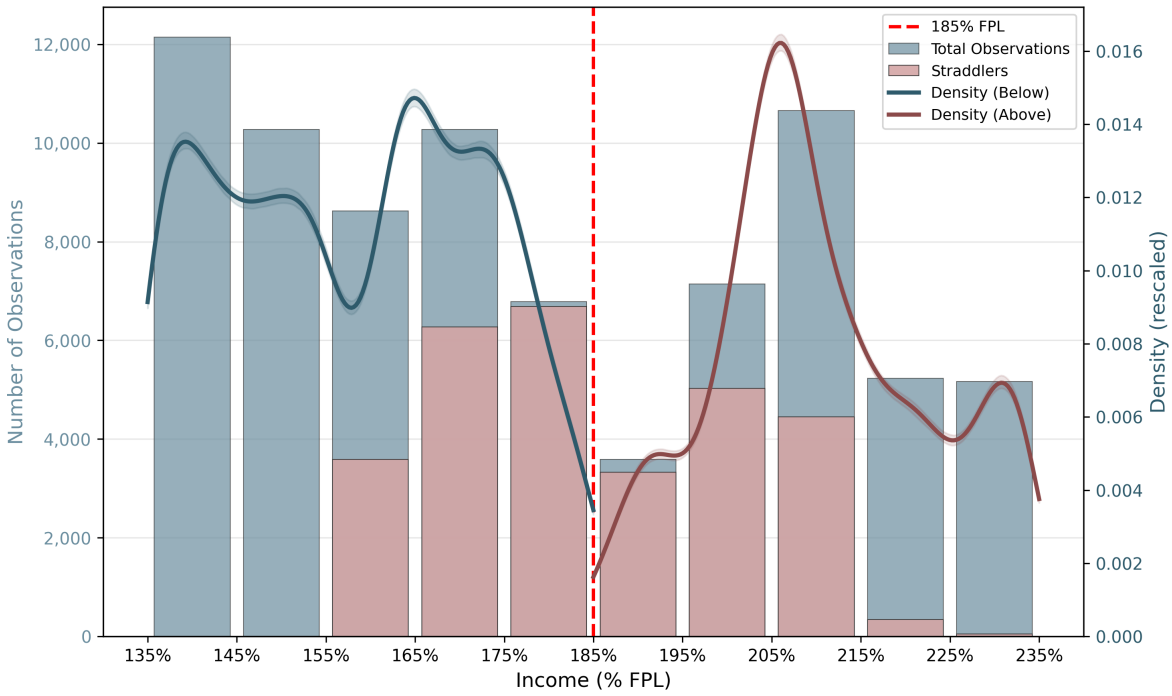
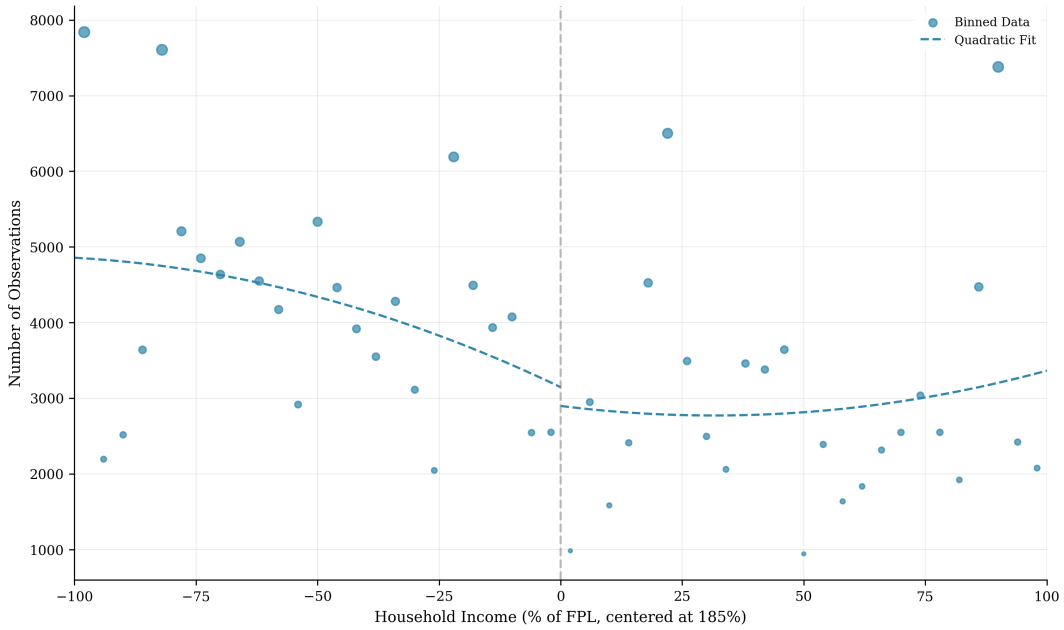
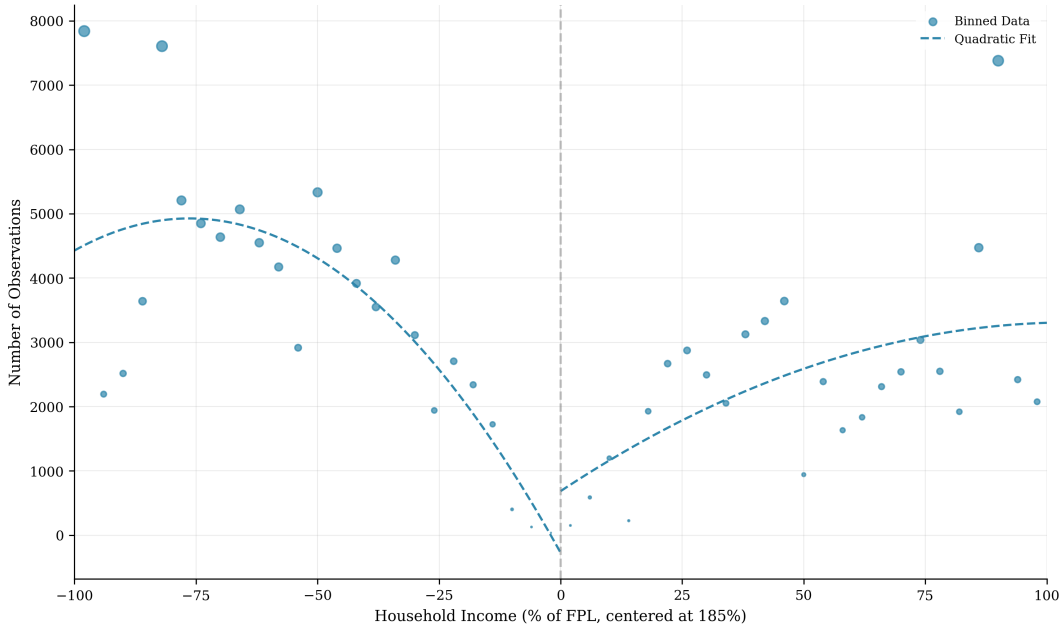


Figure 2: Income Distribution and Straddling Near the Eligibility Threshold

Notes: Histogram bars show total observations (blue) by income bin, with straddling observations—those whose income brackets span the 185% FPL threshold—overlaid (pink). Density curves display kernel density estimates on each side of the threshold, rescaled by sample proportion so that density levels are comparable across the cutoff. The rescaled density shows 60.2% of observations below the threshold and 39.8% above. Among observations within ±50 FPL points of the threshold, 37.3% are straddlers with uncertain eligibility classification. The vertical dashed line indicates the 185% FPL eligibility threshold. Sample: PRAMS 2009–2022.



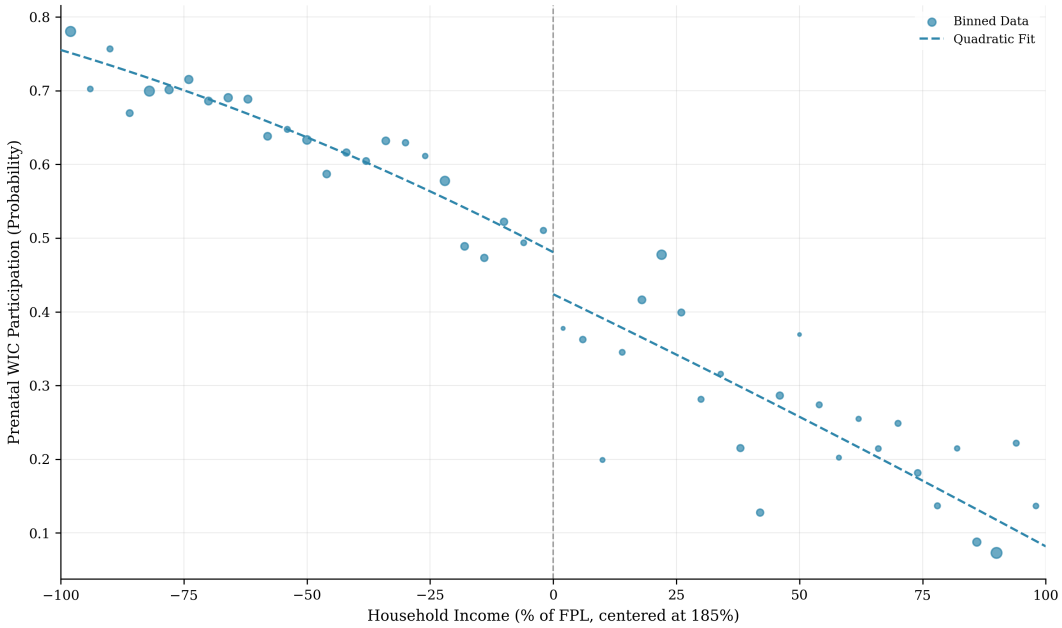
(a) Including Straddling Observations



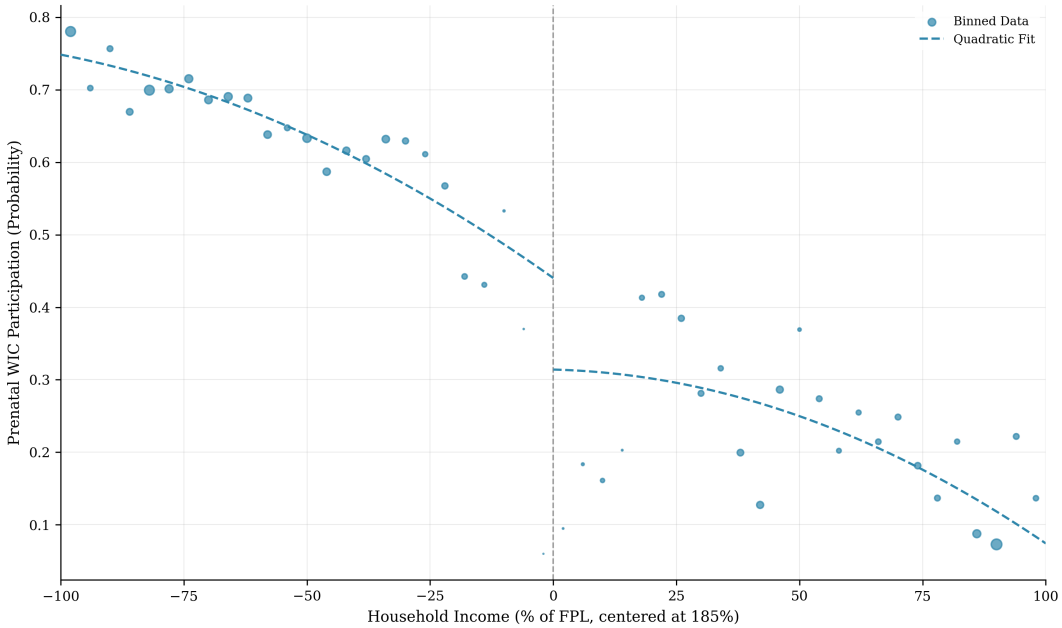
(b) Excluding Straddling Observations

Figure 3: Observation Counts by Household Income

Notes: Binned observation counts by household income as a percentage of the Federal Poverty Level (FPL). Panel (a) includes all observations; panel (b) excludes observations whose income brackets straddle the 185% FPL eligibility threshold. The vertical line indicates the eligibility cutoff. Sample: PRAMS 2009–2022 ($N = 424{,}796$).



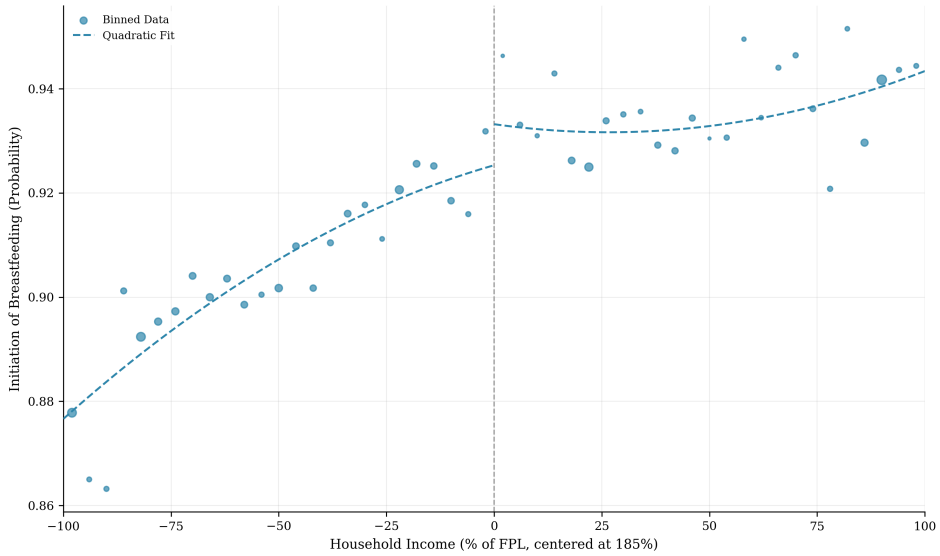
(a) Including Straddling Observations



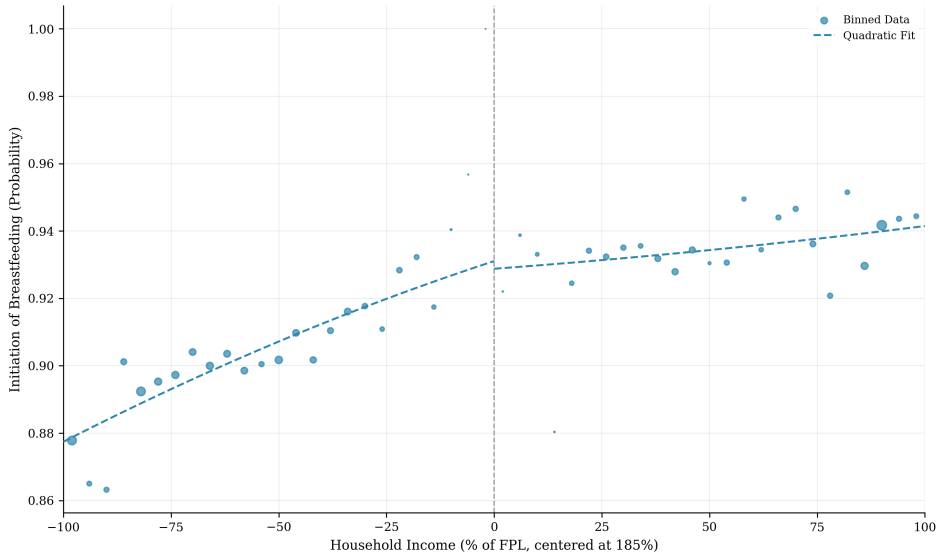
(b) Excluding Straddling Observations

Figure 4: WIC Participation During Pregnancy by Household Income

Notes: Mean WIC participation during pregnancy by household income as a percentage of the Federal Poverty Level (FPL). Panel (a) includes all observations; panel (b) excludes observations whose income brackets straddle the 185% FPL eligibility threshold. The vertical line indicates the eligibility cutoff, points represent bin means, and the dashed lines show separate quadratic fits. Sample: PRAMS 2009–2022 ($N = 424{,}796$).



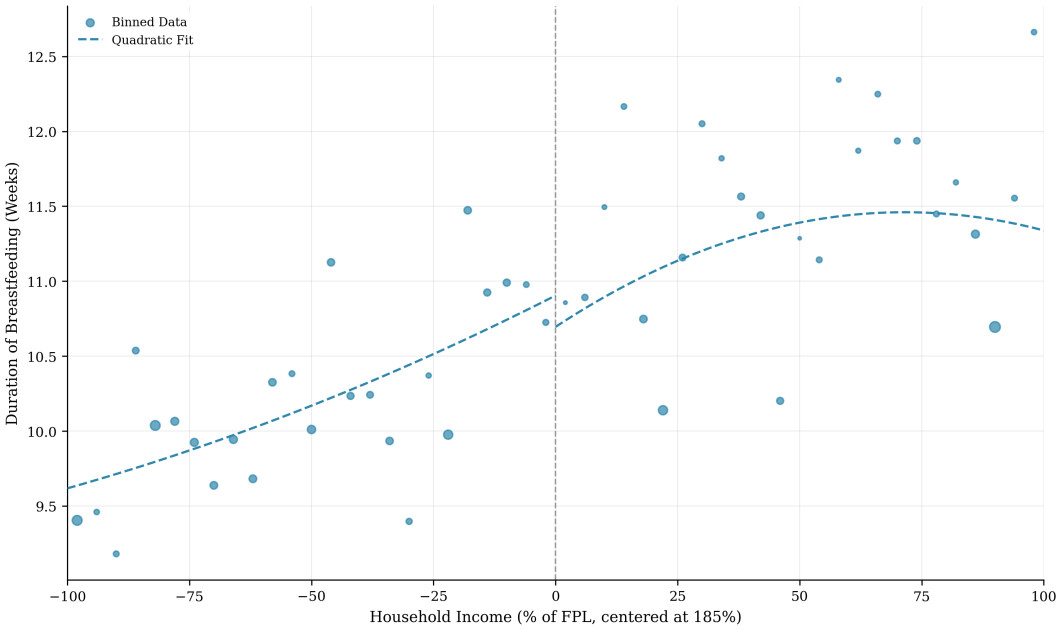
(a) Including Straddling Observations



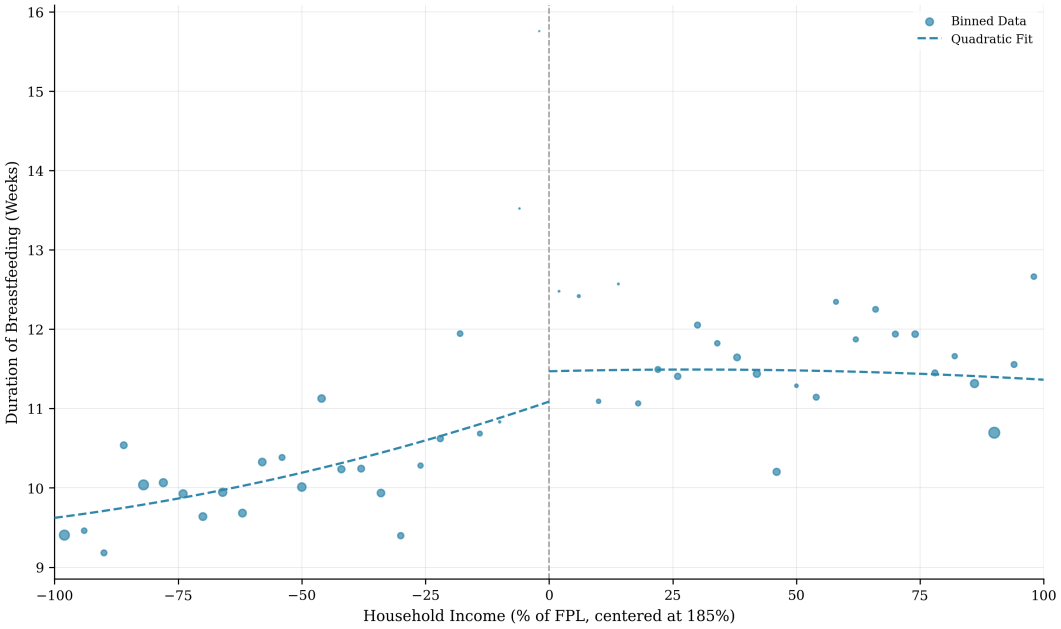
(b) Excluding Straddling Observations

Figure 5: Breastfeeding Initiation by Household Income

Notes: Mean breastfeeding initiation rate by household income as a percentage of the Federal Poverty Level (FPL). Panel (a) includes all observations; panel (b) excludes observations whose income brackets straddle the 185% FPL eligibility threshold. The vertical line indicates the eligibility cutoff, points represent bin means, and the dashed lines show separate quadratic fits. Sample: PRAMS 2009–2022 ($N = 424{,}796$).



(a) Including Straddling Observations



(b) Excluding Straddling Observations

Figure 6: Breastfeeding Duration by Household Income

Notes: Mean breastfeeding duration (weeks) by household income as a percentage of the Federal Poverty Level (FPL). Panel (a) includes all observations; panel (b) excludes observations whose income brackets straddle the 185% FPL eligibility threshold. The vertical line indicates the eligibility cutoff, points represent bin means, and the dashed lines show separate quadratic fits. Sample: PRAMS 2009–2022 ($N = 424{,}796$).

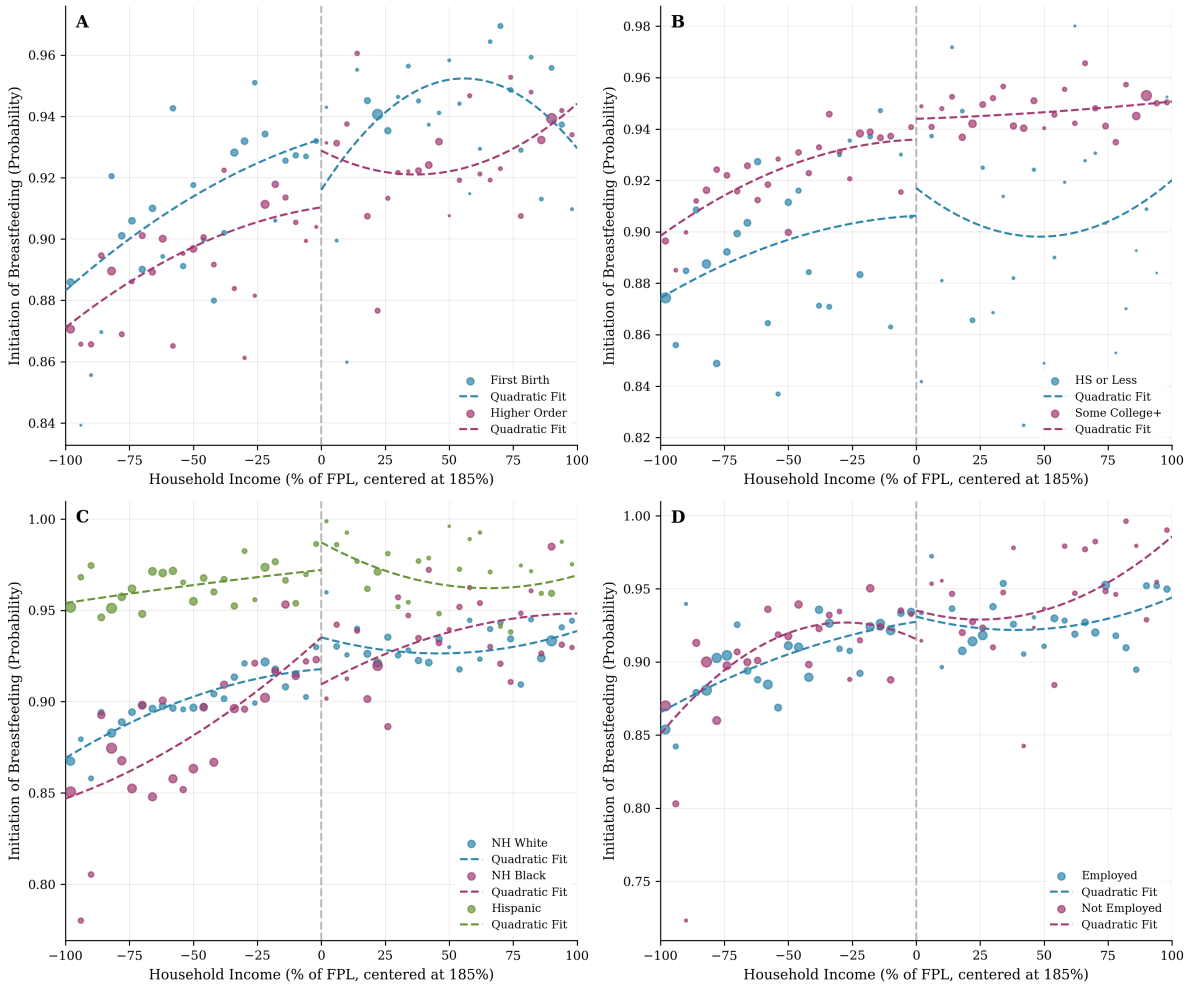


Figure 7: Heterogeneity in Breastfeeding Initiation by Subgroup

Notes: Binned scatter plots of breastfeeding initiation by household income, stratified by maternal characteristics. Each panel shows: A = Parity (first birth vs. higher order); B = Education (high school or less vs. some college or more); C = Race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic); D = Employment status before pregnancy. The vertical line indicates the 185% FPL eligibility threshold, points represent bin means, and the dashed lines show separate quadratic fits. Sample: PRAMS 2009–2022.

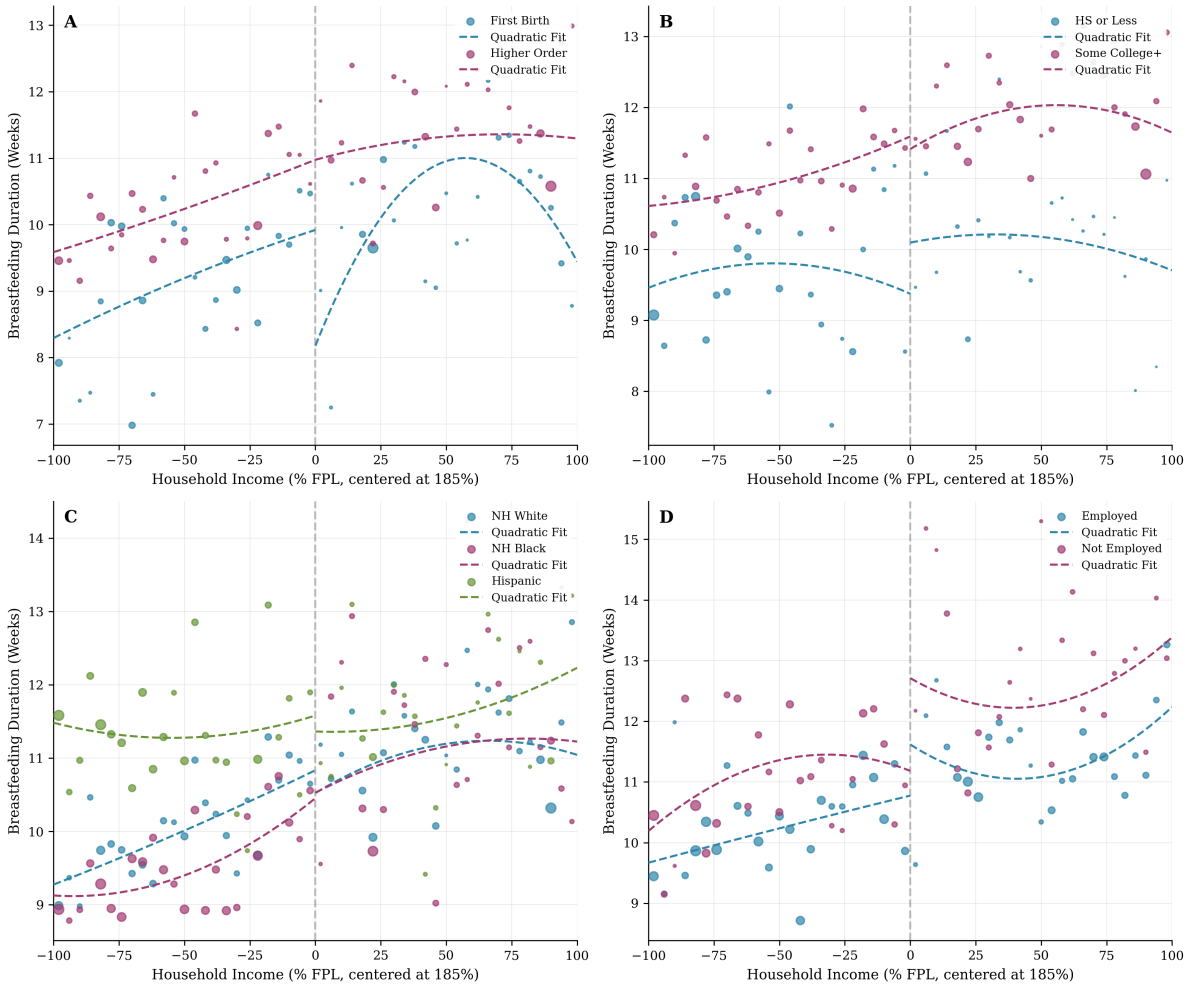


Figure 8: Heterogeneity in Breastfeeding Duration by Subgroup

Notes: Binned scatter plots of breastfeeding duration (weeks) by household income, stratified by maternal characteristics. Each panel shows: A = Parity (first birth vs. higher order); B = Education (high school or less vs. some college or more); C = Race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic); D = Employment status before pregnancy. The vertical line indicates the 185% FPL eligibility threshold, points represent bin means, and the dashed lines show separate quadratic fits. Sample: PRAMS 2009–2022.

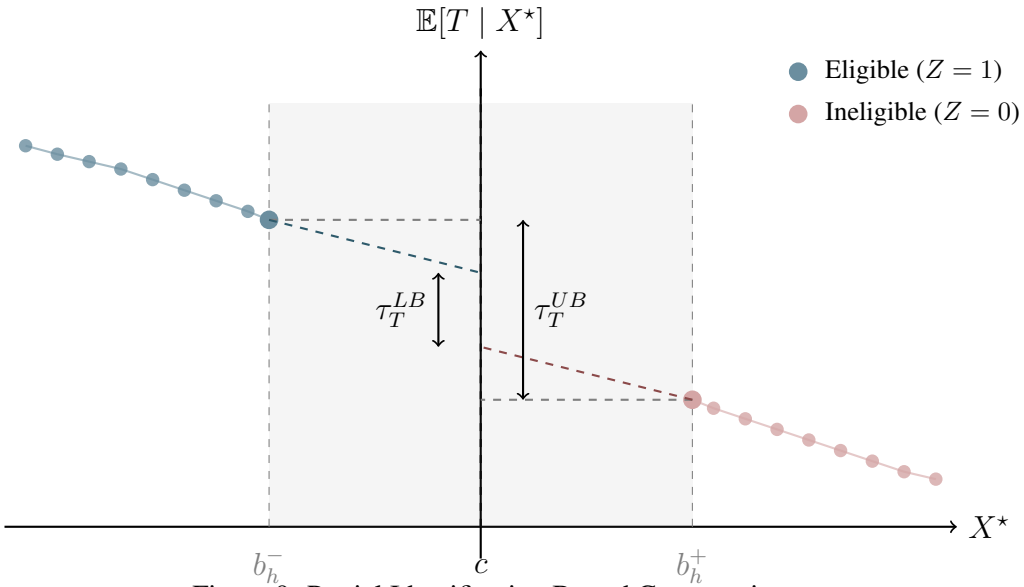
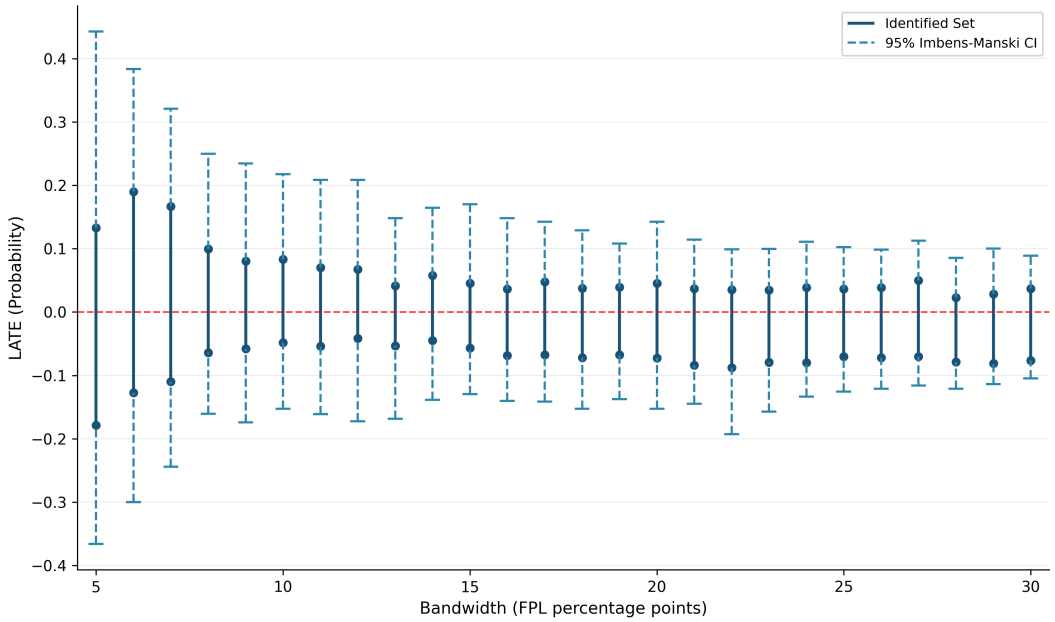
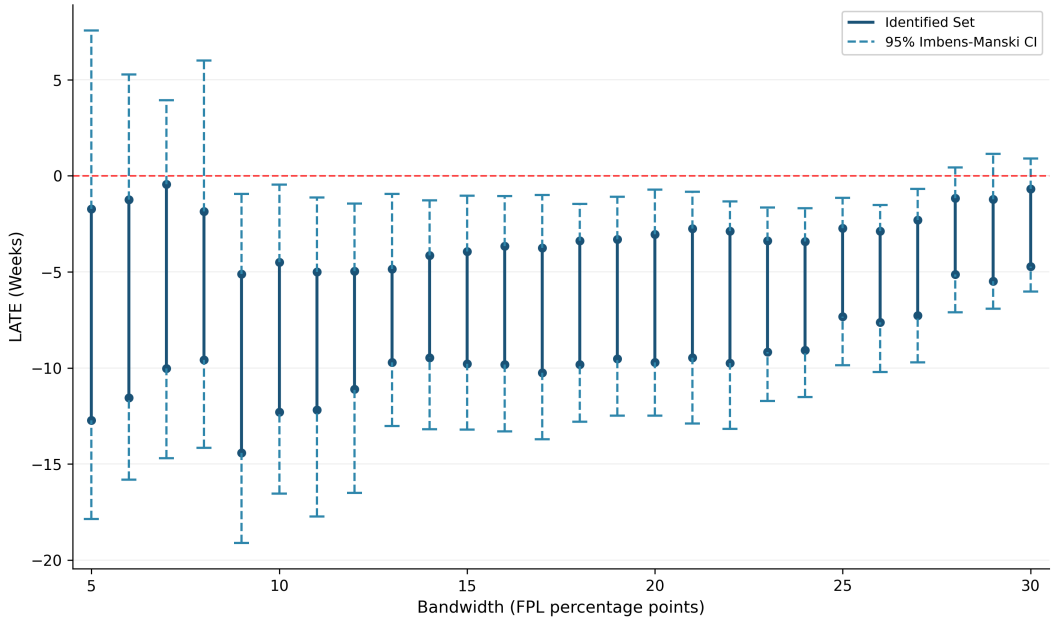


Figure 9: Partial Identification Bound Construction

Notes: This figure illustrates the construction of bounds on the first-stage treatment effect $\tau_T \equiv \lim_{x \uparrow c} \mathbb{E}[T \mid X^\star = x] - \lim_{x \downarrow c} \mathbb{E}[T \mid X^\star = x]$ under partial identification. The running variable $X^\star$ represents true income relative to the eligibility threshold c. Households below the threshold are eligible for WIC ($Z = 1$, blue), and those above the threshold are ineligible ($Z = 0$, pink). The shaded region between the reference bins b_h^- and b_h^+ represents the "straddling zone" where true eligibility status cannot be determined from reported income due to measurement error. Points show binned means of treatment take-up in each certainty region, connected by local linear segments. The dashed gray horizontal lines represent the *monotonicity bounds* and projecting reference bin values to the threshold yields the upper bound τ_T^{UB}. The colored dashed lines represent the *Lipschitz bounds* and projecting with the maximum slope consistent with local curvature in the data yields the tighter lower bound τ_T^{LB}. The Lipschitz constant is calibrated from the empirical distribution of local slopes in each certainty region.



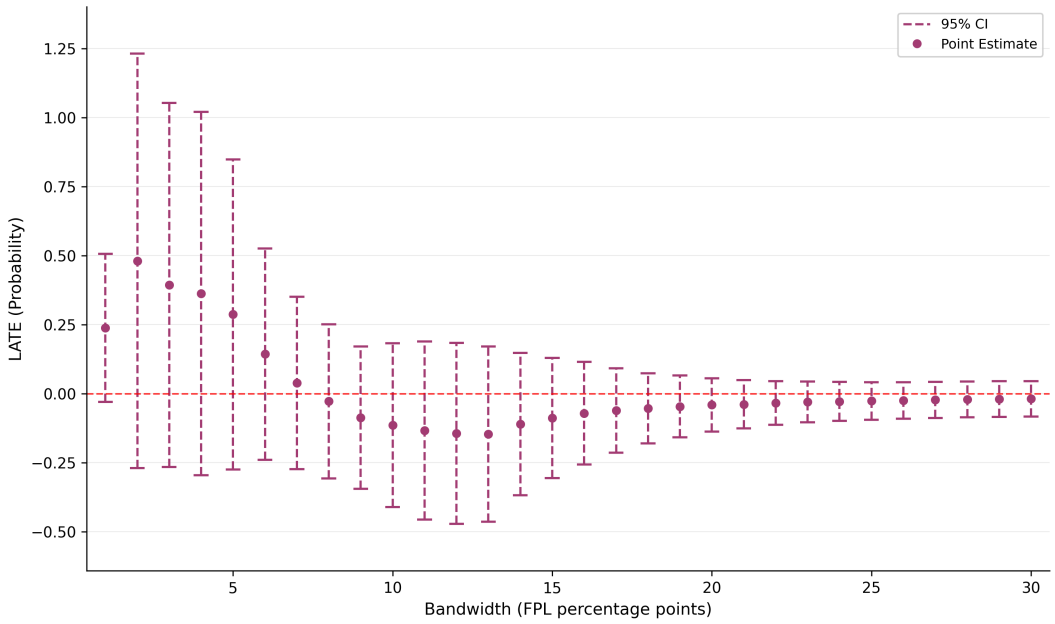
(a) Breastfeeding Initiation (percentage points)



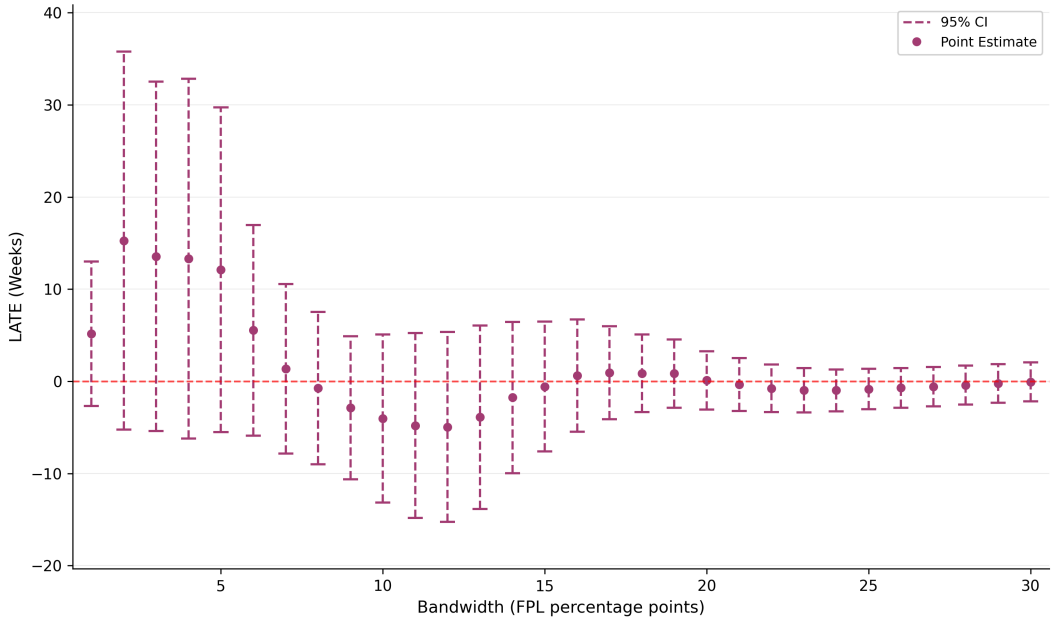
(b) Breastfeeding Duration (weeks)

Figure 10: Partial Identification Bounds: Breastfeeding Outcomes

Notes: Estimated LATE bounds for breastfeeding outcomes under the partial identification framework with calibration quantile $q = 0.50$. Panel (a) displays bounds for breastfeeding initiation (percentage points); panel (b) displays bounds for breastfeeding duration (weeks). Solid lines indicate the upper and lower bounds of the identified set, and shaded regions represent 95% Imbens-Manski confidence intervals. In panel (a), bounds consistently include zero across the bandwidth range, indicating no detectable effect on initiation. In panel (b), bounds are strictly negative and exclude zero at baseline bandwidth ($h^* = 10$ FPL points), providing evidence that WIC participation reduces breastfeeding duration by 4–12 weeks among compliers. Sample: PRAMS 2009–2022, unweighted.



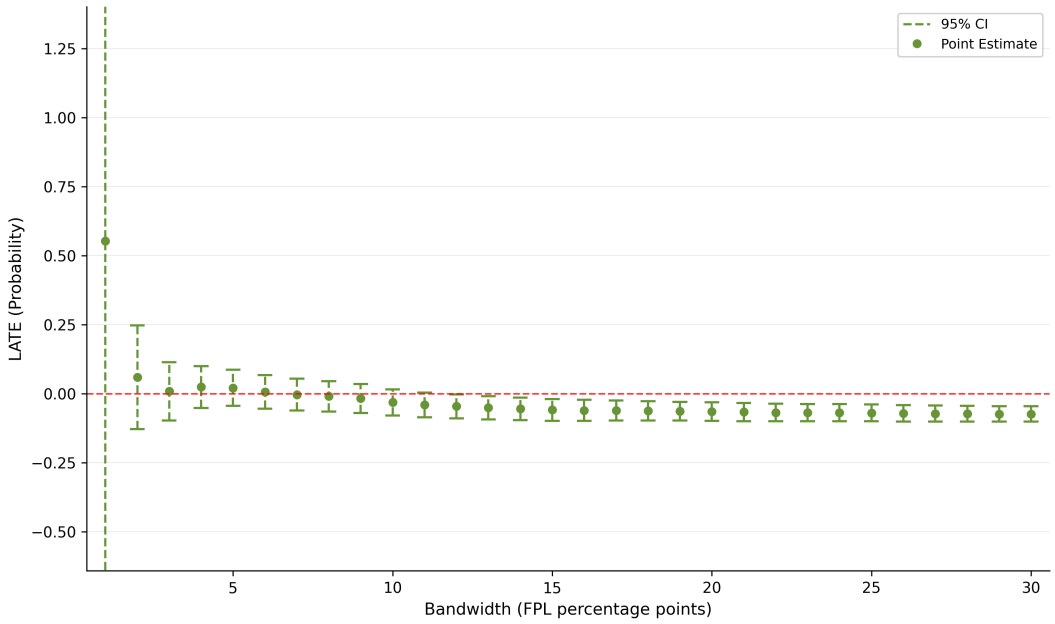
(a) Breastfeeding Initiation



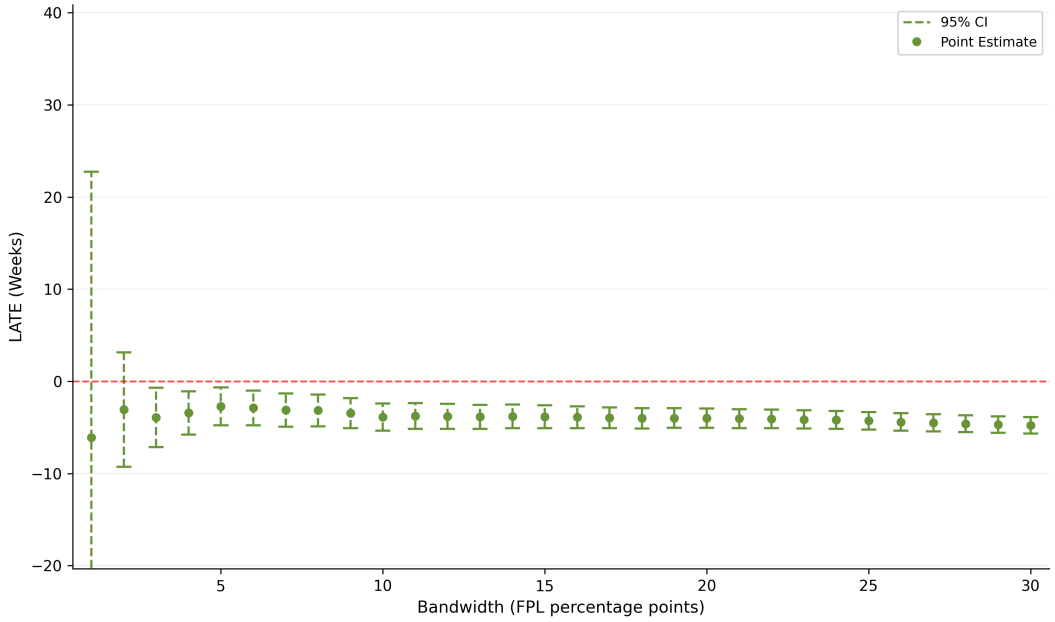
(b) Breastfeeding Duration

Figure 11: Standard RDD Estimates: Breastfeeding Outcomes

Notes: Standard fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 1 to 30 FPL percentage points. Each point is a LATE point estimate, and the dashed lines indicate 95% confidence intervals. The horizontal line marks zero. This approach uses all observations, including those whose income brackets straddle the eligibility threshold, treating bracket midpoints as true income. Sample: PRAMS 2009–2022, unweighted, linear specification.



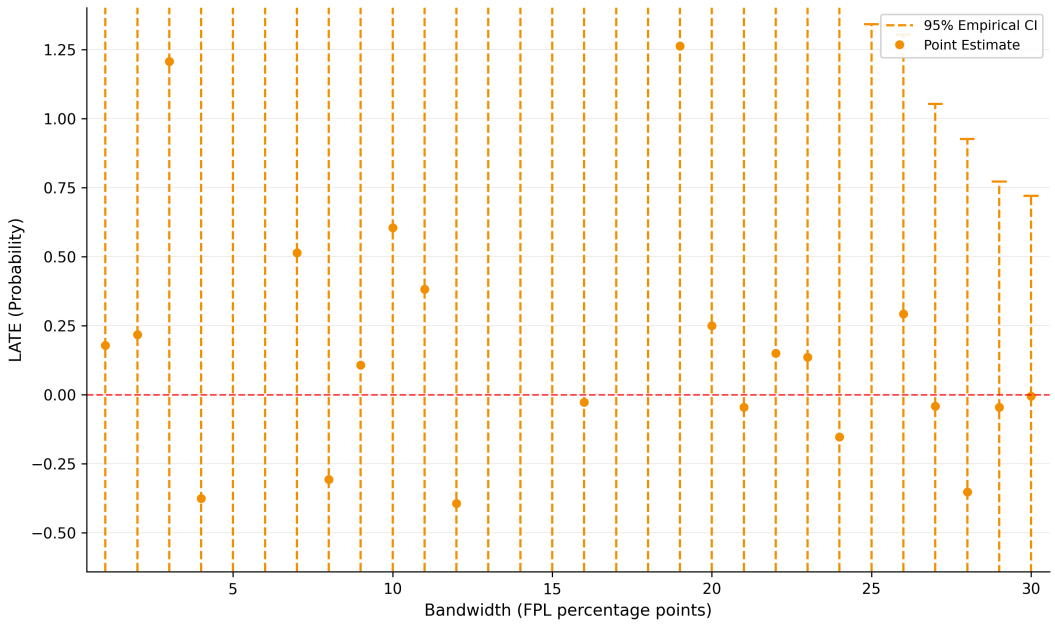
(a) Breastfeeding Initiation



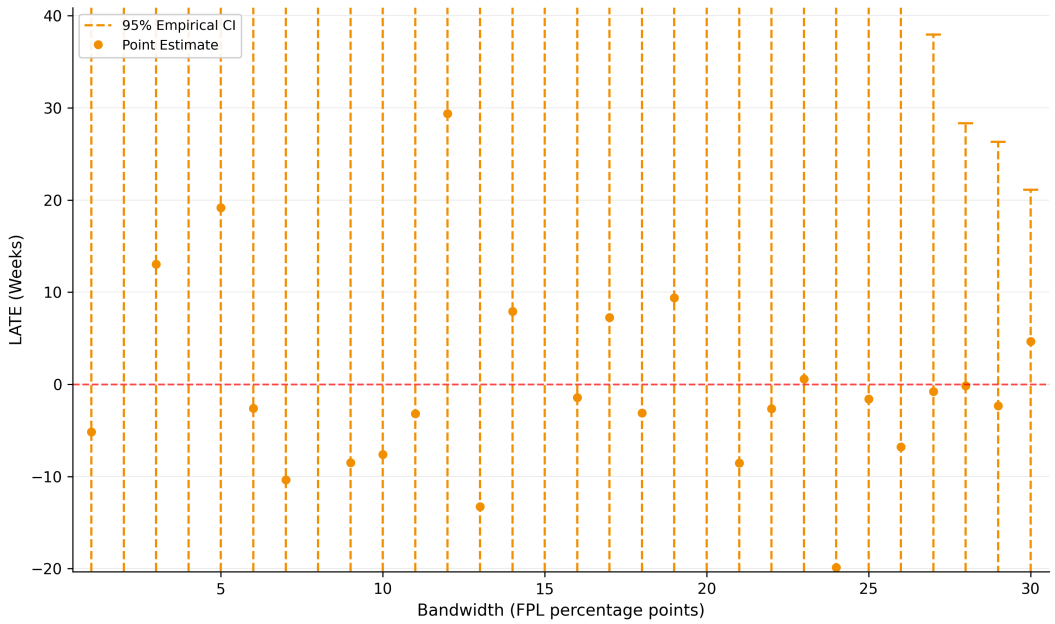
(b) Breastfeeding Duration

Figure 12: Donut RDD Estimates: Breastfeeding Outcomes

Notes: Donut fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 1 to 30 FPL percentage points, linear specification. Each point is a LATE point estimate, and the dashed lines indicate 95% confidence intervals. The horizontal line marks zero. The donut excludes observations whose bracket midpoints fall within $\pm\mu$ of the 185% FPL eligibility threshold, where $\mu \approx 52$ is the average bracket width. Sample: PRAMS 2009–2022, unweighted.



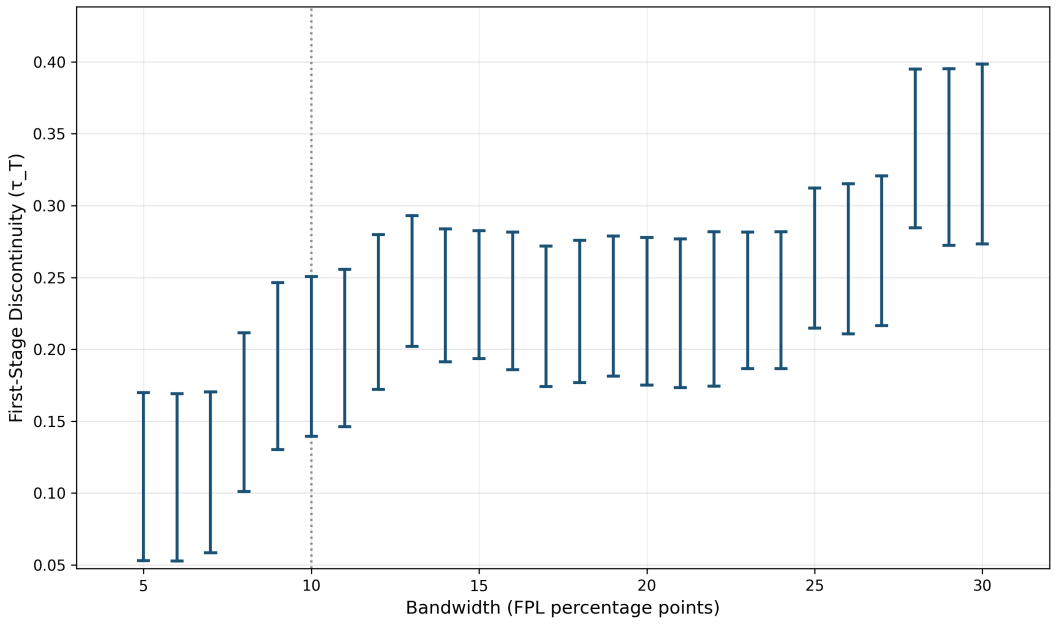
(a) Breastfeeding Initiation



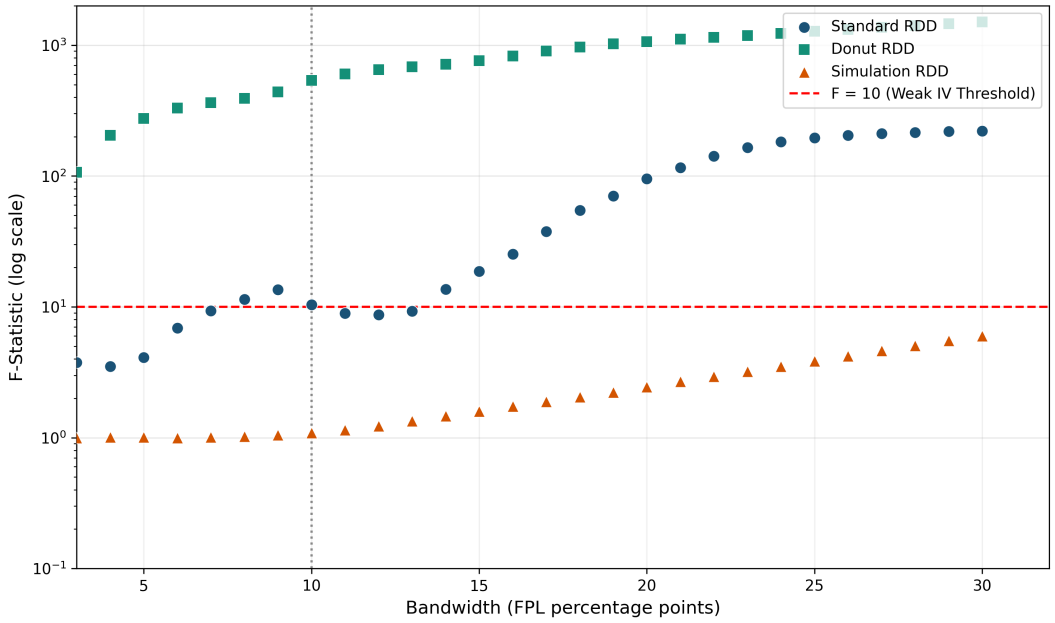
(b) Breastfeeding Duration

Figure 13: Simulation RDD Estimates: Breastfeeding Outcomes

Notes: Simulation-based fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 1 to 30 FPL percentage points. This approach draws 1000 simulated income values uniformly within each household's reported bracket, estimates a fuzzy RDD on each draw, and reports the average point estimate. Each point is the mean LATE across simulations, and the dashed lines indicate 95% empirical confidence intervals. The horizontal line marks zero. Sample: PRAMS 2009–2022, unweighted, linear specification.



(a) Partial Identification



(b) Conventional RDD (Linear)

Figure 14: First-Stage Strength: Partial Identification vs. Conventional RDD

Notes: First-stage strength across bandwidths from 5 to 30 FPL percentage points. Panel (a) shows partial identification first-stage discontinuity bounds $[\tau_T^{LB}, \tau_T^{UB}]$ under Assumption 3 with $q = 0.5$, where each vertical bar spans the identified range of the treatment discontinuity at a given bandwidth. Panel (b) shows first-stage F-statistics (log scale) for the three conventional RDD methods: standard (circles), donut (squares), and simulation (triangles). The horizontal dashed line marks the $F = 10$ weak-instrument threshold. The vertical dotted line marks the baseline bandwidth $h = 10$. Sample: PRAMS 2009–2022, unweighted, linear specification.

B Tables

Table 1: WIC Food Package Allotments by Participant Category and Feeding Practice

Panel A: Infants						
Foods	Fully Formula Fed Packages I & III A: 0–3 mo. B: 4–5 mo.	Fully Formula Fed Packages II & III 6–11 mo.	Partially BF Packages I & III A: 0–3 mo. B: 4–5 mo.	Partially BF Packages II & III 6–11 mo.	Fully BF Package I 0–5 mo.	Fully BF Package II 6–11 mo.
WIC Formula	A: ≤806 fl oz B: ≤884 fl oz	≤624 fl oz	A: ≤364 fl oz B: ≤442 fl oz	≤312 fl oz	N/A	N/A
Infant cereal	N/A	8 oz	N/A	8 oz	N/A	16 oz
Baby food fruits/veg	N/A	128 oz	N/A	128 oz	N/A	128 oz
Baby food meat	N/A	N/A	N/A	N/A	N/A	40 oz
Panel B: Children and Women						
Foods	Children Package IV A: 12–23 mo.	Children Package IV B: 2–4 yrs	Women Package V Pregnant/Mostly BF	Women Package VI Postpartum	Women Package VII Fully Breastfeeding	
Juice, single strength	64 fl oz	64 fl oz	64 fl oz	64 fl oz	64 fl oz	
Milk	A: 12 qt B: 14 qt	14 qt	16 qt	16 qt	16 qt	
Breakfast cereal	36 oz	36 oz	36 oz	36 oz	36 oz	
Eggs	1 dozen	1 dozen	1 dozen	1 dozen	2 dozen	
Fruit/veg CVB	$26	A: $47 B: $52	$47	$47	$52	
Whole wheat bread	24 oz	48 oz	48 oz	48 oz	48 oz	
Fish (canned)	6 oz	A: 10 oz B: 15 oz	10 oz	10 oz	20 oz	
Legumes/Peanut Butter	1 lb dry/64 oz can or 18 oz PB	1 lb/64 oz can and 18 oz PB	1 lb/64 oz can or 18 oz PB	1 lb/64 oz can or 18 oz PB	1 lb/64 oz can and 18 oz PB	

Notes: Monthly food package allotments under WIC following the 2009 food package revisions. Panel A shows infant packages (ages 0–11 months), which vary by feeding practice: fully formula fed infants receive the maximum formula allotment, partially breastfed (BF) infants receive reduced formula, and fully breastfed infants receive no formula but additional complementary foods. Panel B shows packages for children (ages 1–4 years) and women (pregnant, postpartum, and breastfeeding). Fully breastfeeding women receive enhanced food packages. CVB = Cash Value Benefit for fruits and vegetables.

Source: USDA Food and Nutrition Service, WIC Food Packages Regulatory Requirements (7 CFR 246.10).

Table 2: Summary Statistics

	Full Sample (1)	Eligible (2)	Ineligible (3)	Straddlers (4)
Panel A: Outcomes				
Breastfeeding initiation	92.0	88.1	96.1	93.5
Breastfeeding duration (weeks)	10.9 (7.9)	9.2 (8.1)	12.7 (7.0)	11.0 (7.8)
Exclusive BF duration (weeks)	8.2 (7.9)	7.3 (7.8)	9.1 (7.8)	8.1 (7.7)
Panel B: Treatment				
Prenatal WIC participation	45.4	74.7	10.6	50.0
Postpartum WIC participation	45.0	82.4	8.5	57.2
Panel C: Demographics				
Age 25–29	28.5	28.5	28.8	34.1
Age 30–34	27.6	19.4	38.5	24.7
High school graduate	25.2	37.9	9.9	28.1
Some college	28.1	30.9	25.0	41.6
College or more	32.8	8.4	63.5	23.0
Married	59.3	37.4	85.8	62.0
White	59.7	50.4	72.6	61.5
Black	18.0	24.9	8.8	17.6
Hispanic	17.0	23.5	8.0	16.1
Panel D: Household Characteristics				
Household size	3.0 (1.4)	3.1 (1.6)	2.7 (1.0)	2.9 (1.3)
Previous live births	0.6 (0.5)	0.6 (0.5)	0.5 (0.5)	0.6 (0.5)
Panel E: Employment & Leave				
Employed before pregnancy	65.6	61.3	69.1	64.1
Any leave taken	89.0	85.4	91.7	89.0
Paid leave taken	48.0	23.3	61.9	41.1
Panel F: Health & Insurance				
Insured during pregnancy	99.9	99.9	100.0	100.0
Medicaid coverage	55.5	85.9	14.3	60.4
Prenatal care in 1st trimester	84.9	77.6	94.2	86.2
Maternal BMI	27.1 (6.9)	27.9 (7.4)	26.3 (6.3)	27.7 (7.1)
Gestational diabetes	6.6	6.5	6.7	6.8
Panel G: Infant Characteristics				
Male infant	50.5	50.3	50.6	50.8
Low birth weight (<2500g)	23.0	25.2	20.4	22.1
Preterm (<37 weeks)	18.9	20.4	17.3	18.6
NICU admission	20.7	21.9	19.3	20.1
N	508,387	214,740	207,533	29,763

Notes: Standard deviations are presented in parentheses for continuous variables. Binary variables are presented as percentages. "Eligible": FPL upper bound $\leq$185%. "Ineligible": FPL lower bound >185%. "Straddlers": income brackets crossing 185% FPL. Sample sizes vary due to item non-response.
Source: PRAMS 2009–2022.

Table 3: Partial Identification Results: Sensitivity to Calibration Quantile

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: Breastfeeding Initiation						
$q = 0.50$						
LATE Bounds $[\tau^L, \tau^U]$	[-17.9, 13.3]	[-4.8, 8.3]	[-5.7, 4.6]	[-7.3, 4.5]	[-7.0, 3.6]	[-7.6, 3.7]
95% IM CI	[-36.6, 44.3]	[-15.3, 21.8]	[-12.9, 17.0]	[-15.3, 14.3]	[-12.6, 10.2]	[-10.5, 8.9]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.3, 17.0]	[14.0, 25.1]	[19.4, 28.3]	[17.5, 27.8]	[21.5, 31.2]	[27.3, 39.9]
$q = 0.75$						
LATE Bounds $[\tau^L, \tau^U]$	[-18.2, 14.3]	[-5.0, 8.9]	[-5.7, 5.0]	[-7.5, 5.5]	[-7.2, 4.2]	[-7.8, 4.2]
95% IM CI	[-38.9, 28.5]	[-13.7, 26.6]	[-13.9, 14.6]	[-18.6, 13.0]	[-13.3, 11.8]	[-10.9, 11.3]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.2, 17.0]	[13.3, 25.1]	[19.2, 28.3]	[16.8, 27.8]	[20.8, 31.2]	[26.7, 39.9]
$q = 0.90$						
LATE Bounds $[\tau^L, \tau^U]$	[-18.4, 15.0]	[-5.2, 9.2]	[-5.8, 5.2]	[-7.7, 6.0]	[-7.4, 4.5]	[-7.9, 4.5]
95% IM CI	[-31.1, 32.1]	[-14.7, 26.7]	[-13.4, 15.2]	[-15.4, 14.2]	[-14.4, 13.2]	[-11.2, 10.4]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.1, 17.0]	[13.0, 25.1]	[19.1, 28.3]	[16.4, 27.8]	[20.4, 31.2]	[26.2, 39.9]
Panel B: Breastfeeding Duration (Weeks)						
$q = 0.50$						
LATE Bounds $[\tau^L, \tau^U]$	[-12.71, -1.72]	[-12.29, -4.49]	[-9.78, -3.95]	[-9.70, -3.04]	[-7.32, -2.73]	[-4.73, -0.68]
95% IM CI	[-17.87, 7.57]	[-16.53, -0.47]	[-13.21, -1.04]	[-12.48, -0.72]	[-9.85, -1.15]	[-6.03, 0.90]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.3, 17.0]	[14.0, 25.1]	[19.4, 28.3]	[17.5, 27.8]	[21.5, 31.2]	[27.3, 39.9]
$q = 0.75$						
LATE Bounds $[\tau^L, \tau^U]$	[-12.96, -1.45]	[-12.84, -4.32]	[-9.86, -3.61]	[-10.09, -2.86]	[-7.55, -2.66]	[-4.85, -0.50]
95% IM CI	[-19.38, 0.81]	[-17.53, 0.85]	[-13.44, -1.10]	[-13.48, -0.03]	[-10.54, -0.83]	[-6.32, 1.85]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.2, 17.0]	[13.3, 25.1]	[19.2, 28.3]	[16.8, 27.8]	[20.8, 31.2]	[26.7, 39.9]
$q = 0.90$						
LATE Bounds $[\tau^L, \tau^U]$	[-13.12, -1.29]	[-13.20, -4.22]	[-9.90, -3.41]	[-10.34, -2.75]	[-7.70, -2.63]	[-4.93, -0.40]
95% IM CI	[-18.65, 1.17]	[-18.33, 0.42]	[-12.99, -0.93]	[-13.93, 0.11]	[-10.77, -0.88]	[-6.55, 1.12]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.1, 17.0]	[13.0, 25.1]	[19.1, 28.3]	[16.4, 27.8]	[20.4, 31.2]	[26.2, 39.9]

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. LATE bounds computed using partial identification with monotonicity and Lipschitz smoothness constraints. Calibration quantile q controls the slope cap: higher q yields looser caps and wider bounds. 95% Imbens-Manski confidence intervals account for both sampling uncertainty and partial identification. First-stage bounds show the identified set for the discontinuity in WIC participation at the threshold. Panel A reports estimates in percentage points; Panel B in weeks.

Source: PRAMS 2009–2022, unweighted analysis.

Table 4: Method Comparison: Breastfeeding Initiation

	Bandwidth (h/h^*)					
	0.5×	**1.0**×*	**1.5**×	**2.0**×	**2.5**×	**3.0**×
Panel A: Partial Identification ($q = 0.5$)						
LATE Bounds $[\tau^L, \tau^U]$	[-17.9, 13.3]	[-4.8, 8.3]	[-5.7, 4.6]	[-7.3, 4.5]	[-7.0, 3.6]	[-7.6, 3.7]
95% IM CI	[-36.6, 44.3]	[-15.3, 21.8]	[-12.9, 17.0]	[-15.3, 14.3]	[-12.6, 10.2]	[-10.5, 8.9]
Panel B: Conventional RDD (Linear)						
Standard RDD	28.7	-11.4	-8.8	-4.0	-2.6	-1.8
	(28.6)	(15.1)	(11.1)	(4.9)	(3.5)	(3.3)
	[4.1]×	[10.4]	[18.7]	[95.5]	[195.9]	[220.9]
Donut RDD	2.1	-3.1	-5.8	-6.5	-6.9	-7.3
	(3.3)	(2.4)	(2.0)	(1.7)	(1.5)	(1.4)
	[275.7]	[536.6]	[761.6]	[1062.1]	[1278.7]	[1508.6]
Simulation RDD	-193.1	60.5	213.9	25.0	-279.6	-0.5
	(359546.3)	(201735.2)	(946668.9)	(16334.7)	(2350223.5)	(89.1)
	[1.0]×	[1.1]×	[1.6]×	[2.4]×	[3.8]×	[6.0]×

Notes: *Baseline bandwidth $h^* = 10$ FPL percentage points. Partial identification bounds use calibration quantile $q = 0.5$ with Imbens-Manski confidence intervals. Conventional RDD uses local linear regression with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. ×F-statistics below the conventional weak IV threshold of 10. Estimates in percentage points.

Source: PRAMS 2009–2022, unweighted analysis.

Table 5: Method Comparison: Breastfeeding Duration (Weeks)

	Bandwidth (h/h^*)					
	0.5×	**1.0**×*	**1.5**×	**2.0**×	**2.5**×	**3.0**×
Panel A: Partial Identification ($q = 0.5$)						
LATE Bounds $[\tau^L, \tau^U]$	[-12.71, -1.72]	[-12.29, -4.49]	[-9.78, -3.95]	[-9.70, -3.04]	[-7.32, -2.73]	[-4.73, -0.68]
95% IM CI	[-17.87, 7.57]	[-16.53, -0.47]	[-13.21, -1.04]	[-12.48, -0.72]	[-9.85, -1.15]	[-6.03, 0.90]
Panel B: Conventional RDD (Linear)						
Standard RDD	12.11	-4.05	-0.57	0.11	-0.84	-0.06
	(9.00)	(4.65)	(3.59)	(1.62)	(1.13)	(1.07)
	[5.1]$^\times$	[11.1]	[18.5]	[93.8]	[193.0]	[215.3]
Donut RDD	-2.73	-3.87	-3.84	-3.99	-4.29	-4.79
	(1.05)	(0.75)	(0.63)	(0.53)	(0.49)	(0.45)
	[275.6]	[541.4]	[758.0]	[1051.6]	[1266.8]	[1505.2]
Simulation RDD	19.19	-7.62	-29.03	97.63	-1.59	4.65
	(5208.11)	(2205.55)	(26333.68)	(650364.40)	(1364.40)	(5545.29)
	[1.0]$^\times$	[1.1]$^\times$	[1.6]$^\times$	[2.4]$^\times$	[3.8]$^\times$	[5.9]$^\times$

Notes: *Baseline bandwidth $h^* = 10$ FPL percentage points. Partial identification bounds use calibration quantile $q = 0.5$ with Imbens-Manski confidence intervals. Conventional RDD uses local linear regression with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. $^\times$F-statistics below the conventional weak IV threshold of 10. Estimates in weeks.

Source: PRAMS 2009–2022, unweighted analysis.

Table 6: Heterogeneity in WIC Effects on Breastfeeding Initiation

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Parity						
First birth (N = 121,212)						
Bounds	[-5.8, -3.1]	[-5.9, -3.5]	[-8.7, -4.9]	[-9.8, -5.0]	[-13.5, -5.9]	[-11.4, -3.1]
95% IM-CI	[-15.2, +11.8]	[-15.4, +23.7]	[-22.9, +10.8]	[-30.0, +14.3]	[-19.5, +4.6]	[-19.3, +11.4]
Higher order (N = 172,144)						
Bounds	[-325.5, -18.0]	[-49.3, -14.1]	[-53.9, -13.9]	[-44.9, -9.6]	[-9.3, -1.6]	[-10.6, -2.9]
95% IM-CI	[-354.0, +7.9]	[-70.6, +2.2]	[-77.8, +4.9]	[-68.4, +25.1]	[-20.6, +15.6]	[-16.1, +7.5]
Education						
HS or less (N = 152,783)						
Bounds	[-16.7, -9.2]	[-4.9, +3.9]	[-4.2, +1.8]	[-7.2, -0.1]	[-13.6, -3.9]	[-8.4, -1.5]
95% IM-CI	[-21.1, -3.1]	[-14.1, +15.1]	[-23.6, +41.9]	[-28.2, +38.0]	[-35.6, +22.2]	[-24.8, +21.3]
Some college+ (N = 264,347)						
Bounds	[+0.0, +11.4]	[-4.1, +5.1]	[-2.0, +9.4]	[-3.0, +3.3]	[-4.2, +1.0]	[-5.0, +0.2]
95% IM-CI	[-19.1, +24.8]	[-20.9, +24.5]	[-16.7, +25.9]	[-11.0, +13.3]	[-11.6, +10.6]	[-10.7, +6.2]
Race/Ethnicity						
NH White (N = 210,385)						
Bounds	[-14.5, +12.3]	[-12.9, +20.3]	[-14.5, +16.9]	[-3.4, +8.5]	[-7.6, +3.3]	[-10.0, +2.3]
95% IM-CI	[-25.6, +33.9]	[-30.4, +42.8]	[-27.8, +42.8]	[-14.0, +22.2]	[-17.0, +13.2]	[-19.4, +13.3]
NH Black (N = 66,714)						
Bounds	—	[+0.7, +20.4]	[-5.0, +9.5]	[-8.3, +8.1]	[-15.9, +5.3]	[-10.3, +11.8]
95% IM-CI	—	[-15.0, +31.8]	[-32.9, +52.0]	[-31.9, +59.3]	[-35.9, +36.6]	[-35.6, +42.4]
Hispanic (N = 65,056)						
Bounds	—	[-32.0, -2.4]	[-11.2, +0.4]	[-2.5, +4.4]	[-4.4, -0.0]	[-1.9, +4.5]
95% IM-CI	—	[-49.1, +3.5]	[-28.6, +60.2]	[-16.9, +23.5]	[-21.8, +27.4]	[-13.5, +24.8]
Employment						
Employed (N = 89,521)						
Bounds	—	[-101.0, -0.9]	[-13.7, +28.3]	[-6.4, +10.9]	[-10.6, +9.2]	[-15.3, +9.6]
95% IM-CI	—	[-145.5, +8.3]	[-37.9, +56.6]	[-31.0, +43.6]	[-33.9, +33.4]	[-34.7, +28.2]
Not employed (N = 45,829)						
Bounds	—	[+1.2, +11.5]	[-5.1, +7.3]	[-7.9, +3.6]	[-8.8, +4.6]	[-17.3, +0.2]
95% IM-CI	—	[-4.9, +23.2]	[-22.6, +47.2]	[-33.4, +33.2]	[-25.9, +30.8]	[-37.0, +33.6]

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. Partial identification bounds on LATE of WIC on breastfeeding initiation by subgroup. Bounds computed with monotonicity and Lipschitz constraints ($q = 0.50$). "—" = infeasible (first-stage includes zero or insufficient N). IM-CI = Imbens-Manski 95% CI. Estimates in percentage points.
Source: PRAMS 2009–2022, unweighted.

Table 7: Heterogeneity in WIC Effects on Breastfeeding Duration

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Parity						
First birth (N = 121,212)						
Bounds	[-8.87, -5.39]	[-7.43, -4.96]	[-8.22, -5.29]	[-6.92, -3.26]	[-8.07, -3.46]	[-5.43, -1.46]
95% IM-CI	[-13.38, +6.39]	[-12.17, +5.46]	[-13.48, -0.09]	[-12.96, +5.48]	[-10.52, -0.59]	[-7.87, +2.22]
Higher order (N = 172,144)						
Bounds	[-72.80, +5.31]	[-15.37, -1.81]	[-18.10, -2.07]	[-17.68, -1.41]	[-7.53, -2.03]	[-3.43, -0.60]
95% IM-CI	[-82.10, +15.83]	[-22.85, +4.18]	[-25.68, +2.96]	[-26.23, +5.76]	[-12.60, +1.75]	[-5.93, +1.59]
Education						
HS or less (N = 152,783)						
Bounds	[-11.30, -6.45]	[-8.98, -4.24]	[-9.58, -5.81]	[-10.18, -4.46]	[-10.36, -4.48]	[-4.97, -0.64]
95% IM-CI	[-12.60, -4.97]	[-11.10, -1.23]	[-15.58, +1.74]	[-17.03, +1.76]	[-17.70, +0.52]	[-10.34, +3.21]
Some college+ (N = 264,347)						
Bounds	[-17.92, -6.10]	[-13.35, -5.56]	[-9.82, -3.02]	[-6.77, -3.22]	[-6.38, -2.71]	[-2.99, -0.51]
95% IM-CI	[-23.98, +0.74]	[-20.59, -1.23]	[-14.44, +2.07]	[-11.07, -0.15]	[-9.46, -0.21]	[-4.65, +1.59]
Race/Ethnicity						
NH White (N = 210,385)						
Bounds	[-9.18, +4.54]	[-11.85, +1.60]	[-12.04, -0.08]	[-7.45, -1.48]	[-7.81, -1.83]	[-5.46, -0.14]
95% IM-CI	[-14.96, +9.58]	[-17.49, +9.84]	[-17.92, +6.36]	[-12.32, +3.00]	[-10.67, +0.72]	[-8.55, +2.48]
NH Black (N = 66,714)						
Bounds	—	[-13.45, -7.79]	[-11.93, -7.42]	[-12.54, -6.77]	[-12.54, -5.52]	[-10.33, -3.25]
95% IM-CI	—	[-18.74, -5.32]	[-21.74, -1.61]	[-18.94, +1.58]	[-21.10, +6.25]	[-18.14, +2.83]
Hispanic (N = 65,056)						
Bounds	—	[-38.57, -8.72]	[-12.32, -4.70]	[-6.83, -2.60]	[-8.59, -3.85]	[-3.81, -0.40]
95% IM-CI	—	[-40.80, +0.45]	[-20.01, +9.57]	[-12.20, +7.96]	[-16.18, +6.26]	[-9.12, +9.96]
Employment						
Employed (N = 89,521)						
Bounds	—	[-28.16, +30.39]	[-7.55, +10.98]	[-3.31, +4.88]	[-2.89, +5.26]	[-4.61, +4.05]
95% IM-CI	—	[-37.45, +34.17]	[-15.12, +18.24]	[-10.39, +13.22]	[-12.05, +15.80]	[-9.90, +10.95]
Not employed (N = 45,829)						
Bounds	—	[-7.58, -2.87]	[-6.98, -2.71]	[-5.12, -2.58]	[-4.96, -1.43]	[-7.02, -1.74]
95% IM-CI	—	[-9.65, +4.69]	[-14.97, +9.48]	[-16.02, +6.12]	[-11.50, +5.83]	[-14.28, +6.79]

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. Partial identification bounds on LATE of WIC on breastfeeding duration (weeks) by subgroup. Bounds computed with monotonicity and Lipschitz constraints ($q = 0.50$). "—" = infeasible (first-stage includes zero or insufficient N). IM-CI = Imbens-Manski 95% CI.

Source: PRAMS 2009–2022, unweighted.

Table 8: Covariate Balance at 185% FPL Threshold

Covariate	Mean Below (1)	Mean Above (2)	RD Estimate (3)	(SE) (4)	P-value (5)
Age 25–29	33.2	27.4	-3.56	(3.89)	0.359
Age 30–34	22.7	31.8	3.94	(3.61)	0.275
High school graduate	28.8	22.7	6.56*	(3.76)	0.081
Some college	42.9	38.1	0.77	(4.15)	0.853
Married	53.6	77.9	6.53	(4.15)	0.115
White	57.9	68.6	-4.02	(3.99)	0.314
Black	20.3	13.1	1.08	(3.13)	0.730
Hispanic	17.5	12.4	-0.51	(3.39)	0.880
Observations	6,118	5,116			

Notes: Columns (1) and (2) report mean values (in percent) for observations with income bracket midpoints below and above the 185% FPL threshold, respectively, within a bandwidth of 9.0 FPL points. Column (3) reports the regression discontinuity estimate of the jump at the threshold (percentage points) from a local linear regression with triangular kernel weights. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Source: PRAMS 2009–2022.

Table 9: Method Comparison (Pre-Pandemic): Breastfeeding Initiation

	Bandwidth (h/h^*)					
	0.5×	**1.0**×*	**1.5**×	**2.0**×	**2.5**×	**3.0**×
Panel A: Partial Identification ($q = 0.5$)						
LATE Bounds $[\tau^L, \tau^U]$	[-9.2, 9.5]	[-3.3, 8.8]	[-5.1, 4.6]	[-6.4, 4.6]	[-6.6, 4.0]	[-7.3, 3.7]
95% IM CI	[-18.8, 31.0]	[-10.5, 25.0]	[-13.0, 13.9]	[-13.7, 17.1]	[-14.4, 14.1]	[-10.7, 8.3]
Panel B: Conventional RDD (Linear)						
Standard RDD	31.6	-11.6	-10.4	-4.4	-2.8	-1.9
	(29.8)	(16.5)	(12.9)	(5.0)	(3.5)	(3.3)
	[4.0]×	[8.8]×	[14.1]	[91.2]	[190.5]	[215.5]
Donut RDD	-2.0	-5.8	-7.4	-7.4	-7.7	-7.9
	(3.2)	(2.4)	(2.1)	(1.7)	(1.6)	(1.5)
	[338.0]	[566.5]	[734.4]	[1020.8]	[1234.0]	[1452.5]
Simulation RDD	-20.4	-38.5	-20.6	-21.4	0.6	-51.3
	(30179.0)	(9673.3)	(12406.5)	(31043.9)	(2841.2)	(33639.2)
	[1.0]×	[1.1]×	[1.5]×	[2.2]×	[3.4]×	[5.4]×

Notes: *Baseline bandwidth $h^* = 10$ FPL percentage points. Partial identification bounds use calibration quantile $q = 0.5$ with Imbens-Manski confidence intervals. Conventional RDD uses local linear regression (polynomial order = 1) with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. ×F-statistics below the conventional weak IV threshold of 10. Estimates in percentage points.
Source: PRAMS 2009–2020 (pre-pandemic sample), unweighted analysis.

Table 10: Method Comparison (Pre-Pandemic): Breastfeeding Duration (Weeks)

	Bandwidth (h/h^*)					
	0.5×	**1.0**×*	**1.5**×	**2.0**×	**2.5**×	**3.0**×
Panel A: Partial Identification ($q = 0.5$)						
LATE Bounds $[\tau^L, \tau^U]$	[-15.7, -5.0]	[-12.7, -5.2]	[-10.1, -4.2]	[-9.5, -3.1]	[-7.3, -2.8]	[-4.4, -0.5]
95% IM CI	[-19.6, 1.8]	[-16.7, -1.8]	[-13.2, -1.5]	[-12.7, -0.9]	[-10.2, -0.9]	[-5.8, 0.6]
Panel B: Conventional RDD (Linear)						
Standard RDD	12.7	-2.7	-0.6	0.4	-0.3	0.7
	(9.2)	(5.0)	(4.1)	(1.6)	(1.1)	(1.1)
	[5.1]$^\times$	[9.5]$^\times$	[14.2]	[90.4]	[189.2]	[211.4]
Donut RDD	-3.4	-4.2	-4.1	-4.1	-4.4	-4.9
	(1.0)	(0.7)	(0.6)	(0.5)	(0.5)	(0.5)
	[339.0]	[573.0]	[735.5]	[1017.4]	[1231.3]	[1459.6]
Simulation RDD	-14.1	9.2	-4.7	5.6	10.6	-1.4
	(2579.7)	(14908.0)	(1332.9)	(10429.5)	(7324.2)	(25.3)
	[1.0]$^\times$	[1.1]$^\times$	[1.5]$^\times$	[2.2]$^\times$	[3.4]$^\times$	[5.4]$^\times$

Notes: *Baseline bandwidth $h^* = 10$ FPL percentage points. Partial identification bounds use calibration quantile $q = 0.5$ with Imbens-Manski confidence intervals. Conventional RDD uses local linear regression (polynomial order = 1) with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. $^\times$F-statistics below the conventional weak IV threshold of 10. Estimates in weeks.
Source: PRAMS 2009–2020 (pre-pandemic sample), unweighted analysis.

Table 11: Placebo Threshold Tests: Breastfeeding Initiation

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: 185% FPL (True Threshold)						
LATE Bounds $[\tau^L, \tau^U]$	[-17.9, 13.3]	[-4.8, 8.3]	[-5.7, 4.6]	[-7.3, 4.5]	[-7.0, 3.6]	[-7.6, 3.7]
95% IM CI	[-36.6, 44.3]	[-15.3, 21.8]	[-12.9, 17.0]	[-15.3, 14.3]	[-12.6, 10.2]	[-10.5, 8.9]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.3, 17.0]	[14.0, 25.1]	[19.4, 28.3]	[17.5, 27.8]	[21.5, 31.2]	[27.3, 39.9]
Panel B: 135% FPL (Placebo Below)						
LATE Bounds $[\tau^L, \tau^U]$	—	[-254.6, 98.0]	[-93.5, 22.3]	[-139.2, 52.1]	[-74.5, 28.2]	[-63.6, 10.7]
95% IM CI	—	[-277.9, 127.2]	[-114.0, 38.3]	[-162.3, 67.7]	[-94.1, 44.5]	[-85.4, 26.4]
First-Stage $[\tau_T^L, \tau_T^U]$	—	[0.8, 16.4]	[2.8, 18.9]	[1.8, 18.8]	[3.4, 18.8]	[4.6, 20.9]
Feasible %	56%	57%	63%	64%	64%	70%
Panel C: 235% FPL (Placebo Above)						
LATE Bounds $[\tau^L, \tau^U]$	[-0.5, 4.1]	[-1.3, 4.8]	[-4.1, 4.4]	[-6.0, 3.4]	[-6.3, 2.3]	[-13.1, 4.6]
95% IM CI	[-17.8, 11.8]	[-18.6, 23.4]	[-15.9, 19.3]	[-14.3, 17.5]	[-13.9, 13.8]	[-22.8, 13.4]
First-Stage $[\tau_T^L, \tau_T^U]$	[21.9, 33.3]	[20.3, 31.7]	[15.4, 28.1]	[18.5, 31.4]	[21.2, 32.9]	[13.4, 26.5]
Feasible %	65%	81%	85%	90%	94%	92%

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. "—" = infeasible (first-stage includes zero or insufficient N). LATE bounds use partial identification with monotonicity and Lipschitz constraints ($q = 0.50$). 95% IM CI = Imbens-Manski confidence interval. Feasible % = share of subsampling draws with well-defined bounds. Estimates in percentage points.

Source: PRAMS 2009–2022, unweighted analysis.

Table 12: Placebo Threshold Tests: Breastfeeding Duration (Weeks)

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: 185% FPL (True Threshold)						
LATE Bounds $[\tau^L, \tau^U]$	[-12.71, -1.72]	[-12.29, -4.49]	[-9.78, -3.95]	[-9.70, -3.04]	[-7.32, -2.73]	[-4.73, -0.68]
95% IM CI	[-17.87, 7.57]	[-16.53, -0.47]	[-13.21, -1.04]	[-12.48, -0.72]	[-9.85, -1.15]	[-6.03, 0.90]
First-Stage $[\tau_T^L, \tau_T^U]$	[5.3, 17.0]	[14.0, 25.1]	[19.4, 28.3]	[17.5, 27.8]	[21.5, 31.2]	[27.3, 39.9]
Panel B: 135% FPL (Placebo Below)						
LATE Bounds $[\tau^L, \tau^U]$	—	[-60.50, 72.44]	[-28.69, 16.13]	[-45.84, 36.57]	[-26.23, 19.81]	[-27.96, 6.34]
95% IM CI	—	[-66.39, 81.65]	[-34.55, 22.57]	[-52.08, 43.32]	[-31.82, 26.03]	[-35.56, 11.11]
First-Stage $[\tau_T^L, \tau_T^U]$	—	[0.8, 16.4]	[2.8, 18.9]	[1.8, 18.8]	[3.4, 18.8]	[4.6, 20.9]
Feasible %	56%	57%	63%	64%	64%	70%
Panel C: 235% FPL (Placebo Above)						
LATE Bounds $[\tau^L, \tau^U]$	[-4.20, -0.13]	[-4.52, 1.04]	[-4.69, 3.12]	[-2.26, 3.71]	[0.44, 5.02]	[-0.10, 6.61]
95% IM CI	[-5.64, 1.43]	[-9.30, 8.22]	[-10.00, 8.47]	[-6.41, 8.44]	[-2.46, 9.99]	[-2.73, 10.01]
First-Stage $[\tau_T^L, \tau_T^U]$	[21.9, 33.3]	[20.3, 31.7]	[15.4, 28.1]	[18.5, 31.4]	[21.2, 32.9]	[13.4, 26.5]
Feasible %	65%	81%	85%	90%	94%	92%

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. "—" = infeasible (first-stage includes zero or insufficient N). LATE bounds use partial identification with monotonicity and Lipschitz constraints ($q = 0.50$). 95% IM CI = Imbens-Manski confidence interval. Feasible % = share of subsampling draws with well-defined bounds. Estimates in weeks.
Source: PRAMS 2009–2022, unweighted analysis.

C Supplementary Tables

Table 13: Summary Statistics by WIC Participation (Income-Eligible Sample)

	WIC Participants (1)	Non-WIC Non-Participants (2)	Difference (2)-(1) (3)
Panel A: Outcomes			
Breastfeeding initiation	87.1	91.1	4.0***
Breastfeeding duration (weeks)	8.7 (8.0)	10.8 (8.1)	2.1***
Exclusive BF duration (weeks)	7.0 (7.7)	8.5 (8.0)	1.5***
Panel B: Demographics			
Age 25–29	27.7	30.9	3.3***
Age 30–34	18.3	22.4	4.1***
High school graduate	39.2	33.9	-5.3***
Some college	29.6	35.0	5.4***
College or more	6.4	14.5	8.2***
Married	33.8	48.0	14.2***
White	48.4	56.5	8.2***
Black	26.6	19.5	-7.2***
Hispanic	25.5	17.7	-7.7***
Panel C: Household Characteristics			
Household size	3.1 (1.6)	3.3 (1.7)	0.2***
Previous live births	0.6 (0.5)	0.7 (0.5)	0.0***
Panel D: Employment & Leave			
Employed before pregnancy	61.1	61.5	0.4
Any leave taken	84.8	87.2	2.4***
Paid leave taken	21.7	27.6	5.9***
Panel E: Health & Insurance			
Insured during pregnancy	99.9	99.9	-0.0***
Medicaid coverage	90.3	72.5	-17.9***
Prenatal care in 1st trimester	78.2	75.6	-2.6***
Maternal BMI	28.1 (7.5)	27.3 (7.0)	-0.8***
Gestational diabetes	6.6	6.2	-0.4**
Panel F: Infant Characteristics			
Male infant	50.3	50.3	-0.0
Low birth weight (<2500g)	25.3	25.1	-0.2
Preterm (<37 weeks)	20.1	21.3	1.2***
NICU admission	21.7	22.6	0.9**
N	159,753	53,982	

Notes: Sample restricted to income-eligible mothers (FPL upper bound ≤185%). WIC participation defined as prenatal WIC enrollment. Standard deviations in parentheses for continuous variables; percentages for binary variables. Significance levels from two-sample t-tests (continuous) or z-tests (binary): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 14: Partial Identification Results (Weighted): Sensitivity to Calibration Quantile

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: Breastfeeding Initiation						
$q = 0.50$						
LATE Bounds $[\tau^L, \tau^U]$	[-1.4, 17.9]	[-10.4, 3.7]	[-11.2, -0.4]	[-9.5, 2.5]	[-8.1, 2.0]	[-7.2, 2.8]
95% IM CI	[-19.7, 27.2]	[-36.6, 20.6]	[-23.8, 10.9]	[-18.4, 16.1]	[-18.9, 10.6]	[-14.9, 9.5]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.8, 19.2]	[13.7, 24.9]	[20.4, 29.2]	[18.4, 29.0]	[23.0, 32.9]	[28.0, 40.9]
$q = 0.75$						
LATE Bounds $[\tau^L, \tau^U]$	[-1.5, 18.9]	[-11.0, 4.2]	[-11.3, -0.3]	[-9.8, 3.2]	[-8.3, 2.7]	[-7.3, 3.5]
95% IM CI	[-11.6, 27.6]	[-34.7, 21.8]	[-23.7, 11.3]	[-20.0, 13.8]	[-16.7, 11.6]	[-13.2, 11.3]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.6, 19.2]	[13.0, 24.9]	[20.1, 29.2]	[17.9, 29.0]	[22.5, 32.9]	[27.4, 40.9]
$q = 0.90$						
LATE Bounds $[\tau^L, \tau^U]$	[-1.5, 19.5]	[-11.4, 4.6]	[-11.4, -0.2]	[-10.0, 3.7]	[-8.4, 3.1]	[-7.4, 3.9]
95% IM CI	[-12.1, 29.2]	[-30.2, 25.1]	[-24.5, 14.3]	[-18.7, 15.5]	[-18.4, 11.8]	[-13.6, 11.8]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.5, 19.2]	[12.5, 24.9]	[20.0, 29.2]	[17.6, 29.0]	[22.2, 32.9]	[27.1, 40.9]
Panel B: Breastfeeding Duration (Weeks)						
$q = 0.50$						
LATE Bounds $[\tau^L, \tau^U]$	[-4.04, 0.24]	[-13.95, -5.84]	[-10.45, -5.08]	[-9.67, -3.80]	[-7.76, -3.44]	[-5.47, -1.21]
95% IM CI	[-8.38, 4.42]	[-21.13, -1.20]	[-15.21, -2.67]	[-14.40, -1.23]	[-12.38, -1.08]	[-7.87, 0.98]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.8, 19.2]	[13.7, 24.9]	[20.4, 29.2]	[18.4, 29.0]	[23.0, 32.9]	[28.0, 40.9]
$q = 0.75$						
LATE Bounds $[\tau^L, \tau^U]$	[-4.13, 0.43]	[-14.73, -5.56]	[-10.58, -4.61]	[-9.96, -3.52]	[-7.93, -3.28]	[-5.58, -0.99]
95% IM CI	[-6.98, 4.78]	[-20.97, -0.17]	[-16.06, -1.82]	[-14.91, -0.84]	[-12.71, -0.95]	[-8.48, 1.01]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.6, 19.2]	[13.0, 24.9]	[20.1, 29.2]	[17.9, 29.0]	[22.5, 32.9]	[27.4, 40.9]
$q = 0.90$						
LATE Bounds $[\tau^L, \tau^U]$	[-4.19, 0.54]	[-15.24, -5.39]	[-10.66, -4.33]	[-10.14, -3.34]	[-8.05, -3.18]	[-5.65, -0.86]
95% IM CI	[-7.19, 5.56]	[-21.10, 0.26]	[-15.39, -1.07]	[-14.70, -0.79]	[-13.65, -0.58]	[-8.25, 1.52]
First-Stage $[\tau_T^L, \tau_T^U]$	[7.5, 19.2]	[12.5, 24.9]	[20.0, 29.2]	[17.6, 29.0]	[22.2, 32.9]	[27.1, 40.9]

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. LATE bounds computed using partial identification with monotonicity and Lipschitz smoothness constraints. Calibration quantile q controls the slope cap: higher q yields looser caps and wider bounds. 95% Imbens-Manski confidence intervals account for both sampling uncertainty and partial identification. First-stage bounds show the identified set for the discontinuity in WIC participation at the threshold. Panel A reports estimates in percentage points; Panel B in weeks.
Source: PRAMS 2009–2022, weighted analysis.

Table 15: RDD Method Comparison: Breastfeeding Initiation

	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: Linear ($p = 1$)						
Standard RDD	28.7	-11.4	-8.8	-4.0	-2.6	-1.8
	(28.6)	(15.1)	(11.1)	(4.9)	(3.5)	(3.3)
	[4.1]$^{\times}$	[10.4]	[18.7]	[95.5]	[195.9]	[220.9]
Donut RDD	2.1	-3.1	-5.8	-6.5	-6.9	-7.3
	(3.3)	(2.4)	(2.0)	(1.7)	(1.5)	(1.4)
	[275.7]	[536.6]	[761.6]	[1062.1]	[1278.7]	[1508.6]
Simulation RDD	-193.1	60.5	213.9	25.0	-279.6	-0.5
	(359546.3)	(201735.2)	(946668.9)	(16334.7)	(2350223.5)	(89.1)
	[1.0]$^{\times}$	[1.1]$^{\times}$	[1.6]$^{\times}$	[2.4]$^{\times}$	[3.8]$^{\times}$	[6.0]$^{\times}$
Panel B: Quadratic ($p = 2$)						
Standard RDD	59.1	11.7	-9.0	-15.8	-9.3	-6.0
	(53.0)	(17.1)	(21.1)	(28.7)	(12.6)	(6.6)
	[2.2]$^{\times}$	[8.8]$^{\times}$	[5.3]$^{\times}$	[3.0]$^{\times}$	[14.7]	[53.1]
Donut RDD	2.3	3.3	-0.8	-3.6	-4.9	-5.5
	(5.6)	(3.8)	(3.0)	(2.6)	(2.2)	(2.0)
	[94.4]	[214.4]	[356.4]	[470.0]	[624.2]	[751.4]
Simulation RDD	-17.3	60.9	875.2	-268.3	7.3	-1.6
	(257770.2)	(261858.9)	(14585554.5)	(1275200.3)	(24272.4)	(14674.2)
	[1.0]$^{\times}$	[1.0]$^{\times}$	[1.0]$^{\times}$	[1.1]$^{\times}$	[1.2]$^{\times}$	[1.3]$^{\times}$

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. Standard RDD uses all observations within the bandwidth window. Donut RDD excludes observations whose income brackets straddle the 185% FPL threshold. Simulation RDD draws 100 simulated income values uniformly within each bracket. Local polynomial regression with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. $^{\times}$F-statistics below the conventional weak IV threshold of 10. Estimates in percentage points.

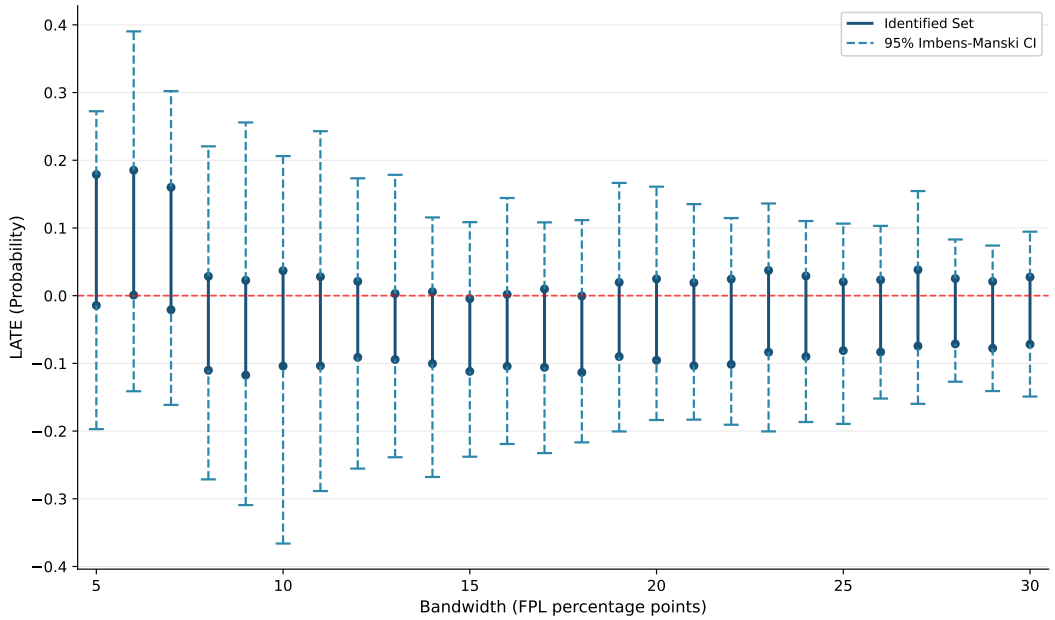
Source: PRAMS 2009–2022, unweighted analysis.

Table 16: RDD Method Comparison: Breastfeeding Duration

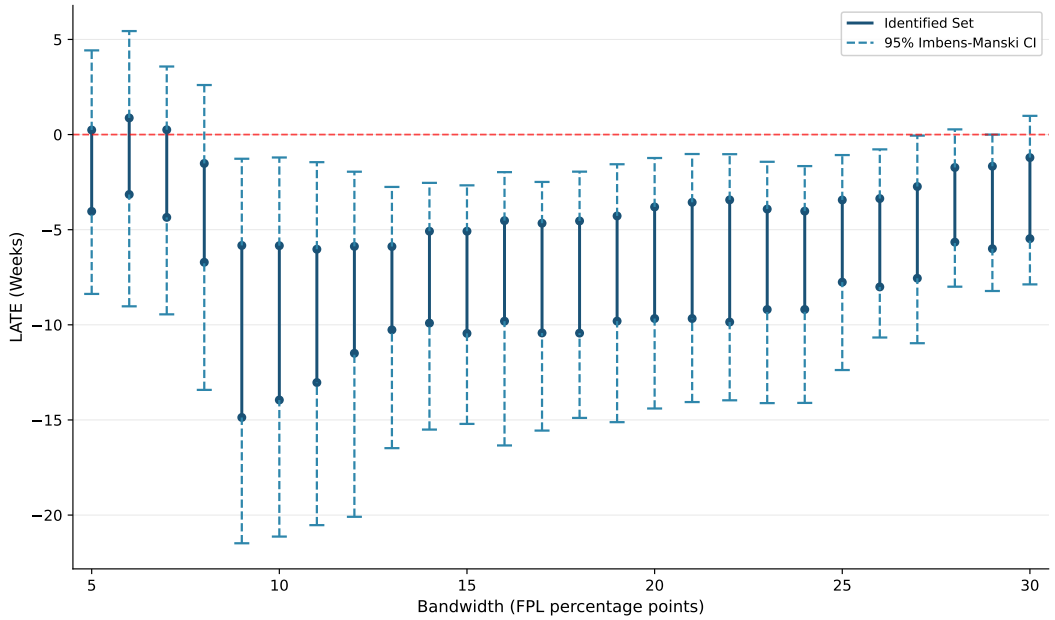
	Bandwidth					
	5	**10***	**15**	**20**	**25**	**30**
Panel A: Linear ($p = 1$)						
Standard RDD	12.11	-4.05	-0.57	0.11	-0.84	-0.06
	(9.00)	(4.65)	(3.59)	(1.62)	(1.13)	(1.07)
	[5.1]$^{\times}$	[11.1]	[18.5]	[93.8]	[193.0]	[215.3]
Donut RDD	-2.73	-3.87	-3.84	-3.99	-4.29	-4.79
	(1.05)	(0.75)	(0.63)	(0.53)	(0.49)	(0.45)
	[275.6]	[541.4]	[758.0]	[1051.6]	[1266.8]	[1505.2]
Simulation RDD	19.19	-7.62	-29.03	97.63	-1.59	4.65
	(5208.11)	(2205.55)	(26333.68)	(650364.40)	(1364.40)	(5545.29)
	[1.0]$^{\times}$	[1.1]$^{\times}$	[1.6]$^{\times}$	[2.4]$^{\times}$	[3.8]$^{\times}$	[5.9]$^{\times}$
Panel B: Quadratic ($p = 2$)						
Standard RDD	18.62	4.44	-4.54	0.52	1.20	-1.86
	(14.19)	(5.00)	(6.33)	(8.45)	(4.06)	(2.12)
	[3.2]$^{\times}$	[10.7]	[6.1]$^{\times}$	[3.3]$^{\times}$	[14.8]	[53.0]
Donut RDD	-4.50	-2.22	-3.41	-3.66	-3.48	-3.35
	(1.73)	(1.19)	(0.92)	(0.80)	(0.70)	(0.64)
	[93.4]	[216.0]	[361.0]	[473.1]	[619.9]	[740.0]
Simulation RDD	-12.82	-22.67	-20.69	12.32	6.43	-3.53
	(21827.78)	(10877.34)	(16337.11)	(3008.82)	(5391.49)	(4177.72)
	[1.0]$^{\times}$	[1.0]$^{\times}$	[1.0]$^{\times}$	[1.1]$^{\times}$	[1.1]$^{\times}$	[1.3]$^{\times}$

Notes: Bandwidth in FPL percentage points. *Baseline bandwidth = 10. Standard RDD uses all observations within the bandwidth window. Donut RDD excludes observations whose income brackets straddle the 185% FPL threshold. Simulation RDD draws 100 simulated income values uniformly within each bracket. Local polynomial regression with triangular kernel weights. Standard errors are in parentheses, and first-stage F-statistics are in brackets. $^{\times}$F-statistics below the conventional weak IV threshold of 10. Estimates in weeks.
Source: PRAMS 2009–2022, unweighted analysis.

D Supplementary Figures



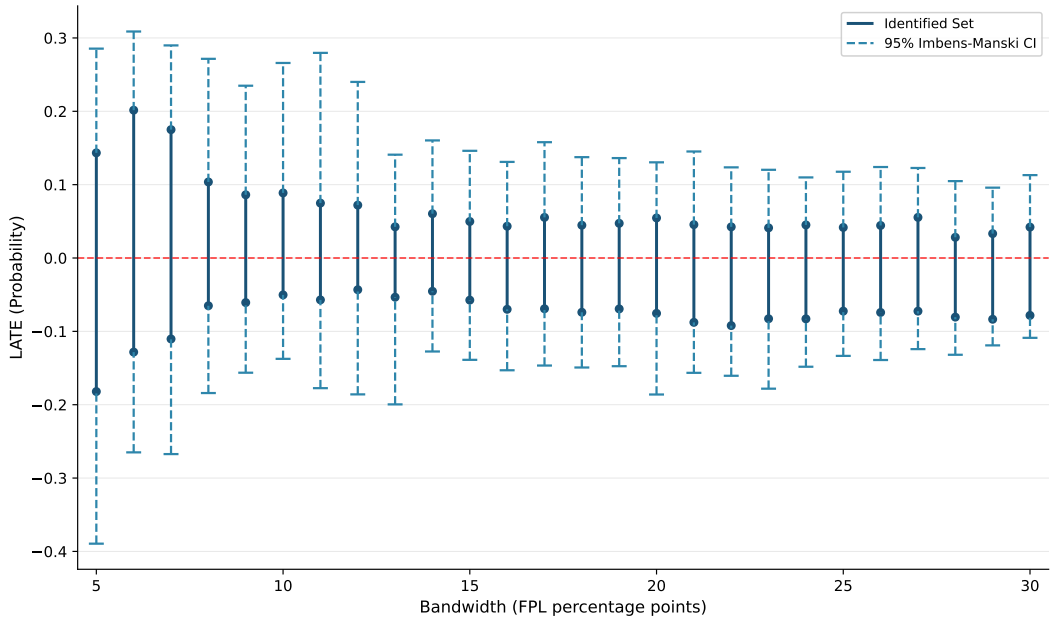
(a) Breastfeeding Initiation (percentage points)



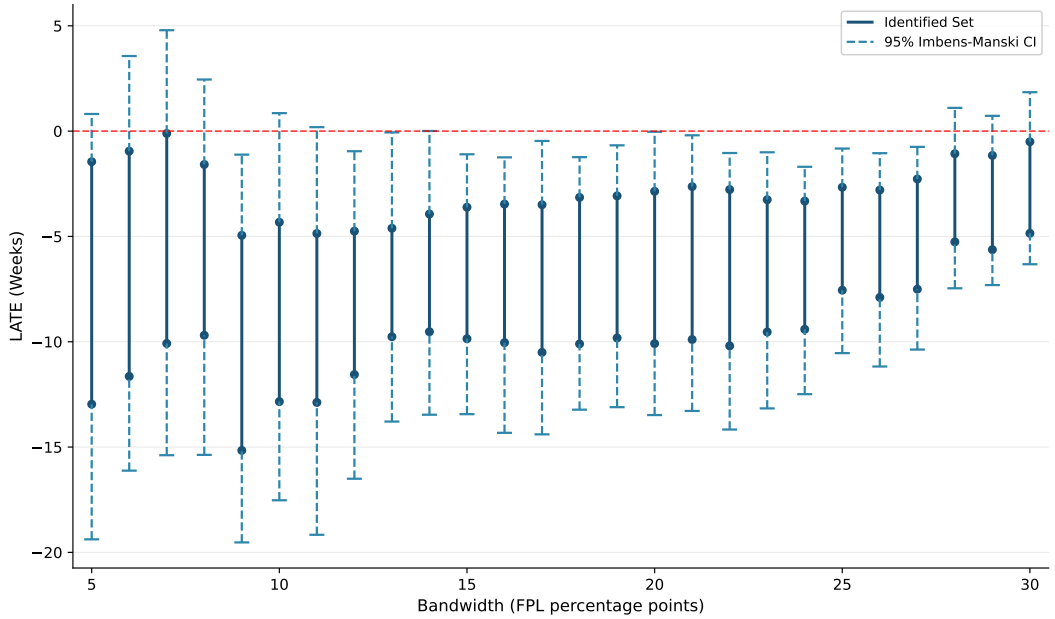
(b) Breastfeeding Duration (weeks)

Figure 15: Partial Identification Bounds: Breastfeeding Outcomes with Survey Weights

Notes: Partial identification LATE bounds for breastfeeding initiation (panel a, percentage points) and duration (panel b, weeks) across bandwidths from 5 to 30 FPL percentage points, using PRAMS survey weights. Calibration quantile $q = 0.50$. Solid lines indicate the upper and lower bounds of the identified set, and shaded regions represent 95% Imbens-Manski confidence intervals. Sample: PRAMS 2009–2022, weighted.



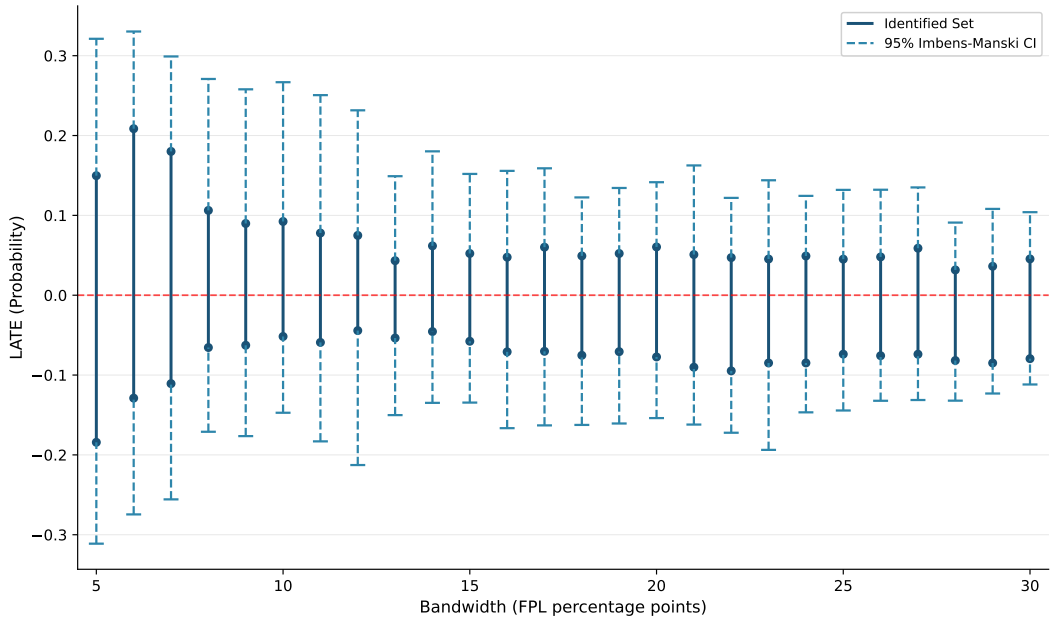
(a) Breastfeeding Initiation (percentage points)



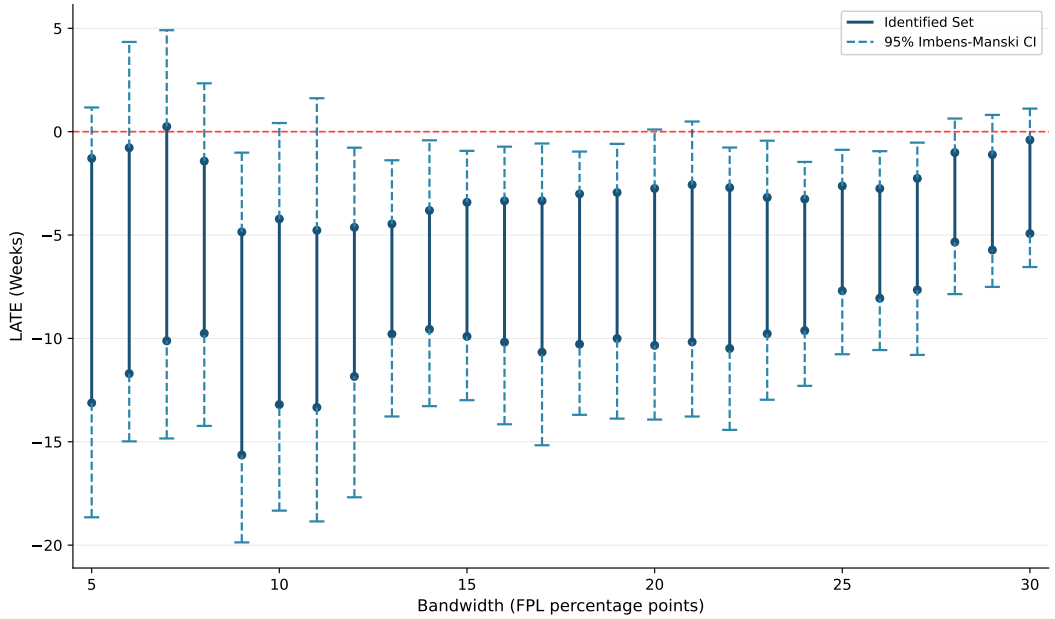
(b) Breastfeeding Duration (weeks)

Figure 16: Partial Identification Bounds: Breastfeeding Outcomes at $q = 0.75$ Calibration

Notes: Partial identification LATE bounds for breastfeeding initiation (panel a, percentage points) and duration (panel b, weeks) across bandwidths from 5 to 30 FPL percentage points. Calibration quantile $q = 0.75$. Solid lines indicate the upper and lower bounds of the identified set, and shaded regions represent 95% Imbens-Manski confidence intervals. Sample: PRAMS 2009–2022, unweighted.



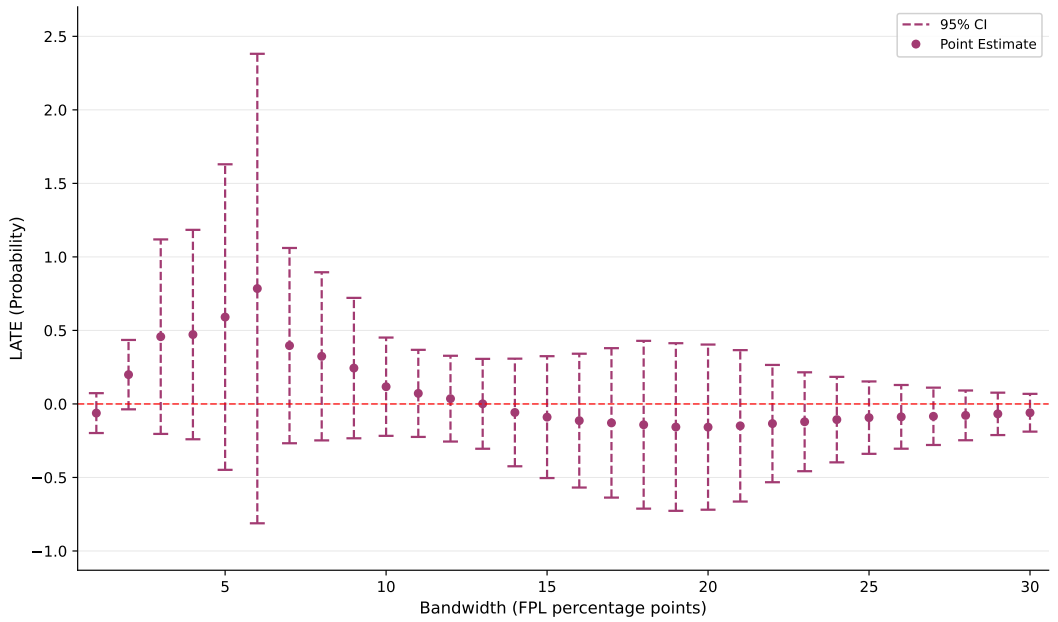
(a) Breastfeeding Initiation (percentage points)



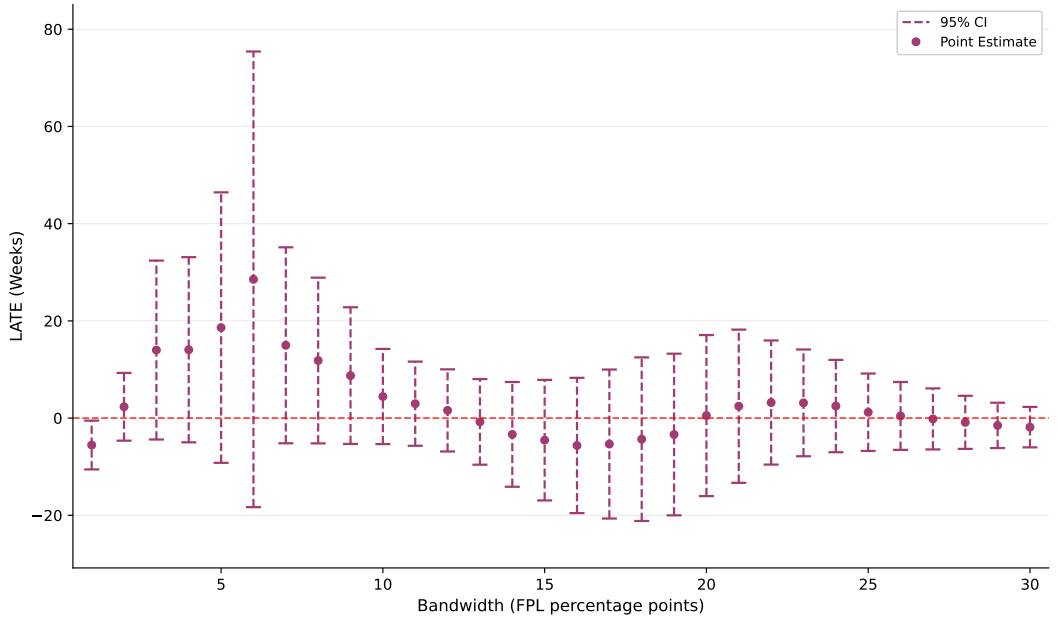
(b) Breastfeeding Duration (weeks)

Figure 17: Partial Identification Bounds: Breastfeeding Outcomes at $q = 0.90$ Calibration

Notes: Partial identification LATE bounds for breastfeeding initiation (panel a, percentage points) and duration (panel b, weeks) across bandwidths from 5 to 30 FPL percentage points. Calibration quantile $q = 0.90$. Solid lines indicate the upper and lower bounds of the identified set, and shaded regions represent 95% Imbens-Manski confidence intervals. Sample: PRAMS 2009–2022, unweighted.



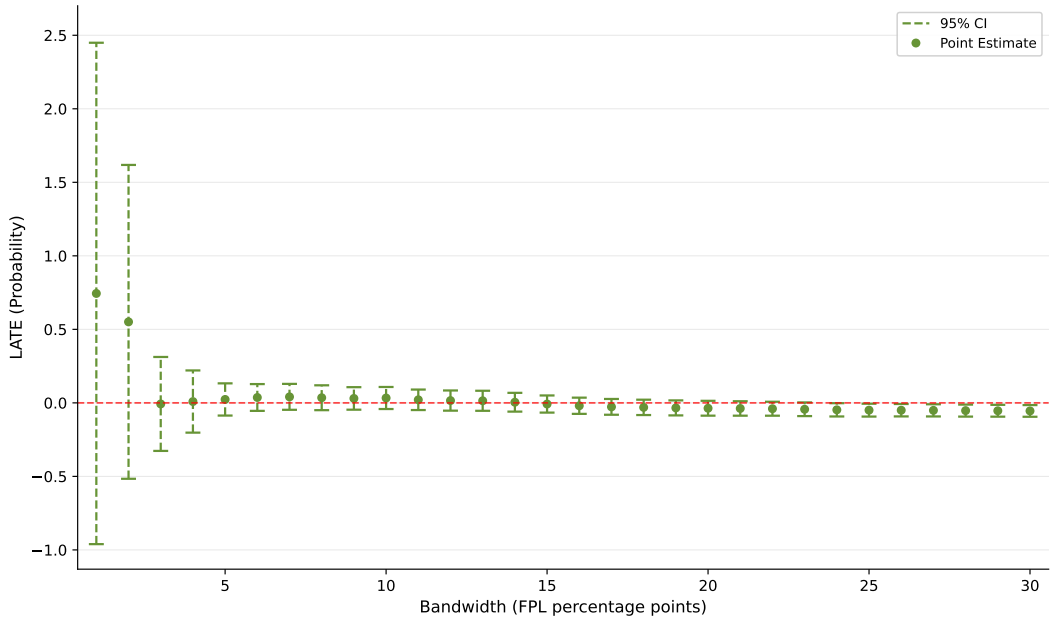
(a) Breastfeeding Initiation



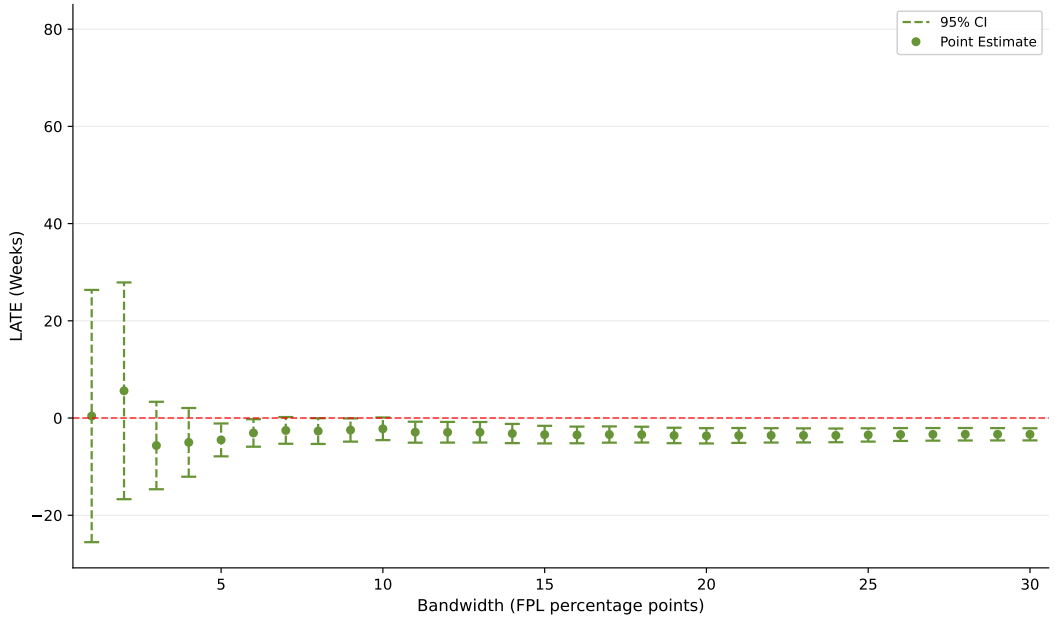
(b) Breastfeeding Duration

Figure 18: Standard RDD Estimates: Breastfeeding Outcomes with Quadratic Polynomial

Notes: Standard fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 1 to 30 FPL percentage points. Each point is a LATE point estimate, and the dashed lines indicate 95% confidence intervals. The horizontal line marks zero. This approach uses all observations, including those whose income brackets straddle the eligibility threshold, treating bracket midpoints as true income. Sample: PRAMS 2009–2022, unweighted, quadratic specification.



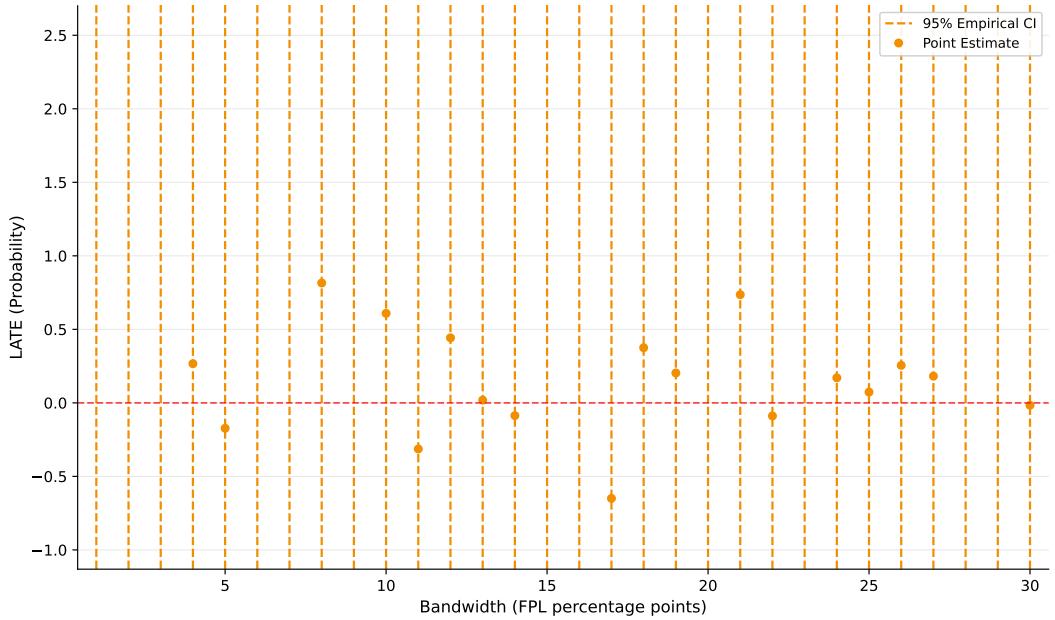
(a) Breastfeeding Initiation



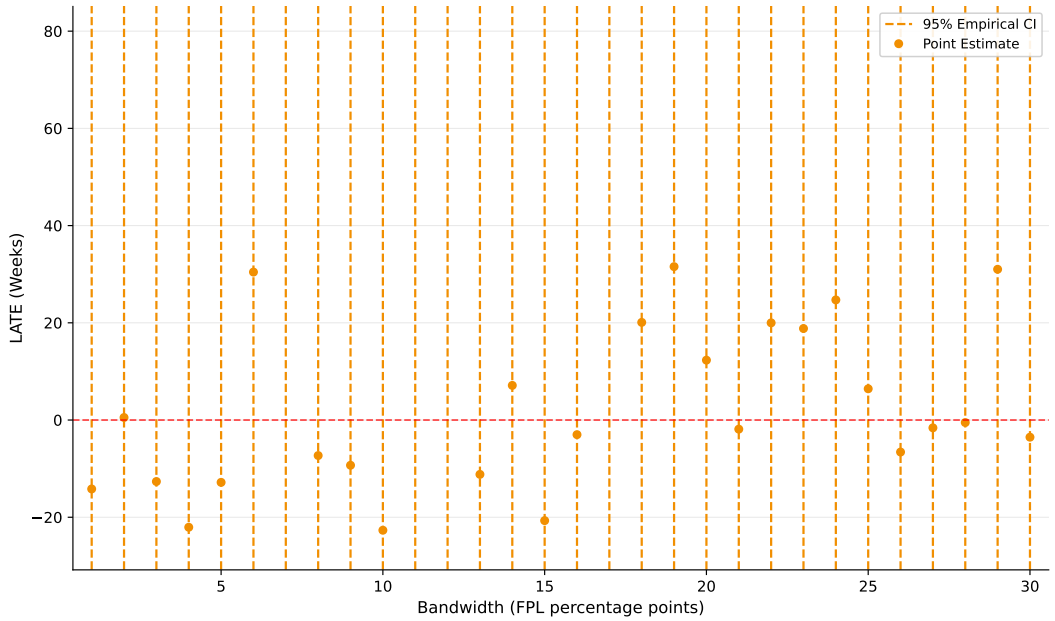
(b) Breastfeeding Duration

Figure 19: Donut RDD Estimates: Breastfeeding Outcomes

Notes: Donut fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 8 to 30 FPL percentage points. Each point is a LATE point estimate, and the dashed lines indicate 95% confidence intervals. The horizontal line marks zero. This approach excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Confidence intervals extend beyond the plotted range at narrow bandwidths, reflecting the loss of near-threshold observations. Sample: PRAMS 2009–2022, unweighted, quadratic specification.



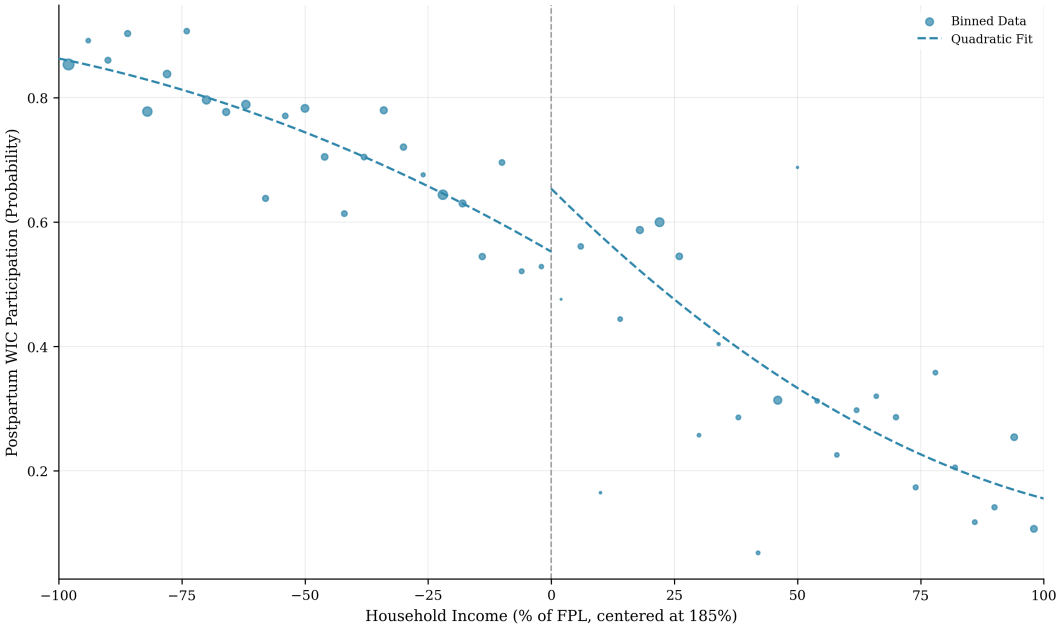
(a) Breastfeeding Initiation



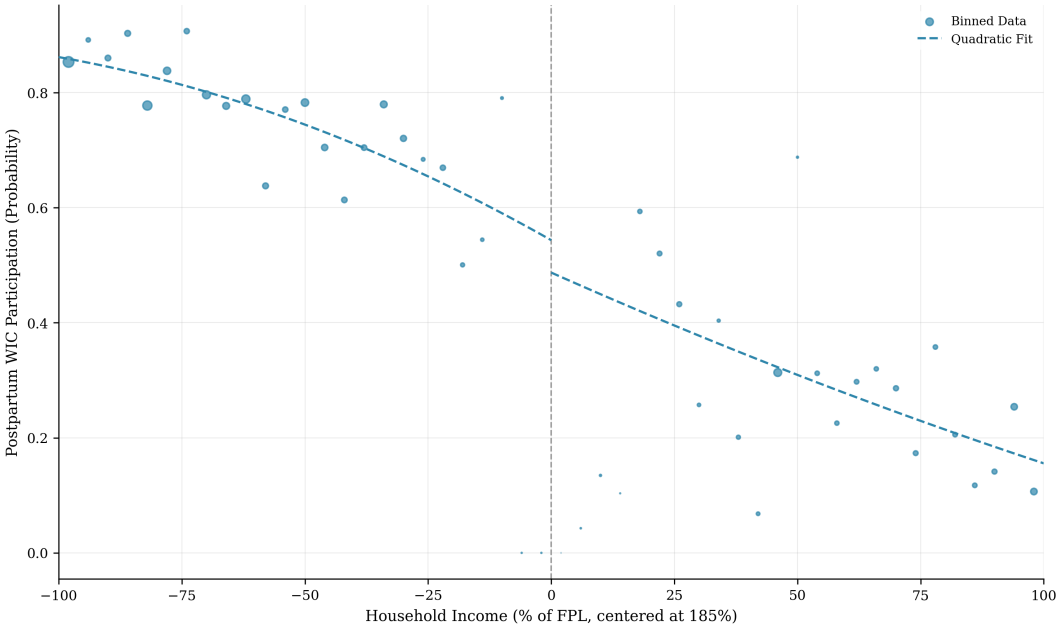
(b) Breastfeeding Duration

Figure 20: Simulation RDD Estimates: Breastfeeding Outcomes

Notes: Simulation-based fuzzy RDD LATE estimates for breastfeeding initiation (panel a) and duration (panel b) across bandwidths from 2 to 30 FPL percentage points. This approach draws 100 simulated income values uniformly within each household's reported bracket, estimates a fuzzy RDD on each draw, and reports the average point estimate. Each point is the mean LATE across simulations, and the dashed lines indicate 95% empirical confidence intervals. The horizontal line marks zero. Sample: PRAMS 2009–2022, unweighted, quadratic specification.



(a) Including Straddling Observations

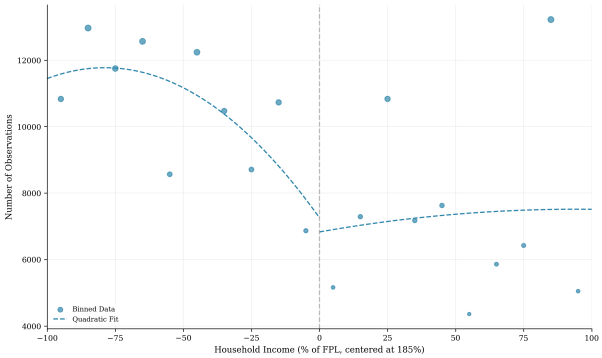


(b) Excluding Straddling Observations

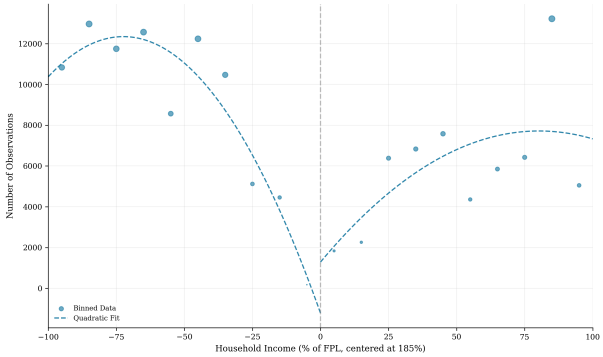
Figure 21: WIC Participation Postpartum by Household Income

Notes: Mean WIC participation in the postpartum period by household income as a percentage of the Federal Poverty Level (FPL). Panel (a) includes all observations; panel (b) excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Points represent bin means, the solid line shows a quadratic fit, and the vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022, 50 bins.

Panel A: 20 Bins

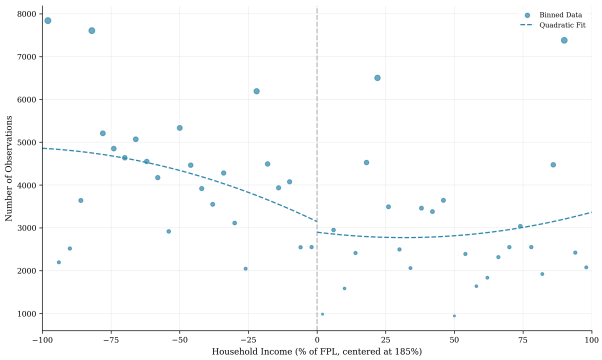


With Straddlers

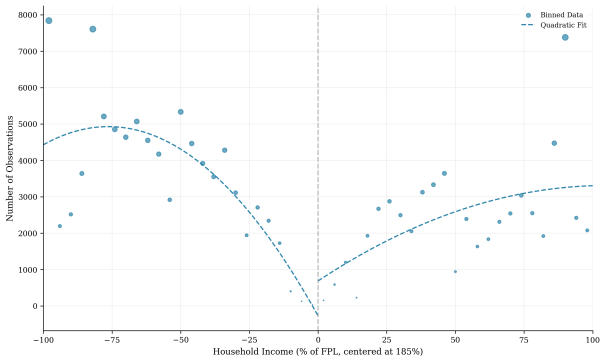


Without Straddlers

Panel B: 50 Bins

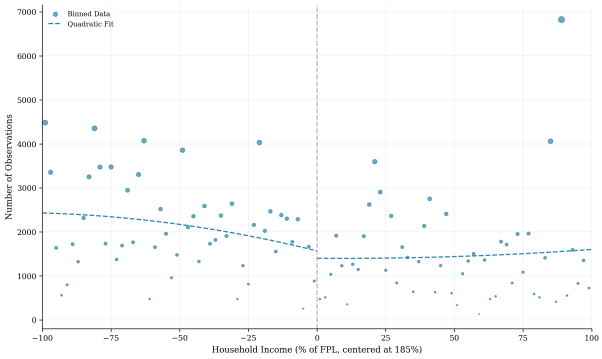


With Straddlers

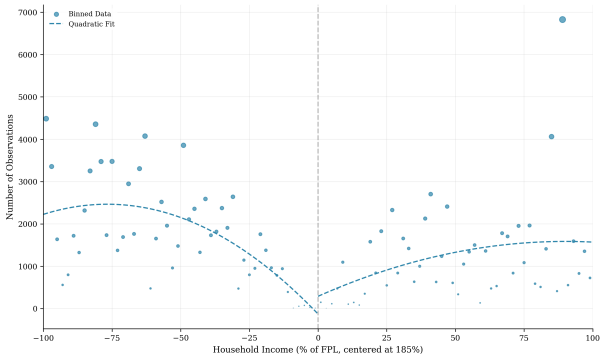


Without Straddlers

Panel C: 100 Bins



With Straddlers

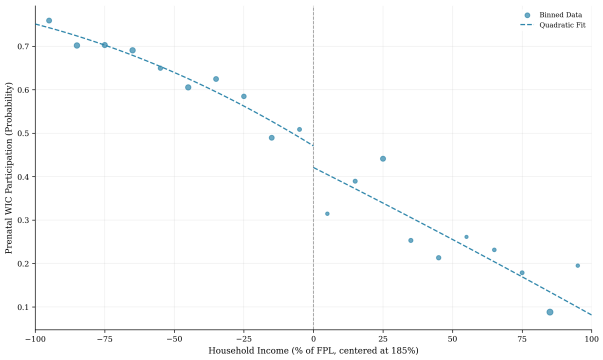


Without Straddlers

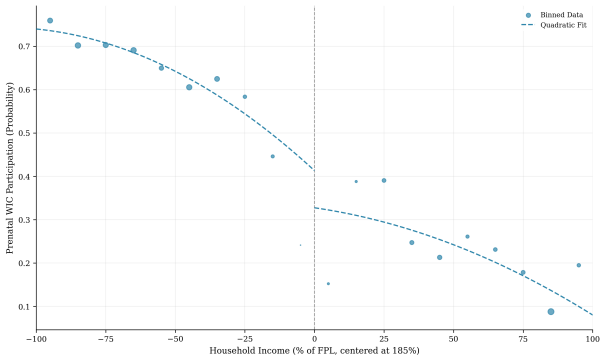
Figure 22: Observation Counts by Household Income: Robustness to Bin Width

Notes: Binned observation counts by household income using 20, 50, and 100 bins. Left column includes all observations; right column excludes observations whose income brackets straddle the 185% FPL eligibility threshold. The vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022.

Panel A: 20 Bins

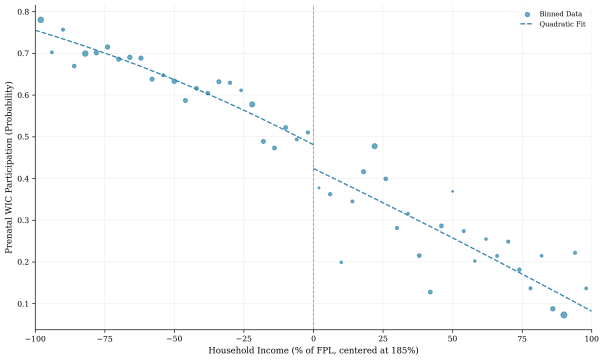


With Straddlers

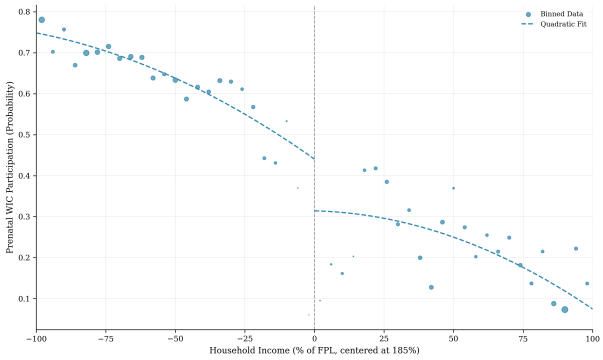


Without Straddlers

Panel B: 50 Bins

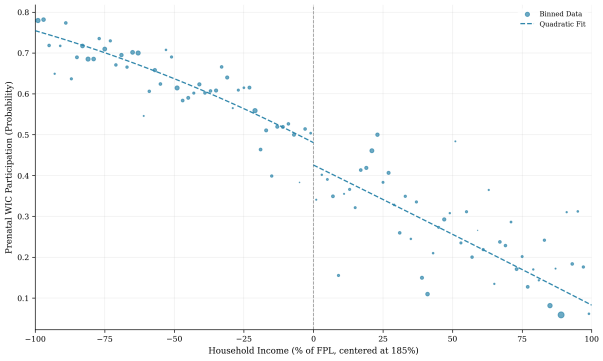


With Straddlers

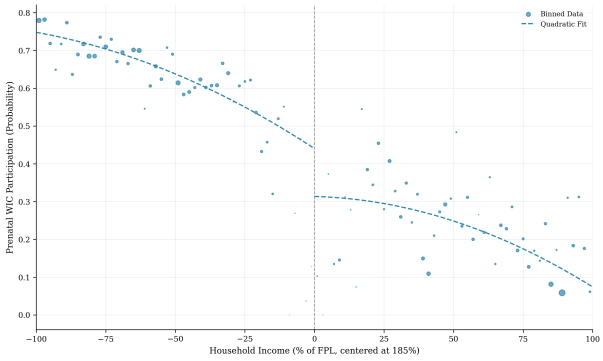


Without Straddlers

Panel C: 100 Bins



With Straddlers

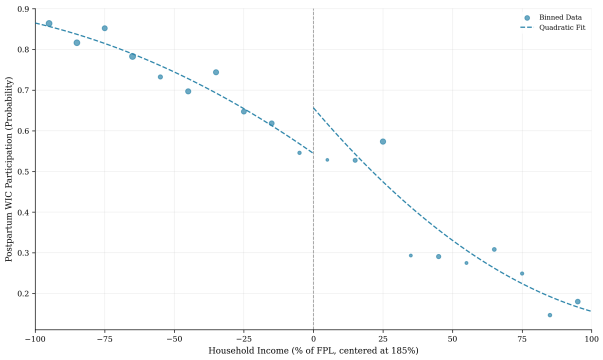


Without Straddlers

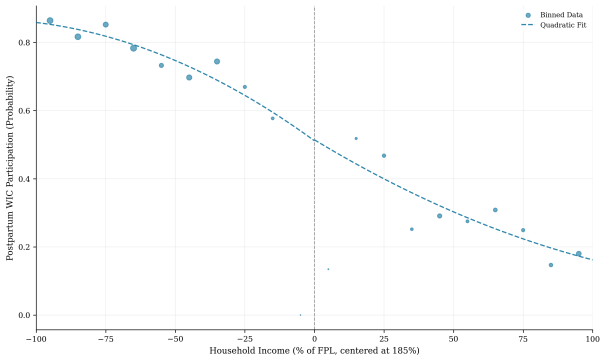
Figure 23: WIC Participation During Pregnancy: Robustness to Bin Width

Notes: Mean WIC participation during pregnancy by household income using 20, 50, and 100 bins. Left column includes all observations; right column excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Points represent bin means, the solid line shows a quadratic fit, and the vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022.

Panel A: 20 Bins

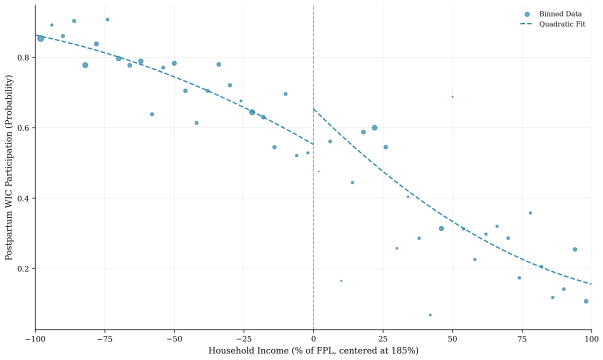


With Straddlers

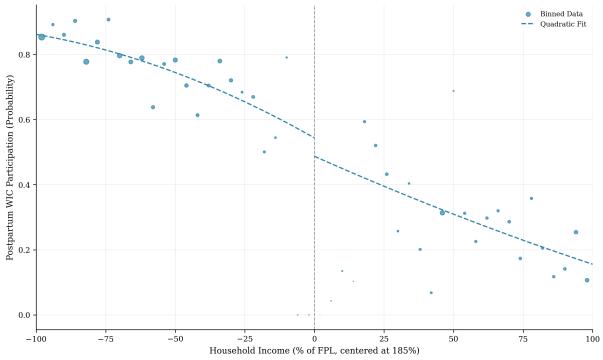


Without Straddlers

Panel B: 50 Bins

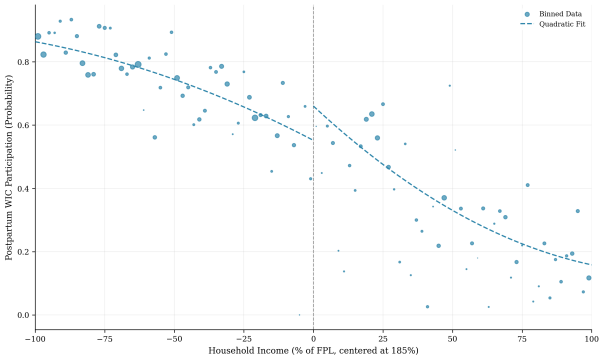


With Straddlers

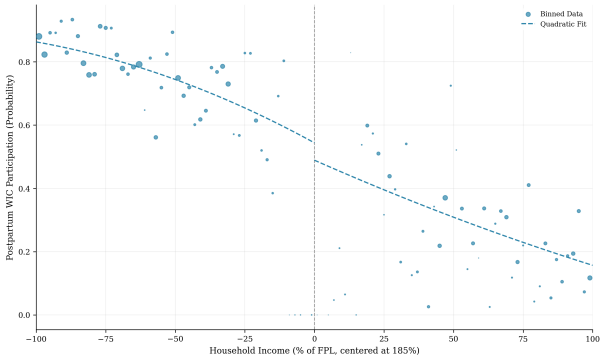


Without Straddlers

Panel C: 100 Bins



With Straddlers

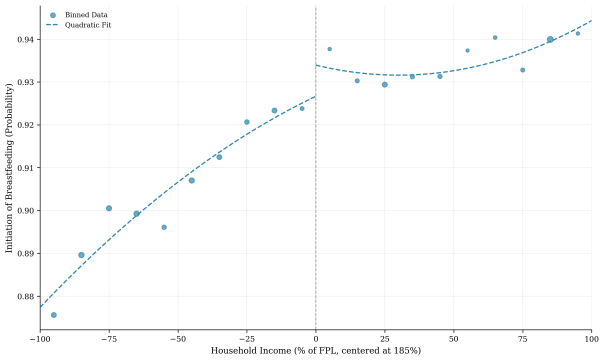


Without Straddlers

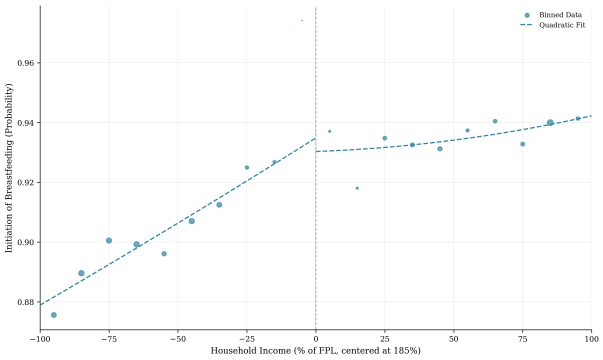
Figure 24: WIC Participation Postpartum: Robustness to Bin Width

Notes: Mean WIC participation in the postpartum period by household income using 20, 50, and 100 bins. Left column includes all observations; right column excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Points represent bin means, the solid line shows a quadratic fit, and the vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022.

Panel A: 20 Bins

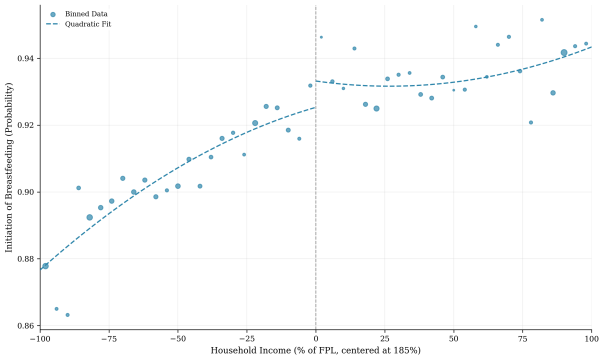


With Straddlers

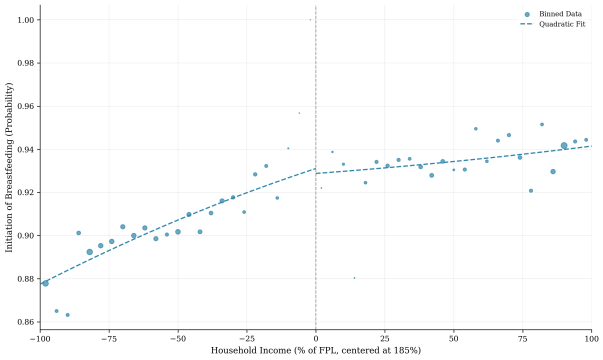


Without Straddlers

Panel B: 50 Bins

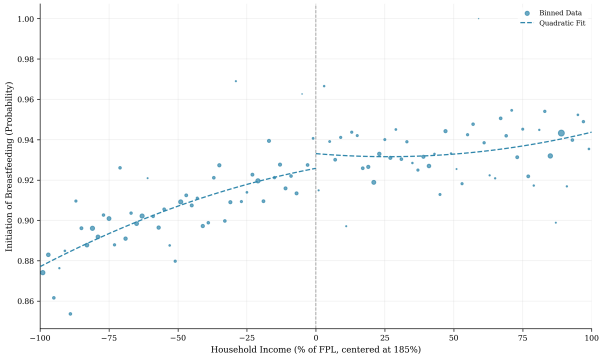


With Straddlers

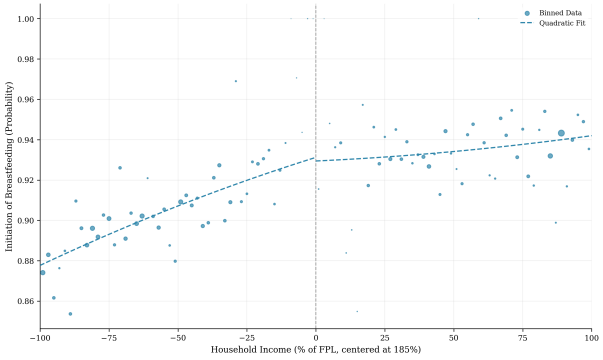


Without Straddlers

Panel C: 100 Bins



With Straddlers

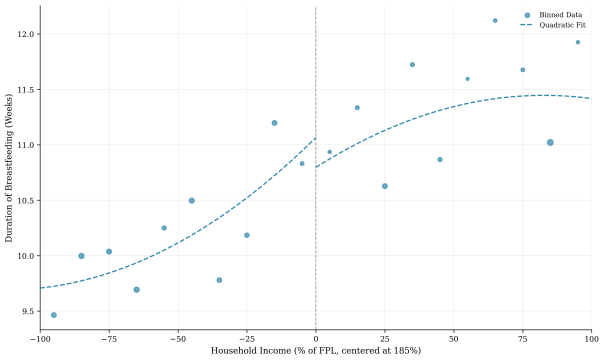


Without Straddlers

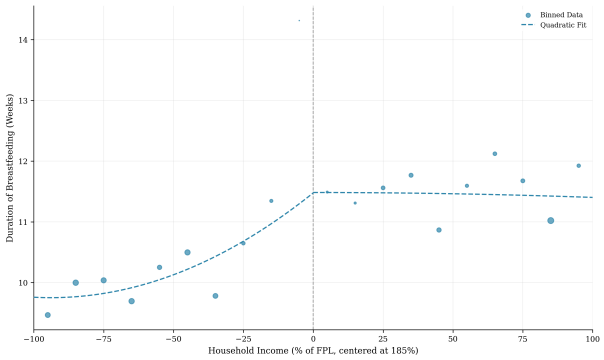
Figure 25: Breastfeeding Initiation: Robustness to Bin Width

Notes: Mean breastfeeding initiation by household income using 20, 50, and 100 bins. Left column includes all observations; right column excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Points represent bin means, the solid line shows a quadratic fit, and the vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022.

Panel A: 20 Bins

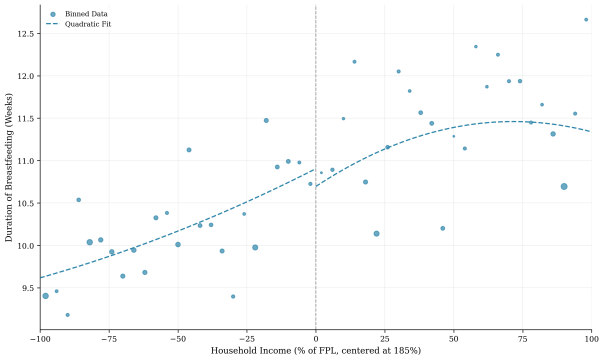


With Straddlers

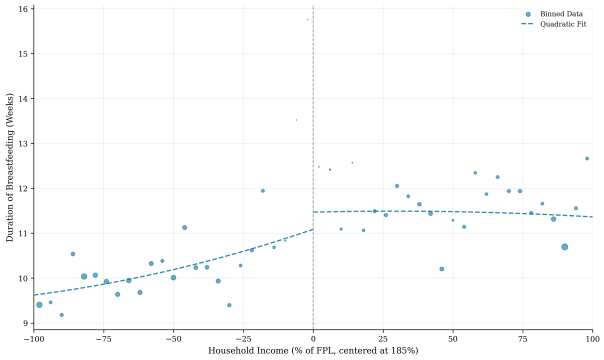


Without Straddlers

Panel B: 50 Bins

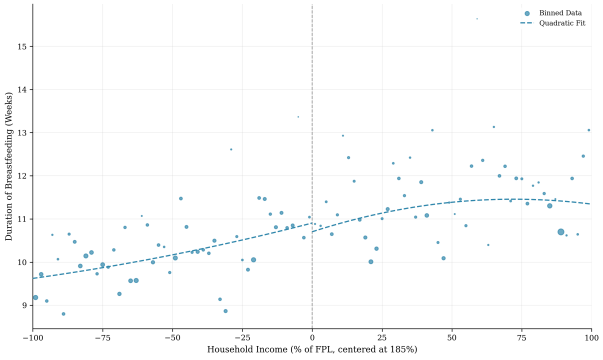


With Straddlers

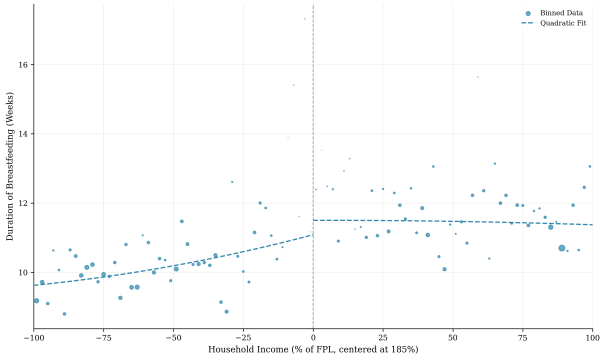


Without Straddlers

Panel C: 100 Bins



With Straddlers



Without Straddlers

Figure 26: Breastfeeding Duration: Robustness to Bin Width

Notes: Mean breastfeeding duration (weeks) by household income using 20, 50, and 100 bins. Left column includes all observations; right column excludes observations whose income brackets straddle the 185% FPL eligibility threshold. Points represent bin means, the solid line shows a quadratic fit, and the vertical line marks the eligibility cutoff. Sample: PRAMS 2009–2022.

E Data Appendix

E.1 Sample Construction

The analytical sample is drawn from the Pregnancy Risk Assessment Monitoring System (PRAMS), a joint surveillance project of the Centers for Disease Control and Prevention and participating state health departments. PRAMS surveys mothers 2–6 months postpartum, linking birth certificate records with questionnaire responses on maternal attitudes, behaviors, and experiences before, during, and shortly after pregnancy.

I focus on PRAMS Phases 6–8, spanning 2009–2022. This period was selected for two reasons: (1) the consistent structure of WIC infant food packages following the October 2009 revision, and (2) the systematic collection of WIC participation data across all participating states. The initial sample comprises 508,387 mothers with live singleton births.

I construct the analytical sample by imposing the following restrictions:

1. Non-missing breastfeeding initiation status (excludes 23,423 observations)
2. Non-missing breastfeeding duration (excludes 21,385 observations)
3. Non-missing WIC participation during pregnancy (excludes 2,411 observations)
4. Non-missing household income relative to the Federal Poverty Level (excludes 55,306 observations)

Observations may be excluded for multiple reasons. The final analytical sample comprises 424,796 mother-infant pairs. Among observations with both upper and lower FPL bounds ($N = 311{,}852$), 196,037 (62.9%) are classified as income-eligible for WIC (upper FPL bound $\leq 185\%$), 87,901 (28.2%) as income-ineligible (lower FPL bound $> 185\%$), and 27,914 (8.9%) as "straddlers" whose income brackets cross the eligibility threshold. The remaining 112,944 observations (26.6% of the analytical sample) have open-ended top income brackets with a lower bound but no upper bound; these are predominantly high-income households classified as ineligible for the purposes of the RDD analysis.

E.2 Variable Construction

Breastfeeding Outcomes. I examine two primary breastfeeding outcomes derived from multiple PRAMS items. *Breastfeeding initiation* is a binary indicator equal to one if the mother ever breastfed or pumped breast milk. I construct this measure using three sources: (1) the PRAMS survey item "Did you ever breastfeed or pump breast milk to feed your new baby?" (BF5EVER), (2) the birth certificate item indicating whether the infant was being breastfed at discharge (BRSTFED), and (3) reported breastfeeding duration greater than zero. A mother is coded as having initiated breastfeeding if any of these sources indicates initiation. I flag observations where sources conflict (2.3% of sample).

Breastfeeding duration is the number of weeks the mother breastfed, derived from the PRAMS item "How many weeks did you breastfeed or pump milk to feed your baby?" (BF5WEEKS). For mothers still breastfeeding at survey completion (BF5STILL), I set duration equal to the infant's age at survey. Duration is thus right-censored by survey timing, which occurs 8–30 weeks postpartum. I flag observations where reported duration exceeds infant age (0.8% of sample) and where duration exceeds 52 weeks (0.1% of sample).

Exclusive breastfeeding duration is defined as the number of weeks until the introduction of any liquid other than breast milk (including formula, water, or juice) or solid food. I construct this measure as the minimum of the reported age at introduction of other liquids (BF6LIQD_WKS) and the age at introduction of solid foods (BF6FOOD_WKS).

Treatment. I define treatment as participation in WIC during pregnancy. The primary source is the birth certificate item indicating whether the mother received WIC food during pregnancy (MAT_WIC). I supplement this with the PRAMS survey item on prenatal WIC participation (WIC_PREG) using a fallback rule: a mother is coded as a WIC participant if either source indicates participation. This approach maximizes sensitivity and reduces misclassification due to underreporting.

Birth certificate–based recording offers advantages over retrospective self-report: information is collected at delivery, reducing recall error, and administrative recording may be more complete than survey responses. I flag observations where the two sources conflict (1.1% of sample). Postpartum WIC participation is available from the PRAMS survey (PPS_WIC6) but is not used in the primary analysis due to potential endogeneity with breastfeeding decisions.

Running Variable: Income Relative to the Federal Poverty Level. PRAMS reports household income in categorical brackets rather than continuous values. The bracket structure varies across survey phases:

- *Phase 6 (2009–2011):* 12–16 categories with bracket widths of $4,000–$12,000
- *Phase 7 (2012–2015):* 12–16 categories with bracket widths of $3,000–$12,000
- *Phase 8 (2016–2022):* 12 categories with bracket widths of $5,000–$14,000

For each observation, I select the highest-resolution income variable available (priority: INCOME8 $>$ INCOME7 $>$ INCOME5) and extract the lower and upper bounds of the reported bracket.

Household size is measured directly using the PRAMS survey item: "During the 12 months before your new baby was born, how many people, including yourself, depended on this income?" (INC_NDEP). This measure captures all household members who share income, including the mother, spouse or partner (if applicable), and children. I then compute the Federal Poverty Level threshold for each observation using HHS poverty guidelines matched by year, state (for Alaska and Hawaii), and household size.

The running variable is household income as a percentage of the FPL. I calculate lower (ℓ_i) and upper (u_i) bounds:

$$\ell_i = \frac{\text{Income Lower Bound}_i}{\text{FPL Threshold}_i} \times 100$$
$$u_i = \frac{\text{Income Upper Bound}_i}{\text{FPL Threshold}_i} \times 100$$

The midpoint $m_i = (\ell_i + u_i)/2$ serves as a proxy for true income in binned scatter plots and descriptive analyses. For observations with open-ended upper brackets (e.g., "$100,000 or more"), the upper bound is undefined; I set the midpoint equal to the lower bound, $m_i = \ell_i$, and assign eligibility based on this value. These observations enter the RDD analysis with a valid running variable and instrument but are excluded from the bracket-based eligibility classification described

below.

Eligibility Classification. For observations with both FPL bounds defined, I classify into three mutually exclusive groups based on their bounds relative to the 185% eligibility threshold:

- *Eligible:* Upper bound at or below threshold ($u_i \leq 185$)
- *Ineligible:* Lower bound above threshold ($\ell_i > 185$)
- *Straddlers:* Bounds cross the threshold ($\ell_i \leq 185 < u_i$)

The average bracket width among observations with both bounds is approximately 52 percentage points of FPL. Among these observations with bracket midpoints within 10 percentage points of the eligibility threshold, 97% are straddlers.

Demographic Covariates. I construct indicator variables for maternal age groups ($\leq$17, 18–19, 20–24, 25–29, 30–34, 35–39, $\geq$40), education levels (less than high school, some high school, high school graduate, some college, college or more), race (White, Black, Asian subgroups, American Indian, Alaska Native, other), and Hispanic ethnicity. Marital status is a binary indicator from the birth certificate.

Employment and Leave. Employment status indicates whether the mother worked for pay during pregnancy. Leave variables indicate whether the mother took paid leave, unpaid leave, or no leave after delivery. I additionally construct indicators for specific leave barriers (could not afford unpaid leave, afraid of losing job, not eligible for FMLA, etc.).

Health and Insurance. Insurance coverage during pregnancy is derived from multiple items indicating payment sources for prenatal care (Medicaid, employer, CHIP, military, other government programs, private purchase, none). I construct a binary indicator for any insurance coverage and a separate indicator for Medicaid coverage. Prenatal care timing is categorized as first trimester initiation versus later or no care.

Infant Characteristics. Birth weight categories follow clinical definitions: very low birth weight ($<$1,500g), low birth weight ($<$2,500g), normal birth weight (2,500–4,000g), high birth weight ($>$4,000g), and macrosomia ($\geq$4,500g). Gestational age categories include preterm ($<$37 weeks), early term (37–38 weeks), full term (39–40 weeks), late term (41 weeks), and post-term ($\geq$42 weeks). Size for gestational age is classified using the 10th and 90th percentiles.

Survey Weights. PRAMS provides analysis weights (WTANAL) that combine three components: (1) base sampling weights reflecting the probability of selection, (2) nonresponse adjustments, and (3) noncoverage adjustments for births not captured by the sampling frame. Weighted estimates are representative of each participating state's population of women with live births. I present results both with and without survey weights to assess sensitivity.

E.3 Income Bracket Structure

Table 17 documents the income bracket structure across PRAMS phases. The number of brackets, bracket boundaries, and bracket widths vary by phase, reflecting periodic revisions to the PRAMS questionnaire.

Table 17: Income Brackets Near the WIC Eligibility Threshold

Source	Income Bracket	FPL Percentage by Household Size			
		2	3	4	5
INCOME5	$25,000–$34,999	145%–203%†	115%–161%	95%–134%	81%–114%
INCOME5	$35,000–$49,999	203%–290%	161%–230%†	134%–191%†	114%–163%
INCOME5	$50,000+	290%+	230%+	191%+	163%+
INCOME7	$26,001–$29,000	151%–168%	120%–134%	99%–111%	85%–95%
INCOME7	$29,001–$37,000	168%–215%†	134%–170%	111%–141%	95%–121%
INCOME7	$37,001–$44,000	215%–255%	170%–203%†	141%–168%	121%–143%
INCOME7	$44,001–$52,000	255%–302%	203%–239%	168%–198%†	143%–169%
INCOME7	$52,001–$56,000	302%–325%	239%–258%	198%–214%	169%–183%
INCOME7	$56,001–$67,000	325%–389%	258%–308%	214%–256%	183%–218%†
INCOME8	$28,001–$32,000	162%–186%†	129%–147%	107%–122%	91%–104%
INCOME8	$32,001–$40,000	186%–232%	147%–184%	122%–153%	104%–130%
INCOME8	$40,001–$48,000	232%–278%	184%–221%†	153%–183%	130%–156%
INCOME8	$48,001–$57,000	278%–331%	221%–262%	183%–218%†	156%–186%†
INCOME8	$57,001–$60,000	331%–348%	262%–276%	218%–229%	186%–196%
INCOME8	$60,001–$73,000	348%–423%	276%–336%	229%–279%	196%–238%

Notes: Income brackets from PRAMS survey questions. FPL percentages calculated using 2020 HHS Poverty Guidelines for the 48 contiguous states. The WIC eligibility threshold is 185% FPL. †Bracket straddles the 185% FPL threshold at this household size, creating uncertainty in eligibility classification. *Source:* PRAMS questionnaire; HHS Poverty Guidelines (2020).

The variation in bracket structure has two implications for the analysis. First, the precision of income measurement varies across phases, with narrower brackets providing more precise FPL calculations. Second, the set of observations classified as straddlers depends on the bracket structure: wider brackets increase the prevalence of straddling near the eligibility threshold. I address this heterogeneity by controlling for survey phase in regression specifications and by presenting results separately by phase in robustness analyses.

E.4 Geographic Coverage

PRAMS participation is voluntary at the state level and has expanded over time. As of 2022, 47 states, the District of Columbia, Puerto Rico, and New York City participate in PRAMS. California, Idaho, North Carolina, and Ohio are non-participating states. The analytical sample draws from participating jurisdictions across Phases 6–8 (2009–2022). Table 18 summarizes the analytical sample composition by phase after applying outcome and covariate restrictions.

Table 18: Analytical Sample Composition by PRAMS Phase

Phase	Years	Sample (Obs)†	% of Total
Phase 6	2009–2011	95,179	22.4%
Phase 7	2012–2015	123,304	29.0%
Phase 8	2016–2022	206,313	48.6%
Total (2009–2022)		**424,796**	**100.0%**

Notes: †Sample counts represent the analytical sample after applying restrictions: (1) non-missing breastfeeding initiation and duration, (2) non-missing WIC participation during pregnancy, and (3) non-missing household income relative to FPL. The initial sample before restrictions comprised 508,387 mothers with live singleton births across PRAMS Phases 6–8. PRAMS participation varies by state and has expanded over time. As of 2022, 47 states plus the District of Columbia, Puerto Rico, and New York City participate in PRAMS. Non-participating states include California, Idaho, North Carolina, and Ohio. The analytical sample includes observations from participating jurisdictions with complete data on key variables.
Source: PRAMS 2009–2022.

State-level variation in sample size reflects both population differences and sampling design choices. States employ stratified sampling that oversamples mothers of low-birth-weight infants, those in high-risk geographic areas, and racial/ethnic minorities (Shulman et al., 2018). Response rates average 65–70% across participating states and years, with nonresponse adjustments incorporated into the survey weights.

F Empirical Appendix

F.1 Bandwidth Selection Justification

The choice of bandwidth in a regression discontinuity design typically involves a trade-off between the bias from polynomial approximation error and the variance from sampling uncertainty. With precisely measured running variables, data-driven procedures select a bandwidth that minimizes an asymptotic approximation to the Mean Squared Error (MSE). These methods are theoretically inappropriate in this setting because household income is reported in intervals rather than as a continuous variable.

With measurement error in the running variable, the core terms of the MSE formula are not identified. The bias term requires derivatives of the conditional expectation function with respect to true latent income, not the observed bracket midpoint. Since the within-bracket distribution of income is unknown, these derivatives cannot be consistently estimated. The variance term reflects sampling variation in the proxy, which may differ from the variance of the target parameter

depending on the within-bracket distribution. An "optimal" bandwidth selected via standard methods therefore lacks a clear theoretical justification.

I use a fixed baseline bandwidth of $h = 10$ percentage points of the Federal Poverty Level, focusing the analysis on households within ± 10 percentage points of the 185% FPL eligibility threshold (175–195% FPL). This window prioritizes the population where the quasi-experimental variation in eligibility is most plausible and minimizes the degree of extrapolation required. To assess sensitivity, I conduct a bandwidth sweep across $h \in \{5, 10, 15, 20, 25, 30\}$, representing $0.5\times$ to $3.0\times$ the baseline bandwidth. The identified bounds for breastfeeding duration remain stable across this range.

F.2 Partial Identification Implementation Details

This section documents the calibration of shape constraints, the construction of first-stage bounds, and the inference procedure for set-identified parameters.

F.2.1 Calibration of Shape Constraints

Bounding the LATE requires Lipschitz constants λ_z and κ_z (Assumption 3) that control the maximum rate of change for the latent treatment probability $p^\star(x)$ and the latent conditional outcome $g^\star(x)$. Rather than imposing these constants a priori, I calibrate them from observed slopes in the certainty regions $\mathcal{X}_E$ and $\mathcal{X}_I$, where treatment status is known.

The calibration proceeds in three steps. First, I partition the certainty regions into bins of width h. Second, I compute the sample means of treatment ($\bar{T}_j$) and outcome ($\bar{Y}_j$) in each bin. Third, I calculate absolute slopes between adjacent bins: $|\Delta_T| = |(\bar{T}_j - \bar{T}_{j-1})/(x_j - x_{j-1})|$ and $|\Delta_Y| = |(\bar{Y}_j - \bar{Y}_{j-1})/(x_j - x_{j-1})|$. These slopes form an empirical distribution of the smoothness of each function.

I set λ_z and κ_z to specific quantiles (q) of these slope distributions. The primary analysis uses the median ($q = 0.50$). I selected a calibration window of $k = 11$ consecutive bins and a bin fraction of 0.25 via grid search over $k \in [5, 20]$ and bin fraction $\in [0.1, 0.5]$, minimizing the width of the Imbens-Manski confidence interval. Higher quantiles ($q = 0.75, 0.90$) appear in sensitivity analyses.

F.2.2 First-Stage Bounds

The first-stage discontinuity $\tau_T = p^\star(0^-) - p^\star(0^+)$ is itself set-identified. I construct the bounds $[\tau_T^{LB}, \tau_T^{UB}]$ by projecting from the reference bins in the certainty regions toward the threshold using the calibrated Lipschitz constants λ_z and the monotonicity restriction (Assumption 2).

The upper bound τ_T^{UB} is the maximum discontinuity consistent with the data and shape restrictions; the lower bound τ_T^{LB} is the minimum. At the baseline bandwidth, the first-stage bounds are $[14.0, 25.1]$ percentage points (Table 3). These bounds are strictly positive across all specifications.

F.2.3 Imbens-Manski Confidence Intervals

I construct confidence intervals for the set-identified LATE using the method of Imbens and Manski (2004), which targets coverage of the true parameter θ_0 at nominal level $1-\alpha$. Standard bootstrap or percentile methods may over-cover in set-identified settings.

The procedure uses a subsampling algorithm with $B = 1{,}000$ draws. For each draw, I select a subsample of size $b = \lfloor n^{0.67} \rfloor$, where n is the full sample size, and re-estimate the entire identification region to obtain subsample bounds $[\hat{\tau}_b^{LB}, \hat{\tau}_b^{UB}]$. The standard errors SE_{LB} and SE_{UB} are the standard deviations of the subsample estimates, scaled to the full-sample size.

The final confidence interval is $[\hat{\tau}^{LB} - C_n \cdot SE_{LB},\ \hat{\tau}^{UB} + C_n \cdot SE_{UB}]$, where C_n adapts to the width of the identified set. When the bounds are wide relative to the standard errors, C_n approaches $z_{1-\alpha} \approx 1.645$. For narrow bounds, C_n approaches $z_{1-\alpha/2} \approx 1.96$. The interval thus maintains valid coverage whether the parameter is partially or nearly point identified.